

Ascent sequences avoiding 000 (OEIS A202058)

Factorial growth constant, finer bounds, logarithmic-square growth and the ratio limit

Four AI-assisted research reports, merged into one ProveIt report

Part I: 1 October 2026 Parts II–IV: 2 October 2026 merged: 2 October 2026

Abstract

Let a_n count ascent sequences in which every value occurs at most twice (OEIS A202058), put $T = 3\pi^2/8$, $\mu = 1/T = 8/(3\pi^2)$, and $c_n = a_n T^n / n!$. This report prints four research manuscripts as Parts I–IV. Part I proves that the exponential generating function has radius T and that $(a_n/n!)^{1/n} \rightarrow \mu$, the constant conjectured by Conway, Conway, Elvey Price and Guttmann, with a coarse error $O(n^{9/13}(\log n)^3)$ in $\log c_n$. Part II proves $-O(n^{2/3}(\log n)^{5/3}) \leq \log c_n \leq O((\log n)^2)$, a controlled Lambert W inverse, exact counts through 400, and the conjecture $\log c_n \sim \frac{2}{3}(\log n)^2$. Part III proves $\log A(t) \sim \frac{2}{3} \log^2 \frac{1}{T-t}$, the cumulative law $\log \sum_{j \leq N} c_j \sim \frac{2}{3}(\log N)^2$ and $\limsup \log c_n / (\log n)^2 = \frac{2}{3}$, so Part II's conjecture holds in limsup and cumulative form. Part IV proves $a_{n+1}/((n+1)a_n) = \mu + O(n^{-1/4}(\log n)^{3/4})$ and improves the lower bound to $-O(\sqrt{n}(\log n)^{3/2})$. The pointwise law $\log c_n \sim \frac{2}{3}(\log n)^2$, normalized log-concavity, a full asymptotic equivalent, an amplitude and an all-orders expansion remain open. The report is AI-assisted and unrefereed, and nothing in it is formalized.

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Guide to this report

The four Parts are four manuscripts of batch 77 (arrival commit `096ee7b87`, 2 October 2026), placed in this directory by commit `f76fcb566`. They form one chain, printed in the order in which they were written and in which each cites the previous ones:

Part	Batch-77 manuscript	Delivered title	Date
I	37 (<code>oeis-a202058-report</code>)	Factorial growth of ascent sequences avoiding 000	1 Oct.
II	36 (<code>oeis-a202058-fine-report</code>)	Finer bounds for ascent sequences avoiding 000	2 Oct.
III	07 (<code>a202058-lognormal-report</code>)	Logarithmic square growth for ascent sequences avoiding 000	2 Oct.
IV	08 (<code>a202058-ratio-report</code>)	Ratio asymptotics for ascent sequences avoiding 000	2 Oct.

Each Part prints its manuscript’s text, with every result, proof, remark, open question and limitation. Its statements keep their delivered wording. Sections, theorems and equations are numbered continuously through the report. Part I keeps exactly the section, statement and equation numbers of its manuscript. In the other Parts a section number is the delivered one plus a fixed offset, statement numbers follow their sections, and equation numbers continue from the preceding Part. The report’s README tabulates the offsets. Each Part has its own symbols, fixed in its own text; Section “Notation across the four Parts” lists the symbols whose meaning changes between Parts. Notes added when the manuscripts were merged are set in small type and tagged **[write]**. They give cross-references between the Parts, mark results proved twice, and mark statements that a later Part supersedes. No delivered statement was changed, and no symbol was renamed.

The neighbouring reports `a202061-ascent-120-deficit` and `a202062-ascent-201-enumeration` in the same collection treat the 120- and 201-avoiding ascent sequences, which Conway, Conway, Elvey Price and Guttmann [1] studied together with this one. Their regimes and methods are different: those two sequences grow exponentially, with common growth constant the largest root $\mu \approx 7.2959$ of $\mu^3 - 8\mu^2 + 5\mu + 1 = 0$, while A202058 grows factorially. The letter μ there is not the $\mu = 8/(3\pi^2)$ of this report.

What is proved, and where

Here $c_n = a_n T^n / n! = a_n / (n! \mu^n)$; Part II writes $h_n = \log c_n$.

Statement	Where
EGF radius T ; $(a_n/n!)^{1/n} \rightarrow \mu$	Part I, Theorem 1.1. Radius again in Part III, Theorem 21.1; root limit again in Part III, Section 30.1 (second route)
$ \log c_n = O(n^{9/13}(\log n)^3)$	Part I, (2)
$-O(n^{2/3}(\log n)^{5/3}) \leq \log c_n \leq O((\log n)^2)$; no equivalent $C_0 n! \mu^n e^{c n^\sigma} n^g (\log n)^h$ with $c, \sigma > 0$	Part II, Theorem 9.1; again in Part III, (146)
$\log c_n \geq -O(\sqrt{n}(\log n)^{3/2})$	Part IV, Theorem 33.1
$\log A'(t(q)) = q^2/6 + O(q^{3/2}\sqrt{\log q})$; $\log A(t) \sim \frac{2}{3} \log^2 \frac{1}{T-t}$	Part III, Theorems 21.1 and 28.1
$\log \sum_{j \leq N} c_j =$ $\frac{2}{3}(\log N)^2 + O((\log N)^{3/2}\sqrt{\log \log N})$; $\limsup \log c_n/(\log n)^2 = \frac{2}{3}$	Part III, Theorem 21.1 and (140)
$a_{n+1}/((n+1)a_n) \rightarrow \mu$	Part IV, (174) (rate $n^{-1/6}(\log n)^{5/6}$, from Part II's bounds) and (153) (rate $n^{-1/4}(\log n)^{3/4}$)
Exact identity $R_{n+1} = R_n + (\text{Var}_n(m) + \mathbb{E}_n(u-k))/R_n$, $R_n = a_{n+1}/a_n$	Part II, (66) (written M_n); Part IV, (166), with the new consequence $R_{n+1} \geq R_n - 1$
Threshold inverses around $x = y/W(y/(eT))$	Part I, Corollary 7.1; Part II, Corollary 9.2; Part III, Section 30; Part IV, (155)
<i>Open in all four Parts</i>	
The pointwise law $\log c_n \sim \frac{2}{3}(\log n)^2$ (Part II, Conjecture 16.1; Part III, (76); Part IV, (160)); eventual normalized monotonicity or log-concavity; a multiplicative equivalent $a_n \sim Cn!\mu^n \dots$, a power prefactor, a limiting amplitude, an all-orders expansion; optimal remainders and effective constants.	

Finite computations in all four Parts (exact counts, residual grids, log-concavity scans) are diagnostics and evidence; no asymptotic or uniform statement rests on them. The literature and repository searches recorded in the Parts are scoped negative evidence, not novelty or priority certificates. The Lambert W approximation $x = y/W(y/(eT))$ used by every inverse solves $x \log(x/(eT)) = y$; after taking logarithms this is the case $a = b = 1$, $L = \log(y/(eT))$ of the exact Lambert-core theorem of the repository's transseries volume (`Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/`, Theorem `p0:thm:lambert-core`). No novelty is claimed for the inversion method; what each Part adds is the error envelope that the inverse inherits from its coefficient bounds.

Notation across the four Parts

Common to all Parts: a_n , $A(t) = \sum a_n t^n/n!$, $T = 3\pi^2/8$, $\mu = 1/T$, the compacted states $x = (s, u, k)$ with $m = s + u$, the four children of (4), the operator $\mathcal{T}$, the continuation series ($\mathcal{F}$ in Parts I and II, F in Parts III and IV; the same function), and the characteristic quantities $p(q)$, $b(q)$, $t(q)$, ψ , $D = s + Ru$ with $R = R(q) = 2 - e^{-q}$ (written $\mathcal{R}$ in Part IV). The symbols below change meaning between Parts; each Part's own text fixes its meaning there.

Symbol	Meanings
c_n, c_j	Parts III and IV: $c_n = a_n T^n / n!$. Part II, Section 16: $c_n = n(g_n - g_{n-1})$ with $g_n = n(h_n - h_{n-1})$, a discrete second difference. Part I, Lemma 6.1, and Part II, Section 14: $c_j = (\mathcal{T}^j \mathbf{1})(x_N)$. <i>Tempting false reading</i> : the column $c_n \approx 1.37\text{--}1.48$ of Table 1 is not $a_n T^n / n!$; it estimates 2α .
h_n, h	Part II: $h_n = \log(a_n / (n! \mu^n))$, which is $\log c_n$ of Parts III and IV. $h(y)$ is the rank function in Parts I and II ($h_0(y)$ in Parts III and IV). In Part IV, Section 38, h is an extension length.
N, y	Everywhere $x_N = (N, 1, N)$ is the growing seed state. Part I: $N(L) = L/W(L/(eT))$ with $L = \log y$, where y is a <i>count</i> and $\nu(y) = \min\{n : a_n \geq y\}$. Parts II–IV: y is the <i>logarithm</i> of a count, $N(y)$ (Part II) = $N_a(y)$ (Parts III, IV) = $\min\{n \geq 1 : \log a_n \geq y\}$. Thus Part I's $\nu(Y)$ is Part II's $N(\log Y)$, and $N(L)$ of Part I is the x of Parts II–IV.
x_0, x_*	Parts I–III: $x_0 = (1, 1, 1)$, $(\mathcal{T}^n \mathbf{1})(x_0) = a_{n+1}$, continuation series $A'(t)$. Part IV: $x_* = (0, 1, 0)$, $(\mathcal{T}^n \mathbf{1})(x_*) = a_n$, continuation series $A(t)$.
E	Part I: $E(q) = \int_0^q C(v)/b(v) dv$, a barrier exponent. Parts III and IV: $E = e^{-q}$. Part IV, Lemma 36.1: $E(n)$, an error envelope. $\mathbb{E}$ is always an expectation.
H	Part II: $H(q)$, the quadratic barrier exponent (Parts III and IV call it $V(q)$). Parts III and IV: $H(x) = s + 2u$, the integer resource. $H_q(v)$ is the Chernoff function in all Parts.
M, R	$M(q) = t(q)p_t(t(q))$ in all Parts. Part II: $M_n = \mathbb{E}_n[m] = a_{n+1}/a_n$, which is Part IV's R_n . $R(q) = 2 - e^{-q}$ in Parts I–III; in Part IV that function is $\mathcal{R}$, $R_n = a_{n+1}/a_n$, and in Section 40 L, R are the two sides of an inequality.
d, D	$d(q) = (\log t(q))'$ in Parts I–III ($\ell(q)$ in Part IV); Part IV: $d(x) = u - k$. $D = s + Ru$ everywhere; Part IV also uses $D_n = R_n - \mu(n+1)$; “ D step” means a duplicate step.
A_0, B_0	Part I: $A_0 = e^{-p}(e^q - 1)$, $B_0 = RA_0$. Part IV: large fixed constants.
L, K	L : Part I $\log y$; Parts II, III padding integer $\lceil Q \rceil$; Part III, (143) a constant; Part IV $1 + \mu$ in Lemma 36.1. K : Part I $NM(q)$; Parts II, III a fixed large constant; Part IV K_j counts paths.
$b_j, C(\cdot), S$	b_j is the characteristic speed; $b_j = (\mathcal{T}^j \mathbf{1})(x_N)$ in Parts III and IV. $C(q) = 8(1+q)e^{5q/8}$ in Part I; $C(z) = A(Tz)$ in Part III. $S = \sum_i g(i)$ in Parts II–IV; $S_N = \sum_{n \leq N} c_n$ in Parts III and IV.
Lemma “slow”	Part III, Lemma 26.1 (slow profile) and Part IV, Lemma 36.1 (slow decrease of ratios) are different lemmas.

The Feynman–Kac representation of the continuation series is printed in all four Parts: (16), (46), (97) and (184) are the same identity.

Provenance and merge decisions

Four manuscripts, all of batch 77 (arrival commit 096ee7b87; placement commit f76fcb566). The pins are the ProveIt commits recorded by each delivery’s repository search.

- **Part I**, manuscript 37, archive `oeis-a202058-report.zip`, 1 October 2026, author line “A proof and reproducible research report”. Pin 0870c19bb (recorded in its literature note, which was delivered only inside manuscript 36’s frozen copy of this report). Contributes the radius, the root limit, the 9/13 error and the first inverse.
- **Part II**, manuscript 36, archive `oeis-a202058-fine-report.zip`, 2 October 2026, author line “A mathematical addendum with reproducible computations”. Pin 13ef7d939. Contributes the asymmetric bounds, the asymmetric inverse, exact counts to 400 and floating values to 1000, the conjecture, the regularity obstructions and the formal route. The archive also contains an unpacked and a zipped copy of manuscript 37.
- **Part III**, manuscript 07, archive `a202058-lognormal-report.zip`, 2 October 2026, author line “A self contained mathematical and reproducibility report”. Pin 4b874cea0. Contributes

the real-axis, cumulative and limsup logarithmic-square theorems and their quantitative refinement.

- **Part IV**, manuscript 08, archive `a202058-ratio-report.zip`, 2 October 2026, author line “A sequel with complete proofs and reproducibility materials”. No pin; it records no repository search. Contributes the ratio theorem, the improved lower bound and inverse, and the childwise counterexample. The archive also contains a byte-identical copy of manuscript 07.

Where the merge had to choose:

1. *Order and base*. The order is the order of writing and citation, which is also the logical order. Manuscript 37 is the base because it is the foundation that the other three use.
2. *Repeated foundations*. Parts II–IV each re-derive the compacted model, the characteristic endpoint $t(\infty) = 3\pi^2/8$, the coarse comparison and the growing-seed Chernoff estimate, so that each manuscript is self-contained. These re-derivations are kept where they stand, because each Part’s later proofs cite their equations; **[write]** notes mark them. The two re-proofs of whole theorems are marked as second routes: Part III’s all-index enclosure and inverse re-prove Part II’s Theorem 9.1 and Corollary 9.2, and Part III re-derives Part I’s root limit.
3. *References between manuscripts*. Manuscript 36 cited manuscript 37, and manuscript 08 cited manuscript 07, as bundled reports. These citations are now references to Parts I and III. Where manuscript 08 named manuscript 07’s sections and labels literally, they are now references. Manuscript 07 did not cite manuscripts 37 or 36; a **[write]** note supplies the attribution.
4. *Front matter*. The four title blocks and abstracts are printed at the head of their Parts. Manuscripts 07 and 08 each had a table of contents; this report has one. Manuscript 36 had an appendix (“Bundled sources and reproduction”); here it is the last section of Part II.
5. *Bibliography*. The four bibliographies are merged into one, keeping every detail of every entry.
6. *Figure*. Part II’s figure is included from `figures/36-fine-correction-diagnostics.pdf`, the delivered file under its shipped name.
7. *Superseded statements*. Statements that a later Part supersedes (the ratio limit “unresolved” or “unproved” in Parts I–III; the coefficient $2/3$ unproved in Part II) are printed as delivered, with a dated note.

Part I

Factorial growth of ascent sequences avoiding 000

Batch-77 manuscript 37, archive `oeis-a202058-report.zip` (delivered as `a202058-report.tex`, 9 pages), dated 1 October 2026. Delivered author line: “A proof and reproducible research report”. Printed as delivered, with `[write]` notes; section, statement and equation numbers are those of the delivery.

Abstract of this Part (as delivered). We prove that the number a_n of ascent sequences in which each value occurs at most twice satisfies

$$\lim_{n \rightarrow \infty} (a_n/n!)^{1/n} = \frac{8}{3\pi^2}.$$

This is the factorial exponential constant conjectured by Conway, Conway, Elvey Price and Guttmann. The proof starts from their exact compacted recurrence. A positive transfer operator admits uniform exponential barriers whose characteristic curve has endpoint $3\pi^2/8$. Stopped Feynman–Kac comparison proves the exact exponential-generating-function radius. A growing-state coefficient argument upgrades the resulting limsup to a full limit. We also obtain a non-optimal logarithmic error bound and a controlled leading Lambert W inverse. The ratio limit, prefactor, and possible stretched-exponential corrections remain unresolved.

1 Result and scope

An ascent sequence is a word $w_1 \cdots w_n$ of nonnegative integers such that $w_1 = 0$ and

$$w_i \leq 1 + \text{asc}(w_1 \cdots w_{i-1}) \quad (i \geq 2), \quad \text{asc}(w) = \#\{i : w_i < w_{i+1}\}.$$

Avoidance of the pattern 000 means that no value occurs three times. Let a_n count these words, with $a_0 = 1$. The sequence is OEIS A202058 [2].

Theorem 1.1. *Put*

$$T = \frac{3\pi^2}{8}, \quad \mu = T^{-1} = 0.27018982304623405718 \dots$$

Then the exponential generating function $A(t) = \sum_{n \geq 0} a_n t^n / n!$ has radius T , and

$$\lim_{n \rightarrow \infty} \left(\frac{a_n}{n!} \right)^{1/n} = \mu. \tag{1}$$

More quantitatively,

$$\log a_n = \log(n!) - n \log T + O(n^{9/13} (\log n)^3). \tag{2}$$

Conway–Conway–Elvey Price–Guttmann [1] conjectured the constant $8/(3\pi^2)$ from long exact series. Their Section 5 also leaves a possible stretched-exponential factor unresolved; its numerical discussion is more cautious than the simplified introduction. The recurrence and its compaction argument are theirs. The contribution proved here is the exact factorial growth constant, including existence of the normalized root limit, together with the coarse error and inverse below. A scoped literature search found no later proof; this is not a comprehensive novelty certification before publication.

The exponent $9/13$ in (2) comes from coarse uniform bounds. It is not a prediction of the actual subexponential scale. In particular, this report does not establish $a_n \sim Cn!\mu^n$, a ratio limit, a power prefactor, or an all-orders expansion.

[write] 2 October 2026. Every quantitative statement of this Part is sharpened in later Parts, and the ratio limit that this paragraph leaves open is proved there. The error in (2) becomes $-O(n^{2/3}(\log n)^{5/3}) \leq \log(a_n/(n!\mu^n)) \leq O((\log n)^2)$ in Part II (Theorem 9.1), and its lower side becomes $-O(\sqrt{n}(\log n)^{3/2})$ in Part IV (Theorem 33.1). Part III proves $\limsup \log(a_n/(n!\mu^n))/(\log n)^2 = 2/3$ and the matching cumulative law (Theorem 21.1). Part IV proves $a_{n+1}/((n+1)a_n) \rightarrow \mu$ with rate $O(n^{-1/4}(\log n)^{3/4})$. A relation $a_n \sim Cn!\mu^n$, a power prefactor and an all-orders expansion remain unproved in every Part. Part III re-derives the radius and the root limit of Theorem 1.1 by its own route (second proofs).

2 The exact compacted recurrence

Twice-used labels can no longer occur, and may be deleted from the active alphabet. The remaining once-used labels can be compacted to an initial segment. In the notation of [1, Sections 2.2 and 2.6], if $F_0(a, \ell, s) = 1$, then

$$F_n(a, \ell, s) = \sum_{i=0}^{s-1} F_{n-1}(a + \mathbf{1}_{i>\ell} - 1, i - 1, s - 1) + \sum_{i=s}^{a+1} F_{n-1}(a + \mathbf{1}_{i>\ell}, i, s + 1), \quad (3)$$

and $a_n = F_{n-1}(0, 0, 1)$ for $n \geq 1$. The adjusted quantities a and ℓ may equal -1 .

For completeness, the compaction mechanism is as follows. If adjacent labels $i, i+1$ have different remaining capacities and $i < \ell$, reverse every maximal suffix block using only these two labels, and interchange the two labels. This interchanges their multiplicities and preserves the total number of ascents in each block, including the comparisons with outside labels. Both labels are already allowed because $i+1 \leq \ell$. Thus the ascent bounds inside a block remain valid, and those after the block see the same number of preceding ascents. The operation is an involution. In the binary-capacity setting needed here, it exchanges a once-used and an unused label below the last value. After appending a previously unused i , bubble it down through unused labels until the once-used set is again an initial segment; after a repetition, delete the exhausted label. This gives (3). The canonical states need not themselves retain every property of an original prefix; they retain its suffix counts.

Set $m = a + 2$, $k = \ell + 1$, and $u = m - s$. The following slightly larger invariant state space is convenient:

$$\mathcal{S} = \{(s, u, k) : s \geq 0, u \geq 1, 0 \leq k < s + u\}, \quad m = s + u.$$

For $x = (s, u, k)$, choose each integer $i \in \{0, \dots, m-1\}$ once and form the child

$$x_i = \begin{cases} (s-1, u, i), & i < s, i < k, \\ (s-1, u+1, i), & i < s, i \geq k, \\ (s+1, u, i+1), & i \geq s, i \geq k, \\ (s+1, u-1, i+1), & i \geq s, i < k. \end{cases} \quad (4)$$

Call the first two types D and the latter two N , for a duplicate or new value. An ascent means $i \geq k$. The last case requires $u \geq 2$, so $\mathcal{S}$ is invariant. Every step changes m by at most one.

Define the positive operator

$$(\mathcal{T}f)(x) = \sum_{i=0}^{m-1} f(x_i), \quad x_0 = (1, 1, 1).$$

Then $\mathcal{T}^n \mathbf{1}(x_0) = a_{n+1}$. Write

$$\mathcal{F}(t, x) = \sum_{n \geq 0} \mathcal{T}^n \mathbf{1}(x) \frac{t^n}{n!}, \quad (5)$$

allowing the value $+\infty$ for $t \geq 0$. We have $\mathcal{F}(t, x_0) = A'(t)$, so the two series have the same radius.

3 A characteristic curve and rank profile

For $q \geq 0$ define

$$\begin{aligned} p(q) &= \frac{1}{2} \log(2e^q - 1), & R(q) &= 2 - e^{-q}, \\ b(q) &= \frac{e^{p(q)}(1 - e^{-q})}{q}, & t(q) &= \int_0^q \frac{dv}{b(v)}, \end{aligned} \quad (6)$$

with $b(0) = 1$. Put $T_* = t(\infty)$ and use $q = q(t)$ for $0 \leq t < T_*$. Along this curve,

$$\frac{dq}{dt} = \frac{e^p(1 - e^{-q})}{q}, \quad \frac{dp}{dt} = \frac{e^{-p}(e^q - 1)}{q}. \quad (7)$$

The identity $e^{2p} = 2e^q - 1$ verifies both equations and their initial values.

For a state $x = (s, u, k)$ let

$$\begin{aligned} D &= s + Ru, & h(y) &= \min(y, s) + R \max(y - s, 0), \\ r(y) &= h(y)/D, & \psi(t, x) &= \exp\{ps + qu - qr(k)\}. \end{aligned} \quad (8)$$

In particular $0 \leq r(k) \leq 1$, $\psi(0, x) = 1$, and

$$e^{ps+qu-q} \leq \psi(t, x) \leq e^{ps+qu}. \quad (9)$$

Write

$$A_0 = e^{-p}(e^q - 1), \quad B_0 = e^p(1 - e^{-q}) = RA_0, \quad \lambda = \frac{A_0 s + B_0 u}{q} = \frac{A_0 D}{q}.$$

The removable value of λ at $q = 0$ is m .

Lemma 3.1. *Uniformly over $x \in \mathcal{S}$ and $0 \leq t < T_*$,*

$$\left| \frac{\partial_t \psi(t, x)}{\psi(t, x)} - \frac{\mathcal{T} \psi(t, x)}{\psi(t, x)} \right| \leq C(q), \quad C(q) = 8(1 + q)e^{5q/8}. \quad (10)$$

Proof. We keep all three sources of error uniform in the state.

The frozen rank integral. For real $y \in [0, m]$ put

$$w(y) = \begin{cases} e^{-p+q\mathbf{1}_{y \geq k}}, & y < s, \\ e^{p-q\mathbf{1}_{y < k}}, & y \geq s, \end{cases} \quad g(y) = w(y)e^{-q(r(y)-r(k))}.$$

Direct integration gives

$$\int_0^m g(y) dy = \lambda. \quad (11)$$

For a verification, set $\phi(y) = e^{-qh(y)/D}$. On the two intervals, $\lambda\phi' = -A_0\phi$ and $\lambda\phi' = -B_0\phi$, respectively, while $\phi(m) = e^{-q}\phi(0)$. If $J(k) = \phi(k) \int g$, these derivative identities show that $J - \lambda\phi$ is constant. At $k = 0$, writing $X = e^{-qs/D}$ and using $e^p/R = e^{q-p}$ gives

$$J(0) = \frac{D}{q} \left[e^{q-p}(1-X) + \frac{e^p}{R}(X - e^{-q}) \right] = \frac{D}{q} e^{-p}(e^q - 1) = \lambda.$$

This proves (11), including the continuous value at $q = 0$.

The inequalities

$$q/2 \leq p \leq q/2 + (\log 2)/2$$

show that $g(y) \leq \sqrt{2}e^{q/2}$. On an ascent interval use $r(y) \geq r(k)$; on a descent interval use $r(k) - r(y) \leq 1$. On each of the at most three intervals separated by the integers s, k , g decreases. The left Riemann sum therefore differs from its integral by at most

$$\left| \sum_{i=0}^{m-1} g(i) - \lambda \right| \leq 3\sqrt{2}e^{q/2}. \quad (12)$$

The shifted states. Let r_i be the rank fraction at the exact child x_i . In the order of the four cases in (4),

$$r_i = \frac{i}{D-1}, \quad \frac{i}{D+R-1}, \quad \frac{h(i)+1}{D+1}, \quad \frac{h(i)+1}{D+1-R}. \quad (13)$$

The relevant denominators are positive. Subtraction yields $|r_i - r(i)| \leq 4/D$. Except in the D -ascent case, $r_i \geq r(i)$, so the exact transition ratio $\psi(x_i)/\psi(x)$ is no greater than its frozen counterpart $g(i)$.

For a D ascent, $k \leq i < s$ and

$$r(k) - r_i \leq \frac{k(R-1)}{D(D+R-1)} \leq \frac{D-2}{D^2} \leq \frac{1}{8}.$$

Here $k \leq s-1$, $D \geq s+1$, and $1 \leq R \leq 2$. Thus every exact or frozen ratio is at most $\sqrt{2}e^{5q/8}$. The exponential mean-value inequality gives

$$\left| \frac{\psi(x_i)}{\psi(x)} - g(i) \right| \leq \frac{4q}{D} \sqrt{2}e^{5q/8}.$$

Since $m \leq D$, its sum is at most $4\sqrt{2}qe^{5q/8}$.

The time derivative. At fixed state, differentiation gives

$$\frac{\partial_t \psi}{\psi} = \lambda - q' \{r(k) + qe^{-q} \partial_R r(k)\}.$$

The two expressions for $\partial_R r(k)$ are $-ku/D^2$ for $k \leq s$ and $-s(m-k)/D^2$ for $k \geq s$. Both lie in $[-1/(4R), 0]$. The expression in braces has absolute value at most one. Also $q' \leq \sqrt{2}e^{q/2}$. Combining these bounds with (12) proves

$$|\partial_t \psi/\psi - \mathcal{T}\psi/\psi| \leq 4\sqrt{2}e^{q/2} + 4\sqrt{2}qe^{5q/8} \leq 8(1+q)e^{5q/8}.$$

□

4 Comparison for the infinite state space

Define

$$E(q) = \int_0^q \frac{C(v)}{b(v)} dv, \quad G_{\pm}(t, x) = e^{\pm E(q(t))} \psi(t, x). \quad (14)$$

Lemma 3.1 gives $\partial_t G_- \leq \mathcal{T}G_-$ and $\partial_t G_+ \geq \mathcal{T}G_+$, with initial values one.

Lemma 4.1. *For every state and $0 \leq t < T_*$,*

$$e^{-E(q)} \psi(t, x) \leq \mathcal{F}(t, x) \leq e^{E(q)} \psi(t, x). \quad (15)$$

Proof. Consider the continuous-time chain which takes each of the m transitions in (4) at rate one. Its generator is $\mathcal{L} = \mathcal{T} - mI$. Its jump rate is m , and each jump increases m by at most one, so a Yule-process domination proves nonexplosion. Let τ_N be the first exit from $m \leq N$. Each stopped state space is finite.

Summing over its finite jump paths gives the extended Feynman–Kac identity

$$\mathcal{F}(t, x) = \mathbb{E}_x \exp \left\{ \int_0^t m(X_v) dv \right\}. \quad (16)$$

Indeed, the holding-time exponential is canceled by the displayed weight. Each n -step path then contributes the simplex volume $t^n/n!$, exactly as in (5). Nonexplosion and Tonelli justify the sum even when it is infinite.

For the supersolution, stopped Dynkin comparison applied to

$$\exp \left\{ \int_0^v m(X_z) dz \right\} G_+(t-v, X_v)$$

gives

$$\mathbb{E}_x \left[\exp \left\{ \int_0^t m(X_v) dv \right\}; \tau_N > t \right] \leq G_+(t, x).$$

Letting $N \rightarrow \infty$ proves the upper bound and finiteness before T_* .

The lower comparison requires control of exit terms. Choose $\varepsilon > 0$ such that $t + \varepsilon < T_*$. Uniformly in $v \in [0, t]$ and states y ,

$$\frac{G_-(v, y)}{G_+(v + \varepsilon, y)} \leq e^{q(t+\varepsilon)} e^{-\eta m(y)}, \quad (17)$$

where $\eta > 0$ is the minimum, on this compact interval, of both $p(v + \varepsilon) - p(v)$ and $q(v + \varepsilon) - q(v)$. These functions are strictly increasing. Inequality (17) follows directly from (9) and $E \geq 0$.

Stopped Dynkin comparison for G_- bounds $G_-(t, x)$ above by the terminal expectation in (16), restricted to $\tau_N > t$, plus the exit contribution. Use (17) and then the later supersolution to bound the latter by

$$e^{q(t+\varepsilon)} e^{-\eta(N+1)} G_+(t + \varepsilon, x).$$

It tends to zero. Passing to the limit gives the lower bound. This step avoids an unjustified comparison of an arbitrary subsolution with an unbounded positive operator. $\square$

For $q \geq 1$,

$$b(q) \geq (1 - e^{-1})e^{q/2}/q, \quad E(q) = O((1 + q)^3 e^{q/8}) = o(e^p).$$

This deliberately loose estimate suffices for all subsequent arguments.

5 The exact radius

The upper comparison proves that the radius of $\mathcal{F}(t, x_0)$ is at least T_* . Positivity and Tonelli give the semigroup identity

$$\mathcal{F}(t + \delta, x_0) = \sum_{j \geq 0} \frac{\delta^j}{j!} (\mathcal{T}^j \mathcal{F}(t, \cdot))(x_0),$$

with infinity allowed. There is a path of j successive new-maximal-label steps from x_0 to $(j+1, 1, j+1)$. At this state the $qu - qr$ term is nonnegative. Hence

$$\mathcal{F}(t + \delta, x_0) \geq e^{p-E(q)} \sum_{j \geq 0} \frac{(\delta e^p)^j}{j!} = \exp\{p - E(q) + \delta e^p\}.$$

For any fixed $\delta > 0$ the right side tends to infinity as $t \uparrow T_*$, because $E(q) = o(e^p)$. Monotonicity gives $\mathcal{F}(T_* + \delta, x_0) = \infty$. Its radius is therefore exactly T_* .

To evaluate the endpoint, put $y = \sqrt{2e^q - 1}$ in (6):

$$T_* = \int_1^\infty \frac{2 \log((y^2 + 1)/2)}{y^2 - 1} dy.$$

Integrate by parts using $\log((y - 1)/(y + 1))$ as an antiderivative of $2/(y^2 - 1)$, and then set $z = (y - 1)/(y + 1)$. Both endpoint terms vanish, and

$$\begin{aligned} T_* &= -2 \int_0^1 \log z \frac{1 + z}{(1 - z)(1 + z^2)} dz \\ &= -2 \int_0^1 \frac{\log z}{1 - z} dz - 2 \int_0^1 \frac{z \log z}{1 + z^2} dz = \frac{\pi^2}{3} + \frac{\pi^2}{24} = \frac{3\pi^2}{8}. \end{aligned}$$

The geometric-series integrations are absolutely integrable. We have proved the radius assertion and the corresponding factorial-normalized limsup. The next section establishes the full limit.

6 From the radius to the full coefficient limit

For $x_N = (N, 1, N)$, $N \geq 1$, the uniform comparison becomes

$$\log \mathcal{F}(t(q), x_N) = Np(q) + O(E(q) + q), \quad (18)$$

uniformly in N , since the extra rank term is $qR/(N + R) \in [0, q]$. Put

$$d(q) = \frac{d}{dq} \log t(q) = \frac{1}{t(q)b(q)}, \quad M(q) = \frac{p_q(q)}{d(q)} = t(q)p_t(t(q)).$$

As $q \rightarrow \infty$,

$$p_q = \frac{1}{2} + O(e^{-q}), \quad d(q) \sim \frac{qe^{-q/2}}{\sqrt{2T}}, \quad M(q) \sim \frac{Te^{q/2}}{\sqrt{2q}}. \quad (19)$$

Lemma 6.1. *Let $c_j = \mathcal{T}^j \mathbf{1}(x_N)$, and define*

$$\mathbb{P}(J = j) = \frac{c_j t(q)^j}{j! \mathcal{F}(t(q), x_N)}, \quad K = NM(q).$$

For all sufficiently large q and $0 < \epsilon < 1/2$,

$$\mathbb{P}(|J - K| > \epsilon K) \leq 2 \exp \left\{ -\frac{3N\epsilon^2}{128} + 2E(q+1) + O(q+1) \right\}, \quad (20)$$

uniformly in N and ϵ .

Proof. For $|v| \leq 1$ set

$$H_q(v) = p(q+v) - p(q) - M(q)\{\log t(q+v) - \log t(q)\}.$$

We have $H_q(0) = H'_q(0) = 0$. The exact formulas

$$p_q = 1/R, \quad \frac{d'}{d} = -d - \frac{1}{R} - \frac{1}{e^q - 1} + \frac{1}{q}$$

show, uniformly for bounded v , that

$$H''_q(v) \longrightarrow \frac{1}{4}e^{-v/2}, \quad M(q)d(q+v) \longrightarrow \frac{1}{2}e^{-v/2}.$$

Thus for all sufficiently large q , $|H_q(v)| \leq v^2/2$ on $[-1, 1]$. Also, for $0 < \eta < 1/2$,

$$M(q)|\log t(q \pm \eta) - \log t(q)| \geq \eta/4.$$

Choose $\eta = \epsilon/8$. The Chernoff upper-tail bound, with the generating function tilted from $t(q)$ to $t(q + \eta)$, has logarithm at most

$$NH_q(\eta) - \epsilon NM(q)\{\log t(q + \eta) - \log t(q)\} + 2E(q+1) + O(q+1).$$

It is at most $-3N\epsilon^2/128 + 2E(q+1) + O(q+1)$. The tilt $q - \eta$ gives the same bound for the lower tail. All error constants follow uniformly from (18). $\square$

For each sufficiently large integer n , take

$$q = \log n, \quad \epsilon = (\log n)^{-2}, \quad N = \left\lfloor \frac{(1 - 4\epsilon)n}{M(q)} \right\rfloor. \quad (21)$$

Then

$$N \sim \frac{\sqrt{2}}{T} \sqrt{n} \log n, \quad K = (1 - 4\epsilon)n + O(\sqrt{n}/\log n),$$

while $N\epsilon^2$ has order $\sqrt{n}/(\log n)^3$ and $E(q + 1) = O(n^{1/8}(\log n)^3) = o(N\epsilon^2)$. Lemma 6.1 puts at least half the probability in $|j - K| \leq \epsilon K$. Some integer j in this interval therefore satisfies

$$\log c_j \geq \log(j!) - j \log t(q) + Np(q) - E(q) - O(\log n). \quad (22)$$

Indeed, there are at most $2\epsilon K + 3$ integers in the interval, and the lower comparison gives $\log \mathcal{F} \geq Np - E$.

The floor error in (21) is $o(\epsilon n)$, and $N = o(\epsilon n)$. Consequently

$$(1 - 6\epsilon)n \leq j, \quad N + j \leq n$$

for large n . The initial state reaches x_N after $N - 1$ new-maximal-label steps, so $a_{N+j} \geq c_j$. Every valid ascent sequence extends injectively by appending one plus its current number of ascents. The appended label is new: induction gives $\max w \leq \text{asc}(w)$ for every ascent sequence. Hence $a_n \geq a_{N+j} \geq c_j$.

It follows that

$$\log(a_n/n!) \geq -j \log t(q) - \log(n!/j!) + Np(q) - E(q) - O(\log n) \geq -n \log T - o(n).$$

Here $j/n \rightarrow 1$, $t(q) \rightarrow T$, and $\log(n!/j!) \leq 6\epsilon n \log n = o(n)$. The radius theorem supplies the matching limsup, proving (1).

7 A coarse error and a threshold inverse

For a quantitative version, repeat the coefficient argument with

$$q = \frac{8}{13} \log n, \quad \epsilon = n^{-4/13}(\log n)^2.$$

Now N has order $n^{9/13} \log n$, $E(q + 1) = O(n^{1/13}(\log n)^3)$, and $N\epsilon^2$ has order $n^{1/13}(\log n)^5$. Thus the same Chernoff estimate applies, and $N = o(\epsilon n)$. The factorial loss is

$$O(\epsilon n \log n) = O(n^{9/13}(\log n)^3),$$

which proves the lower half of (2).

For the upper half, coefficient positivity and monotonicity give

$$\frac{a_n}{n!} \leq \frac{A(t)}{t^n}, \quad A(t) = 1 + \int_0^t \mathcal{F}(v, x_0) dv \leq 1 + t\mathcal{F}(t, x_0).$$

The superbarrier yields $\log A(t(q)) \leq E(q) + O(q + 1)$, whereas $T - t(q) = O(qe^{-q/2})$. At the same choice of q this gives

$$\log(a_n/n!) + n \log T \leq O\{E(q) + q + nqe^{-q/2}\} = O(n^{9/13} \log n).$$

This completes Theorem 1.1.

Corollary 7.1. For $y \rightarrow \infty$ let $L = \log y$ and

$$N(L) = \frac{L}{W(L/(eT))},$$

where W is the positive real Lambert function. The integer threshold $\nu(y) = \min\{n \geq 0 : a_n \geq y\}$ satisfies

$$\nu(y) = N(L) + O(N(L)^{9/13}(\log N(L))^2). \quad (23)$$

Proof. Stirling's formula and (2) give

$$\log a_n = f(n) + O(n^{9/13}(\log n)^3), \quad f(x) = x \log \frac{x}{eT}.$$

The chosen $N(L)$ is the positive solution of $f(N) = L$. Since $f'(x) = \log(x/T) \sim \log N$ near N , evaluating at $N \pm CN^{9/13}(\log N)^2$ brackets L for a sufficiently large fixed C . The error envelope is of the same order throughout this $o(N)$ interval. Monotonicity of a_n and integer rounding prove (23). $\square$

The inverse is a controlled leading approximation. This does not provide unit-level rounding accuracy or an all-orders inverse, nor a multiplicative asymptotic equivalent for a_n .

[write] **2 October 2026.** The threshold variable y of Corollary 7.1 is a count; Parts II–IV use the logarithm of a count, and $\nu(Y)$ here is $N(\log Y)$ of Corollary 9.2. That corollary improves the error to $-O(\log x)$ and $+O(x^{2/3}(\log x)^{2/3})$ around $x = N(\log Y)$, and (155) improves the upper displacement to $O(\sqrt{x \log x})$.

8 Verification and remaining questions

The accompanying programs provide independent finite checks of the exact operator, its agreement with (3), and direct word enumeration through length ten. They verify 21 initial coefficients. The uniform residual was also checked on 287820 state-parameter cases with $m \leq 40$ and thirteen positive q values up to 80. These finite checks are diagnostics; the proof of uniformity is Lemma 3.1. A separate numerical script checks the bounded-shift calculus and the characteristic integral at high precision.

[write] **2 October 2026.** The programs and recorded outputs of this Part are shipped as `code/37-foundation-verify_radius.py`, `code/37-foundation-verify_root_limit.py`, `data/37-foundation-verify_radius.json` and `data/37-foundation-verify_root_limit.json`. The radius and root-limit proof drafts, their three independent audits and the 1 October literature note were not in this Part's own archive; they were delivered inside manuscript 36's copy of it and are shipped as `37-foundation-*.md`. The report README describes them and the state of the replay at intake.

The remaining asymptotic questions are substantial. Neither these barriers nor the growing-state coefficient extraction determine whether $a_n/(n!\mu^n)$ has a power or stretched-exponential factor. The method does not prove $a_{n+1}/((n+1)a_n) \rightarrow \mu$. Extracting a prefactor or a full transseries would require sharper control near the singular endpoint and individual-coefficient transfer. The published numerical uncertainty about a weak stretched factor is therefore preserved.

[write] **2 October 2026.** Superseded in part. The ratio limit is proved in Part IV, Theorem 33.1, with rate $O(n^{-1/4}(\log n)^{3/4})$. The qualitative limit already follows from the root limit proved here: it gives $\log c_n = o(n)$, and the qualitative clause of Lemma 36.1 with the exact inequalities (167) turns that into the ratio limit. Combined with Part II's enclosure, the same lemma gives the rate $O(n^{-1/6}(\log n)^{5/6})$ of (174). A positive stretched-exponential factor is excluded by Part II. The power or prefactor question, the transseries question and the full equivalent remain open.

Part II

Finer bounds for ascent sequences avoiding 000

Batch-77 manuscript 36, archive `oeis-a202058-fine-report.zip` (delivered as `a202058-fine-addendum.tex`, 18 pages), dated 2 October 2026. Delivered author line: “A mathematical addendum with reproducible computations”. The delivered “1 October report” and “frozen foundation” are Part I. Delivered Section k is Section $k + 8$ here; the delivered Appendix A is Section 20.

Abstract of this Part (as delivered). Let a_n count ascent sequences in which each value occurs at most twice, and let $\mu = 8/(3\pi^2)$. Building on the exact characteristic endpoint and normalized root limit in the bundled 1 October report, we prove

$$-O(n^{2/3}(\log n)^{5/3}) \leq \log \frac{a_n}{n!\mu^n} \leq O((\log n)^2).$$

A signed, weighted residual estimate and an exact state-padding inequality give uniform quadratic bounds along the characteristic curve. Positivity yields the coefficient upper bound; a growing-state Chernoff argument yields the weaker lower bound. We give a controlled asymmetric Lambert W inverse. Exact counts through 400, floating computations through 1000, and a formal frozen-kernel calculation motivate a separate conjecture with leading coefficient $2/3$ on the logarithmic-square scale. That coefficient, a matching fine lower bound, a ratio limit, and a full asymptotic equivalent remain unproved here.

9 Statements and scope

An ascent sequence of length $n \geq 1$ is a word $w_1 \cdots w_n$ of nonnegative integers with $w_1 = 0$ and

$$w_i \leq 1 + \text{asc}(w_1 \cdots w_{i-1}) \quad (i \geq 2), \quad \text{asc}(w) = \#\{i : w_i < w_{i+1}\}.$$

The classical pattern 000 is present when a value appears in three positions, not necessarily consecutive. Thus avoidance means that every value occurs at most twice. Let a_n count these words and put $a_0 = 1$. This is OEIS A202058 [2]; its first terms are

$$1, 1, 2, 4, 10, 27, 83, 277, 1015, 4007, 17047.$$

Throughout this addendum all logarithms are natural. Set

$$T = \frac{3\pi^2}{8}, \quad \mu = T^{-1}, \quad \beta_n = \frac{a_n}{n!}, \quad h_n = \log \frac{\beta_n}{\mu^n}.$$

The symbol $b(q)$ below is a characteristic speed and is unrelated to β_n .

Theorem 9.1 (Asymmetric coefficient bounds). *There are constants $C_-, C_+ > 0$ and an integer n_0 such that for every $n \geq n_0$,*

$$-C_- n^{2/3}(\log n)^{5/3} \leq h_n \leq C_+ (\log(n+2))^2. \quad (24)$$

In particular, there is no asymptotic equivalent of the form

$$a_n \sim C_0 n! \mu^n \exp(cn^\sigma) n^g (\log n)^h$$

with $C_0, c, \sigma > 0$ and fixed real g, h .

The last assertion concerns a *positive* stretched-exponential correction. The theorem does not exclude all negative stretched-exponential corrections and does not identify the true size or sign of h_n . The exponent $2/3$ on the lower side of (24) is a loss in a broad-window extraction argument, not a proposed stretched-exponential exponent.

Corollary 9.2 (Controlled inverse). *For real $y \rightarrow \infty$, define*

$$N(y) = \min\{n \geq 1 : \log a_n \geq y\}, \quad x = x(y) = \frac{y}{W(y/(eT))},$$

where W is the positive real Lambert function. Then

$$x - O(\log x) \leq N(y) \leq x + O(x^{2/3}(\log x)^{2/3}). \quad (25)$$

Here the threshold variable is the *logarithm* of a count. For the threshold $\min\{n : a_n \geq Y\}$, substitute $y = \log Y$. This convention avoids confusion with the differently named variables in the frozen report. Neither side gives unit-level rounding accuracy or an all-orders inverse.

The 1 October report (Part I of this report) established the exact exponential-generating-function radius T , the root limit $\beta_n^{1/n} \rightarrow \mu$, and a symmetric, non-optimal logarithmic error $O(n^{9/13}(\log n)^3)$. The present argument improves that error asymmetrically. The original report, its TeX, and its proof notes are bundled unchanged. The exact compacted recurrence is due to Conway, Conway, Elvey Price and Guttmann [1]. The analysis here is a research addendum, not a claim of publication, exhaustive priority, or a settled full equivalent.

[write] 2 October 2026. The “1 October report” and the “frozen foundation” of this Part are Part I. Its PDF, TeX and verification files, bundled with this manuscript, are not shipped twice; the proof notes, audits and literature note that only this bundle carried are shipped as `37-foundation-*.md`. Theorem 9.1 and Corollary 9.2 are proved a second time in Part III, Section 30.1, as (146) and (150), by the same growing-seed argument. Part IV improves both: (152) raises the lower side to $-O(\sqrt{n}(\log n)^{3/2})$, and (155) lowers the right displacement of the inverse to $O(\sqrt{x \log x})$. Of the items listed as unproved in this Part’s abstract, the coefficient $2/3$ is proved in Part III in limsup and cumulative form (not pointwise), and the ratio limit is proved in Part IV. A matching fine lower bound and a full asymptotic equivalent remain unproved. The scope remark after Theorem 9.1 also stays true of that theorem. Part III’s limsup theorem (75) additionally excludes an equivalent of the displayed form with $c < 0$, since such an equivalent would force $\log c_n/(\log n)^2 \rightarrow -\infty$. This last observation was added at the merge.

10 Exact states and the frozen characteristic foundation

10.1 The compacted positive operator

Once a label has been used twice it cannot recur. Delete such labels from the active alphabet and compact once-used labels into an initial segment, as justified in [1, Sections 2.2 and 2.6] and explained in Section 2 (Part I). The resulting suffix-count states lie in the invariant space

$$\mathcal{S} = \{(s, u, k) : s \geq 0, u \geq 1, 0 \leq k < s + u\}, \quad m = s + u.$$

Here s is the number of once-used active labels, u the number of unused active labels, and k is one plus the adjusted last label. Canonical states retain suffix counts; they need not retain every property of an original uncompactd prefix.

For $x = (s, u, k)$ and each integer $0 \leq i < m$, form exactly one child

$$x_i = \begin{cases} (s-1, u, i), & i < s, i < k, \quad D \text{ descent}, \\ (s-1, u+1, i), & i < s, i \geq k, \quad D \text{ ascent}, \\ (s+1, u, i+1), & i \geq s, i \geq k, \quad N \text{ ascent}, \\ (s+1, u-1, i+1), & i \geq s, i < k, \quad N \text{ descent}. \end{cases} \quad (26)$$

The letters D, N mean duplicate and new. The last case requires $u \geq 2$: if $u = 1$ then $i \geq s$ forces $i = s$ and $k \leq s$. Thus every child remains in $\mathcal{S}$. The value of m changes by at most one at a step.

Define

$$(\mathcal{T}f)(x) = \sum_{i=0}^{m-1} f(x_i), \quad x_0 = (1, 1, 1), \quad \mathcal{F}(t, x) = \sum_{j \geq 0} (\mathcal{T}^j \mathbf{1})(x) \frac{t^j}{j!}. \quad (27)$$

For $t \geq 0$ the series initially permits $+\infty$. Exactly,

$$(\mathcal{T}^j \mathbf{1})(x_0) = a_{j+1}, \quad \mathcal{F}(t, x_0) = A'(t), \quad A(t) = \sum_{n \geq 0} \frac{a_n}{n!} t^n.$$

These identities are combinatorial; no limiting approximation has entered.

10.2 Characteristic curve and rank

For $q \geq 0$ set

$$\begin{aligned} p(q) &= \frac{1}{2} \log(2e^q - 1), & R(q) &= 2 - e^{-q}, \\ b(q) &= \frac{e^{p(q)}(1 - e^{-q})}{q}, & t(q) &= \int_0^q \frac{dv}{b(v)}, \end{aligned} \quad (28)$$

where $b(0) = 1$ by continuity. For $0 \leq t < T$ write $q = q(t)$. Direct differentiation gives

$$q_t = b, \quad p_q = \frac{1}{R}, \quad p_t = \frac{b}{R}, \quad R_q = e^{-q}. \quad (29)$$

For a fixed state define

$$\begin{aligned} D &= s + Ru, & h(y) &= \min(y, s) + R \max(y - s, 0), \\ r(y) &= h(y)/D, & \psi(q, x) &= \exp\{ps + qu - qr(k)\}, \quad \lambda = Db/R. \end{aligned} \quad (30)$$

Thus $m \leq D \leq 2m$, $0 \leq r(k) \leq 1$, and

$$e^{ps+qu-q} \leq \psi(q, x) \leq e^{ps+qu}, \quad \psi(0, x) = 1. \quad (31)$$

At $q = 0$ all formulas with a removable denominator are interpreted continuously, and $\lambda = m$.

The characteristic endpoint from the frozen foundation is

$$t(\infty) = T = \frac{3\pi^2}{8}, \quad T - t(q) \sim \sqrt{2}(q+2)e^{-q/2}. \quad (32)$$

We recall enough of its verification to specify exactly what is being used. Setting $v = \sqrt{2e^q - 1}$ gives

$$T = \int_1^\infty \frac{2 \log((v^2 + 1)/2)}{v^2 - 1} dv = -2 \int_0^1 \log z \left(\frac{1}{1-z} + \frac{z}{1+z^2} \right) dz.$$

Integration by parts and $z = (v - 1)/(v + 1)$ give the second expression; the boundary terms vanish. Geometric-series integration yields $\pi^2/3$ and $\pi^2/24$ for its two terms. This proves the endpoint. Moreover,

$$t_q = \frac{q}{\sqrt{2}} e^{-q/2} \{1 + O(e^{-q})\},$$

whose integral from q to infinity gives the tail in (32). The frozen radius and root-limit proofs use this same curve and operator. Those proofs are accessible in the bundled foundation; none is inferred here merely from a coefficient fit.

[write] **2 October 2026.** The frozen radius and root-limit proofs are Sections 5 and 6 of Part I. This subsection's endpoint calculation repeats Part I, Section 5; Parts III and IV repeat it once more.

10.3 The exact frozen integral

For real $y \in [0, m]$ define

$$w(y) = \begin{cases} e^{-p+q} \mathbf{1}_{y \geq k}, & y < s, \\ e^{p-q} \mathbf{1}_{y < k}, & y \geq s, \end{cases} \quad g(y) = w(y) e^{-q(r(y)-r(k))}, \quad S = \sum_{i=0}^{m-1} g(i). \quad (33)$$

The identity

$$\int_0^m g(y) dy = \lambda \quad (34)$$

has a useful direct verification. Put $r' = h(y)/D$ and $r = r(k)$. On duplicate labels $dy = D dr'$ and on new labels $dy = (D/R) dr'$. The relation $e^p/R = e^{q-p}$ makes both pieces agree, giving

$$\frac{g(y) dy}{\lambda} = \kappa_q((r' - r) \bmod 1) dr', \quad \kappa_q(v) = \frac{q e^{-qv}}{1 - e^{-q}} \quad (0 \leq v < 1). \quad (35)$$

The density integrates to one. At $q = 0$ it is the uniform density. This proves (34) exactly for the continuous frozen kernel, without asserting that it is the evolving discrete chain.

11 Signed weighted residual estimates

The improvement over the frozen report is to retain signs and normalize by the characteristic speed before bounding the residual.

Lemma 11.1 (Signed quadrature and shifted ranks). *For every $q \geq 0$ and $x \in \mathcal{S}$,*

$$0 \leq S - \lambda \leq 3\sqrt{2}e^{q/2}. \quad (36)$$

Let r_i denote the rank fraction at the actual child x_i . Then

$$r_i = \frac{i}{D-1}, \quad \frac{i}{D+R-1}, \quad \frac{h(i)+1}{D+1}, \quad \frac{h(i)+1}{D+1-R} \quad (37)$$

in the four cases of (26). In every case $|r_i - r(i)| \leq 4/D$. Except for a duplicate ascent, $r_i \geq r(i)$; in that exceptional case

$$0 \leq r(i) - r_i = \frac{i(R-1)}{D(D+R-1)} \leq \frac{1}{D}. \quad (38)$$

Proof. The inequalities $q/2 \leq p \leq q/2 + (\log 2)/2$ imply $g(y) \leq \sqrt{2}e^{q/2}$. For ascents use $r(y) \geq r(k)$; for descents use $r(k) - r(y) \leq 1$. On every interval separated by the integer breakpoints s, k , the function g decreases. Each unit interval has its corresponding summand at its left endpoint. Hence its sum dominates its integral. On each of the at most three nonempty pieces the excess is at most the starting value, proving (36). At a breakpoint the formula for the following interval is used; a value at one integration endpoint has no effect on the integral.

Substitution of each child into (30) gives (37). The denominators in the cases that occur are positive. For a duplicate descent,

$$r_i - r(i) = \frac{i}{D(D-1)} \in [0, 1/D].$$

For a duplicate ascent, subtraction gives (38), since $i < D$ and $0 \leq R-1 \leq 1$. For a new ascent,

$$r_i - r(i) = \frac{D - h(i)}{D(D+1)} \in [0, 1/D].$$

For a new descent the difference is

$$\frac{D + (R-1)h(i)}{D(D+1-R)} \geq 0.$$

Here $u \geq 2$, so $D \geq 2R$ and $D+1-R \geq D/2$; also $h(i) \leq D$ and $R \leq 2$. The last difference is therefore at most $4/D$. $\square$

Differentiating the profile at a fixed state gives

$$\frac{\partial_t \psi}{\psi} = \lambda - bB, \quad B = r(k) + qe^{-q}\partial_R r(k), \quad |B| \leq 1. \quad (39)$$

Indeed,

$$\partial_R r(k) = \begin{cases} -ku/D^2, & k \leq s, \\ -s(m-k)/D^2, & k \geq s, \end{cases} \quad -\frac{1}{4R} \leq \partial_R r(k) \leq 0.$$

The bound follows from $sRu/(s+Ru)^2 \leq 1/4$, $k \leq s$ in the first case, and $m-k \leq u$ in the second. Since $0 \leq r(k) \leq 1$ and $qe^{-q} \leq 1/e$, it also implies $|B| \leq 1$.

Proposition 11.2 (Weighted residual). *Uniformly over all states and all $q \geq 0$,*

$$\frac{\mathcal{T}\psi - \partial_t \psi}{b\psi} \leq qe^{q/D} + 3\sqrt{2}(q+1)e^{q/D} + 1, \quad (40)$$

$$\frac{\partial_t \psi - \mathcal{T}\psi}{b\psi} \leq 4q + 1. \quad (41)$$

Proof. The exact ratio is

$$\frac{\psi(q, x_i)}{\psi(q, x)} = g(i)e^{-q(r_i - r(i))}.$$

Lemma 11.1 and positivity give the two signed estimates

$$\frac{\mathcal{T}\psi}{\psi} \leq e^{q/D} S, \quad \frac{\mathcal{T}\psi}{\psi} \geq e^{-4q/D} S \geq e^{-4q/D} \lambda. \quad (42)$$

Because $\lambda/b = D/R$, (39) implies

$$\begin{aligned} \frac{\mathcal{T}\psi - \partial_t\psi}{b\psi} &\leq \frac{D}{R}(e^{q/D} - 1) + 3\sqrt{2}\frac{e^{q/2}}{b}e^{q/D} + 1 \\ &\leq qe^{q/D} + 3\sqrt{2}(q+1)e^{q/D} + 1. \end{aligned}$$

We used $e^z - 1 \leq ze^z$, $R \geq 1$, and

$$\frac{e^{q/2}}{b} = \frac{q}{\sqrt{2 - e^{-q}}(1 - e^{-q})} \leq \frac{q}{1 - e^{-q}} \leq q + 1.$$

The last inequality is equivalent to $e^q \geq 1 + q$; its limiting value at zero is valid. Similarly,

$$\frac{\partial_t\psi - \mathcal{T}\psi}{b\psi} \leq \frac{D}{R}(1 - e^{-4q/D}) + 1 \leq 4q + 1,$$

using $1 - e^{-z} \leq z$. No lower bound on the state size was used for this second estimate. $\square$

12 Padding and comparison on the infinite state space

12.1 An exact padding inequality

For an integer $L \geq 0$ put

$$I_L(s, u, k) = (s + L, u, k + L).$$

This map takes $\mathcal{S}$ into itself. Replacing i by $i + L$ preserves the comparisons with s, k , and direct substitution into each case of (26) shows that the corresponding child of $I_L x$ is $I_L x_i$. There are additionally L children indexed by $0, \dots, L - 1$. Therefore every nonnegative f satisfies

$$\mathcal{T}(f \circ I_L)(x) \leq (\mathcal{T}f)(I_L x). \quad (43)$$

This is an inequality, not a commutation identity.

Fix $Q > 0$ and, for the entire interval $0 \leq q \leq Q$, fix the integer $L = \lceil Q \rceil$. Then $D(I_L x) \geq L + 1 \geq Q$, so $e^{q/D(I_L x)} \leq e$. Define

$$\begin{aligned} H(q) &= \frac{e(1 + 3\sqrt{2})}{2}q^2 + (3\sqrt{2}e + 1)q, \\ G_Q(t, x) &= e^{H(q(t))}\psi(q(t), I_L x), \quad g_-(t, x) = e^{-2q(t)^2 - q(t)}\psi(q(t), x). \end{aligned} \quad (44)$$

The derivative $H'(q)$ dominates the right side of (40) at every padded state throughout this interval. Since $dq/dt = b$, Proposition 11.2 and (43) give

$$\partial_t G_Q \geq \mathcal{T}G_Q, \quad \partial_t g_- \leq \mathcal{T}g_-, \quad G_Q(0, x) = g_-(0, x) = 1. \quad (45)$$

The lower derivative follows because $(2q^2 + q)' = 4q + 1$.

12.2 Stopped upper comparison

Formal comparison with an unbounded positive operator is insufficient. We give the stopping argument, including the boundary term needed for the lower bound.

Let X_v be the continuous-time chain with one rate-one transition for every child of its current state. Its total jump rate is m and its generator is $\mathcal{L} = \mathcal{T} - mI$. Since each jump increases m

by at most one, its successive holding rates are bounded above by those of a pure-birth process with rates $m(x), m(x) + 1, \dots$. That process is nonexplosive, as $\sum_j 1/(m(x) + j) = \infty$; hence X is nonexplosive. Let

$$\tau_M = \inf\{v : m(X_v) > M\}, \quad V_v = \int_0^v m(X_w) dw.$$

States with $m \leq M$ form a finite set. By expansion over jump paths and holding times,

$$\mathcal{F}(t, x) = \mathbb{E}_x e^{V_t}, \quad (46)$$

with infinity allowed: the holding exponentials cancel the factor e^{V_t} , and a fixed j -step path contributes the time-simplex volume $t^j/j!$. Nonexplosion accounts for all paths; Tonelli justifies the nonnegative sums.

For $t \leq t(Q)$, stopped Dynkin comparison applied to $e^{V_v} G_Q(t - v, X_v)$ gives

$$\mathbb{E}_x [e^{V_t}; \tau_M > t] \leq G_Q(t, x).$$

One may obtain this by stopping at $t \wedge \tau_M$ and discarding the nonnegative exit term. The drift is $e^{V_v}(\mathcal{T}G_Q - \partial_t G_Q)(t - v, X_v) \leq 0$. Let $M \rightarrow \infty$. Nonexplosion and monotone convergence, followed by (46), give

$$\mathcal{F}(t(q), x) \leq e^{H(q)} \psi(q, I_L x), \quad 0 \leq q \leq Q. \quad (47)$$

In particular the minimal positive series is finite before T .

12.3 The lower comparison and its exit term

Fix $t < T$ and choose $\delta > 0$ with $t + \delta < T$. Choose one Q with $t + \delta \leq t(Q)$ and keep its padding integer L fixed. From the opposite drift in (45),

$$g_-(t, x) \leq \mathbb{E}_x[e^{V_t}; \tau_M > t] + \mathbb{E}_x[e^{V_{\tau_M}} g_-(t - \tau_M, X_{\tau_M}); \tau_M \leq t]. \quad (48)$$

Write $p(v) = p(q(v))$ in this paragraph, and similarly for the other time-dependent functions. Uniformly for $0 \leq v \leq t$ and all $y = (s, u, k)$,

$$\frac{g_-(v, y)}{G_Q(v + \delta, y)} \leq C_{t, \delta, Q} \exp\{-[p(v + \delta) - p(v)]s - [q(v + \delta) - q(v)]u\} \leq C_{t, \delta, Q} e^{-\eta m(y)} \quad (49)$$

for some $\eta > 0$. To see this, use (31) in the numerator and denominator. Rank exponents have bounded magnitude on the compact time interval; the extra term $Lp(v + \delta)$ is harmless since L is fixed. Both p, q are strictly increasing and continuous, so the two positive increments have positive compact minima.

On exit, $m(X_{\tau_M}) = M + 1$, because an upward jump has size at most one. Apply the supersolution once more, now to the time horizon $t + \delta$, stopping at $t \wedge \tau_M$. It bounds the exit part with the later function by

$$\mathbb{E}_x[e^{V_{\tau_M}} G_Q(t + \delta - \tau_M, X_{\tau_M}); \tau_M \leq t] \leq G_Q(t + \delta, x).$$

Thus the second term in (48) is at most $C_{t, \delta, Q} e^{-\eta(M+1)} G_Q(t + \delta, x)$, which tends to zero. Letting $M \rightarrow \infty$ proves

$$\mathcal{F}(t(q), x) \geq e^{-2q^2 - q} \psi(q, x). \quad (50)$$

This is the essential justification of the infinite-state lower comparison.

Corollary 12.1 (Uniform characteristic bounds). *For every $q \geq 0$ and every state,*

$$-2q^2 - q \leq \log \mathcal{F}(t(q), x) - \log \psi(q, x) \leq H(q) + \lceil q \rceil p(q) + q. \quad (51)$$

In particular the difference is $O((q+1)^2)$ uniformly in x .

Proof. Use (47) with $Q = q$ and

$$\log \psi(q, I_L x) - \log \psi(q, x) = Lp(q) - q\{r(I_L x) - r(x)\} \leq Lp(q) + q.$$

For $q = 0$ use the initial values. The lower assertion is (50). $\square$

13 The logarithmic square coefficient upper bound

At the root, $I_L x_0 = (L+1, 1, L+1)$, so

$$\log \mathcal{F}(t(Q), x_0) \leq H(Q) + (L+1)p(Q) + Q = O((Q+1)^2).$$

The constants are independent of Q since $L = \lceil Q \rceil$ and $p(Q) \leq Q/2 + (\log 2)/2$. Nonnegative coefficients imply

$$\beta_n \leq \frac{A(t)}{t^n}, \quad A(t) = 1 + \int_0^t \mathcal{F}(v, x_0) dv \leq 1 + t\mathcal{F}(t, x_0). \quad (52)$$

Choose $Q = 2 \log(n+2)$. Then (32) gives

$$n \log \frac{T}{t(Q)} = O(n(Q+2)e^{-Q/2}) = O(\log(n+2)),$$

where $t(Q)$ is bounded away from zero for large n . Also $\log A(t(Q)) = O((\log(n+2))^2)$. Taking logarithms in (52) proves the upper side of (24). If the equivalent excluded in Theorem 9.1 held, then

$$h_n = cn^\sigma + g \log n + h \log \log n + \log C_0 + o(1),$$

contradicting the upper bound because $c, \sigma > 0$.

14 The growing state lower bound

14.1 A uniform tilted coefficient estimate

Let $x_N = (N, 1, N)$ for an integer $N \geq 1$. Since $r(k) = N/(N+R)$ at this state,

$$\log \psi(q, x_N) = Np(q) + \frac{qR}{N+R}.$$

Corollary 12.1 therefore gives an absolute constant C_0 such that

$$|\log \mathcal{F}(t(q), x_N) - Np(q)| \leq C_0(q+1)^2 \quad (N \geq 1, q \geq 0). \quad (53)$$

Put

$$d(q) = \frac{d}{dq} \log t(q), \quad M(q) = \frac{p_q(q)}{d(q)} = t(q)p_t(t(q)) = \frac{t(q)b(q)}{R(q)}.$$

These formulas are used for $q > 0$. As $q \rightarrow \infty$,

$$p_q(q) = \frac{1}{2} + O(e^{-q}), \quad d(q) \sim \frac{qe^{-q/2}}{\sqrt{2T}}, \quad M(q) \sim \frac{Te^{q/2}}{\sqrt{2}q}. \quad (54)$$

Differentiating the explicit formulas also gives $d'(q)/d(q) \rightarrow -1/2$ and $M'(q)/M(q) \rightarrow 1/2$. In particular M is eventually strictly increasing and tends to infinity.

For fixed N, q , write $c_j = (\mathcal{T}^j \mathbf{1})(x_N)$ and define a probability law

$$\mathbb{P}(J = j) = \frac{c_j t(q)^j}{j! \mathcal{F}(t(q), x_N)}, \quad j \geq 0. \quad (55)$$

Its reference location $NM(q)$ need not be its exact mean.

Lemma 14.1 (Uniform Chernoff window). *There are absolute constants $c, C > 0$ and q_0 such that for every $N \geq 1$, $q \geq q_0$, and $0 < \epsilon < 1/2$,*

$$\mathbb{P}(|J - NM(q)| > \epsilon NM(q)) \leq 2 \exp\{-cN\epsilon^2 + C(q+1)^2\}. \quad (56)$$

One may take $c = 3/128$ after increasing C and q_0 .

Proof. For $|v| \leq 1$ define

$$H_q(v) = p(q+v) - p(q) - M(q)\{\log t(q+v) - \log t(q)\}.$$

Then $H_q(0) = H'_q(0) = 0$. Uniformly for bounded v , explicit differentiation of (28) and (54) yields

$$H''_q(v) \rightarrow \frac{1}{4}e^{-v/2}, \quad M(q)d(q+v) \rightarrow \frac{1}{2}e^{-v/2}.$$

Consequently, by taking one sufficiently large q_0 ,

$$|H_q(v)| \leq v^2/2 \quad (|v| \leq 1), \quad M(q)|\log t(q \pm \eta) - \log t(q)| \geq \eta/4 \quad (57)$$

for $0 \leq \eta \leq 1$. The first inequality follows by integrating the uniform second-derivative bound twice; the second follows by integrating the positive lower bound for $M(q)d$.

Set $K_q = NM(q)$ and $\eta = \epsilon/8$. Markov's inequality applied to $(t(q+\eta)/t(q))^J$ bounds the upper tail by

$$\frac{\mathcal{F}(t(q+\eta), x_N)}{\mathcal{F}(t(q), x_N)} \left(\frac{t(q)}{t(q+\eta)} \right)^{(1+\epsilon)K_q}.$$

By (53), the logarithm is at most

$$\begin{aligned} & NH_q(\eta) - \epsilon NM(q)\{\log t(q+\eta) - \log t(q)\} + C(q+1)^2 \\ & \leq N\eta^2/2 - \epsilon N\eta/4 + C(q+1)^2 = -\frac{3}{128}N\epsilon^2 + C(q+1)^2. \end{aligned}$$

The bounded shifts change $(q+1)^2$ by only an absolute factor. For the lower tail use $q-\eta$ and the decreasing random variable $(t(q-\eta)/t(q))^J$. Its logarithmic bound is

$$NH_q(-\eta) + \epsilon NM(q)\{\log t(q-\eta) - \log t(q)\} + C(q+1)^2,$$

which satisfies the same inequality. Adding the two tail bounds proves the lemma. $\square$

14.2 A fixed choice of the growing state

Choose once and for all a sufficiently large constant $K > 0$, and for each sufficiently large integer n set

$$N = \left\lceil Kn^{2/3}(\log n)^{2/3} \right\rceil, \quad \epsilon = \frac{2N}{n}. \quad (58)$$

Then $\epsilon \rightarrow 0$. Choose q on the eventually increasing branch of M by

$$NM(q) = (1 - 4\epsilon)n. \quad (59)$$

The right side divided by N tends to infinity, so the choice exists uniquely on that branch for all large n . Taking logarithms in (54) gives

$$q = \frac{2}{3} \log n + \frac{2}{3} \log \log n + O_K(1). \quad (60)$$

For clarity, first $q \sim (2/3) \log n$ follows from $q/2 - \log q = \log(n/N) + O(1)$; inserting this estimate into $\log q$ gives (60). In particular, for any fixed K and all sufficiently large n , $q \leq \log n$ and $(q+1)^2 \leq 4(\log n)^2$.

There is no n -dependent choice of K hidden in the tail estimate. Indeed,

$$N\epsilon^2 = \frac{4N^3}{n^2} \geq 4K^3(\log n)^2.$$

Choose the fixed K so that $4cK^3 > 4C + 1$, where c, C are from Lemma 14.1. Its bound is then at most $2e^{-(\log n)^2}$ for all sufficiently large n , hence at most $1/2$. Thus probability at least $1/2$ lies in

$$|j - NM(q)| \leq \epsilon NM(q).$$

This interval has at most $2\epsilon NM(q) + 3$ integer points. Some such j therefore satisfies

$$\frac{c_j t(q)^j}{j!} \geq \frac{\mathcal{F}(t(q), x_N)}{4\epsilon NM(q) + 6}. \quad (61)$$

Using (59) and (58), every j in the window has

$$\begin{aligned} j &\geq (1 - \epsilon)(1 - 4\epsilon)n \geq n - 6\epsilon n, \\ j + N &\leq (1 + \epsilon)(1 - 4\epsilon)n + N \leq n - 3\epsilon n + N \leq n. \end{aligned} \quad (62)$$

The last inequality uses $N = \epsilon n/2$. There is thus ample space for the initial path and for extension to the desired length.

14.3 Moving the extracted coefficient to the target index

The root $x_0 = x_1$ reaches x_N by $N - 1$ successive new-maximal-label transitions. Hence $a_{N+j} \geq c_j$. Every cap-two ascent word can be extended injectively by appending one plus its current number of ascents. This label is new, since the largest label already used is at most the current number of ascents, and it obeys the next ascent bound. Repeating this deterministic extension gives $a_n \geq a_{N+j}$ whenever $n \geq N + j$. Thus (62) gives $a_n \geq c_j$.

By (61) and (53),

$$\begin{aligned} h_n &\geq -\log \frac{n!}{j!} - j \log t(q) + n \log T + Np(q) - C_0(q+1)^2 - O(\log n) \\ &\geq -6\epsilon n \log n - C_0(q+1)^2 - O(\log n). \end{aligned}$$

For the second line, $n - j \leq 6\epsilon n$ bounds the factorial loss, $Np(q) \geq 0$, and eventually $1 < t(q) < T$ with $j \leq n$ gives $-j \log t(q) \geq -n \log T$. The denominator in (61) is $O(n)$ and contributes only $O(\log n)$. Finally, $\epsilon n = 2N = O(n^{2/3}(\log n)^{2/3})$ and $q = O(\log n)$. This proves

$$h_n \geq -O(n^{2/3}(\log n)^{5/3}),$$

completing Theorem 9.1.

15 Proof of the asymmetric inverse

Let $f(v) = v \log(v/(eT))$. Stirling's formula and Theorem 9.1 imply, after adjusting constants, that for all sufficiently large integers n ,

$$f(n) - C_1 n^{2/3}(\log n)^{5/3} \leq \log a_n \leq f(n) + C_2(\log n)^2. \quad (63)$$

The term $\frac{1}{2} \log(2\pi n) + O(1/n)$ is absorbed into the respective envelopes. The value $x = y/W(y/(eT))$ solves $f(x) = y$: if $w = W(y/(eT))$, then $x = eTe^w$ and $xw = y$.

For v in any $o(x)$ neighborhood of x ,

$$f'(v) = \log(v/T) \sim \log x.$$

Take $n_- = \lfloor x - A \log x \rfloor$ for a sufficiently large fixed A . The mean value theorem gives

$$f(x) - f(n_-) \geq \frac{A}{2}(\log x)^2$$

for large x . The upper envelope in (63) is at most $2C_2(\log x)^2$ there. Choose $A > 4C_2$ to obtain $\log a_{n_-} < y$. Monotonicity then gives $N(y) > n_-$, proving the lower side of (25).

Next take

$$n_+ = \lceil x + Bx^{2/3}(\log x)^{2/3} \rceil$$

with a sufficiently large fixed B . This displacement is $o(x)$, and

$$f(n_+) - f(x) \geq \frac{B}{2}x^{2/3}(\log x)^{5/3}$$

for large x . The lower error envelope in (63) is at most $2C_1x^{2/3}(\log x)^{5/3}$ there. Choose $B > 4C_1$ to obtain $\log a_{n_+} \geq y$. Thus $N(y) \leq n_+$. Integer rounding is absorbed in the stated errors, proving Corollary 9.2.

16 Exact computation and numerical evidence

The numerical investigation is independent of the analytic proof. The bundled sources and data distinguish exact integers from floating approximations explicitly.

16.1 A reproducible positive recurrence

Let $P_n(s, u, k)$ count compacted states after n letters, starting from $P_1(1, 1, 1) = 1$. At fixed (s, u) , put

$$A_i = \sum_{k \leq i} P_n(s, u, k), \quad B_i = \sum_{k > i} P_n(s, u, k), \quad 0 \leq i < s + u.$$

Using prefix and suffix sums, the four updates are

$$\begin{aligned} P_{n+1}(s-1, u+1, i) &+= A_i, & P_{n+1}(s-1, u, i) &+= B_i & (i < s), \\ P_{n+1}(s+1, u, i+1) &+= A_i, & P_{n+1}(s+1, u-1, i+1) &+= B_i & (i \geq s). \end{aligned}$$

Invalid target contributions vanish. Reachability implies

$$0 \leq s \leq n, \quad s \equiv n \pmod{2}, \quad 1 \leq u \leq 1 + (n-s)/2, \quad 0 \leq k < s+u.$$

Summing target states gives a_{n+1} . Through length N this requires $O(N^4)$ big-integer additions and $O(N^3)$ stored integers; the operation count does not include the growth of integer bit lengths.

The C++17/GMP implementation reproduces all 177 published b-file terms, indexed 0 through 176, and extends the exact counts through 400. The positive-sum long-double implementation rescales each transition by $1/((n+1)\mu)$, builds suffix sums without subtraction, and uses Kahan summation for its scalar totals. It reaches 1000 and agrees with exact normalized values through 400 to about 3.22×10^{-17} maximum relative error. This comparison is an empirical floating-point diagnostic, not an interval certificate for terms 401 through 1000.

16.2 Diagnostics and their interpretation

Define

$$g_n = n(h_n - h_{n-1}), \quad c_n = n(g_n - g_{n-1}).$$

If $h_n = \alpha(\log n)^2 + \beta \log n + \gamma$ with sufficiently regular lower-order corrections, one expects $c_n \rightarrow 2\alpha$. Such regularity is an assumption in using this diagnostic; an asymptotic relation for h_n alone cannot be differentiated discretely without care.

n	Count source	h_n	g_n	c_n
100	Exact integers	22.0137088785	8.0262005526	1.4825921960
176	Exact integers	26.7698812146	8.8501413019	1.4449635811
400	Exact integers	34.5036804481	10.0154282936	1.4020458798
800	Long double	41.7744627482	10.9771569991	1.3770655779
1000	Long double	44.2569370449	11.2835625443	1.3709052117

Table 1: Rounded real diagnostics. Even in rows with exact integer counts, logarithms and μ are evaluated numerically.

Conditional fits to

$$h_n = \alpha(\log n)^2 + \beta \log n + \gamma + \frac{P_2(\log n)}{n} + \dots \quad (64)$$

give $\alpha = 0.665913, 0.666559, 0.666678$ on the respective intervals 100–200, 200–400, and 400–800. Adding another quadratic logarithmic polynomial divided by n^2 gives $\alpha \approx 0.66666764$ on 500–1000, with $\beta \approx 2.12357$. The fitted amplitude $e^\gamma \approx 0.1008$ and the value of β are conditional numerical observations only. High-order fits are ill-conditioned; small residuals are not rigorous error bounds.

A fit on 200–400 using $\alpha L^2 + \beta L + \gamma + (dL + e)/n$, $L = \log n$, predicts h_n through 800 with maximum error about 3.18×10^{-5} . Three-parameter models $An^\sigma + B \log n + C$ with $\sigma = 0.06, 0.08, 0.10$ miss by approximately 0.023, 0.028, 0.033, respectively. These are finite-range comparisons, not exclusions of every alternative expansion. In particular, extrapolating c_n by low-degree polynomials in $1/\log n$ is unstable: degrees 1, 2, 3 on 500–1000 give limiting estimates about 1.174, 1.357, 1.598.

A202058 correction diagnostics

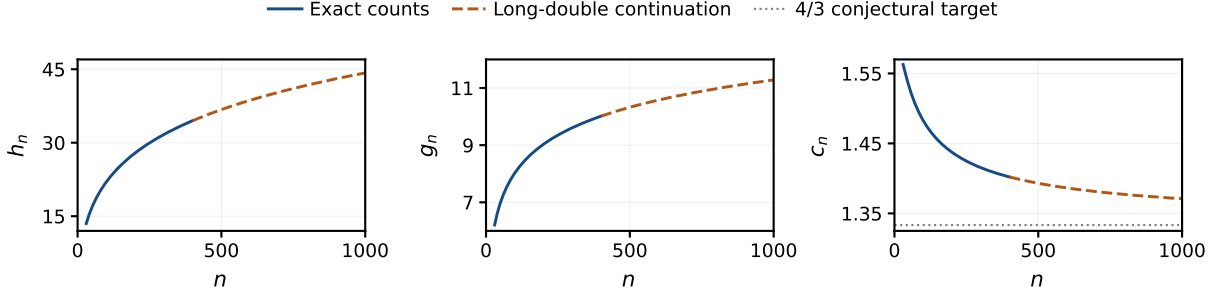


Figure 1: Illustrative diagnostics from the bundled computation. Exact and floating ranges are marked in the source figure. Trend lines and candidate limits are exploratory and play no role in Theorem 9.1.

Conjecture 16.1 (Logarithmic-square leading correction). The sequence satisfies

$$\log \frac{a_n}{n! \mu^n} \sim \frac{2}{3} (\log n)^2.$$

This conjecture is strictly stronger than the proved bounds. It does not by itself assert the additional terms or constants in (64), nor a multiplicative equivalent for a_n . If it were proved, the mean-value argument in Section 15, applied to $f(n) + (2/3)(\log n)^2 + o((\log n)^2)$, would give the conditional refinement

$$N(y) = x - \frac{2}{3} \log x + o(\log x).$$

This conditional formula is not part of Corollary 9.2.

[write] 2 October 2026. Status of Conjecture 16.1: proved in two weaker forms, open as stated. Part III, Theorem 21.1, proves $\limsup_n \log c_n / (\log n)^2 = 2/3$ and $\log \sum_{j \leq N} c_j \sim \frac{2}{3} (\log N)^2$, where $c_n = a_n / (n! \mu^n) = e^{h_n}$. The pointwise asymptotic stated here remains open: Part III, (76), and Part IV, (160). The conditional inverse refinement above therefore remains conditional. Part III, (143), and Part IV, (204), each give a one-sided ratio bound that would suffice.

17 The coefficient regularity obstruction

Uniform function estimates such as (51) bound a positive weighted sum of coefficients. They give upper bounds on every individual coefficient but provide no comparable lower bound at a prescribed index. The growing-state proof extracts one coefficient in a broad window, then pays an $O(N \log n)$ factorial loss to move to the requested index. This is exactly why its lower estimate is much weaker than $O((\log n)^2)$.

One possible additional input is log-concavity of $\beta_n = a_n / n!$. Under the compacted-state distribution of a uniformly selected length- n word, let $\mathbb{E}_n$ denote expectation and put

$$M_n = \mathbb{E}_n[m] = a_{n+1} / a_n.$$

Directly summing (26) gives

$$\begin{aligned}\mathcal{T}m &= m^2 + m - s - k = m^2 + u - k, \\ \mathcal{T}s &= ms + m - 2s, \quad \mathcal{T}k = \frac{m(m+1)}{2} - s, \\ \mathcal{T}(m^2) &= m^3 + 2m(m - s - k) + m - |s - k|.\end{aligned}\tag{65}$$

Consequently,

$$M_{n+1} = M_n + \frac{\text{Var}_n(m) + \mathbb{E}_n(u - k)}{M_n}.\tag{66}$$

Indeed, the next-step expectation of m is $\mathbb{E}_n[\mathcal{T}m]/\mathbb{E}_n[m]$, because a state is weighted by its number of children. Log-concavity of β_j at $j = n + 1$ is therefore equivalent to

$$(n + 1)\{\text{Var}_n(m) + \mathbb{E}_n(u - k)\} \leq M_n^2.\tag{67}$$

The absolute value in the last identity of (65) obstructs a straightforward closed polynomial moment recursion. No proof of (67) is supplied here.

[write] 2 October 2026. Part IV restates the identity (66) as (166), writing R_n for M_n , without citing this Part. It deduces the exact inequality $R_{n+1} \geq R_n - 1$ and from it the ratio limit. The target (67) reappears there as (205) and is still unproved. Part IV, Section 40, records two further exact scans: factorial-normalized and rising-factorial log-concavity of $(\mathcal{T}^r \mathbf{1})(x)$ at every state with $H = s + 2u$ and $1 \leq r \leq 179$, $H + r + 1 \leq 200$, each over 48,510,450 inequalities with no violation. Together they extend the all-state check below. Part IV's counterexample concerns a stronger transition-wise inequality that this Part does not state.

Exact integer tests give

$$(j + 1)a_j^2 > ja_{j-1}a_{j+1} \quad (2 \leq j \leq 399),$$

with equality at $j = 1$. The stronger all-state property that $(\mathcal{T}^j \mathbf{1})(x)/j!$ is log-concave also passes all 3060 tested centers $1 \leq j \leq 15$ across the 204 states with $s + u \leq 8$. Neither finite check is a proof.

If global normalized log-concavity were established, the ratios β_{n+1}/β_n would decrease. The known root limit forces their limit to be μ : the arithmetic means of their logarithms tend to the logarithm of their decreasing limit, and are $\log(\beta_n)/n$. Thus each ratio would be at least μ , giving $\beta_n \geq \mu^n$ and

$$0 \leq h_n \leq O((\log n)^2).$$

This conditional consequence would exclude negative as well as positive stretched-exponential factors, but would still not identify $2/3$.

Several stronger shortcuts are demonstrably unavailable:

1. The normalized sequence is not a Pólya-frequency sequence of order three. Since $\beta_0 = \beta_1 = \beta_2 = 1$ and $\beta_3 = 2/3$,

$$\det \begin{pmatrix} \beta_1 & \beta_2 & \beta_3 \\ \beta_0 & \beta_1 & \beta_2 \\ 0 & \beta_0 & \beta_1 \end{pmatrix} = -\frac{1}{3}.$$

Thus full Toeplitz total positivity is false. A PF_2 -specific argument remains possible.

2. The length-seven doubled-letter polynomial is $1 + 21z + 126z^2 + 129z^3$. Its discriminant is -84159 , so it has two nonreal roots. Real-rootedness of this natural refinement cannot be assumed.

3. The words 001 and 010 are cap-two ascent words but their coordinatewise minimum 000 is not. The raw-word class is not closed under the usual lattice meet.

These exact counterexamples concern particular methods; they do not disprove normalized log-concavity or Conjecture 16.1.

18 A formal route toward the conjectured constant

This section describes a research mechanism. The exact frozen identities are distinguished from asymptotic transport assumptions; no additional coefficient theorem is claimed.

18.1 Exact frozen averaging

By (35), the frozen Markov operator on the unit circle is

$$(K_q f)(r) = \int_0^1 \kappa_q((r' - r) \bmod 1) f(r') dr'.$$

Uniform measure is exactly stationary. On the Fourier mode $e^{2\pi i \ell r}$ its multiplier is $q/(q - 2\pi i \ell)$. On mean-zero periodic functions the corresponding inverse is

$$(K_q - I)^{-1} = q\partial_r^{-1} - I, \quad (68)$$

where ∂_r^{-1} is the mean-zero periodic antiderivative. Since $\lambda/b = D/R$, the inverse frozen generator on the q time scale is $(R/D)(q\partial_r^{-1} - I)$. A forcing of size q thus has a frozen rank corrector of size $O(q^2/D)$. This does not imply a uniform bound for the actual discrete evolving chain.

For some additional precision, let $z = s/D$ and write $V = \mathcal{T}\psi/\psi - \partial_t \log \psi$. In the limit $D \rightarrow \infty$ at fixed q , with limiting ranks away from coincident breakpoints, the first-order continuum residual has rank average

$$\langle V/b \rangle = qI(q) + \frac{q(z-1)}{2R} + \frac{1}{2}, \quad I(q) = \frac{\frac{1}{2} - [1 - (1+q)e^{-q}]/q^2}{1 - e^{-q}}. \quad (69)$$

The identity is for that first-order continuum limit, not for a finite state. Its derivation retains the rank-boundary layers: the Euler–Maclaurin contribution is

$$\frac{q}{2R} \mathbf{1}_{r < z} + \frac{q}{2} \mathbf{1}_{r > z} + \frac{R-1}{2R} \{ \kappa_q((z-r) \bmod 1) - \kappa_q((-r) \bmod 1) \},$$

and the last two kernel terms integrate to zero. If $a(r, r')$ is the leading value of $D(r_i - r(i))$, its four values are

$$r', \quad -(R-1)r', \quad 1-r', \quad 1+(R-1)r'$$

in the order of (26). Under stationary rank measure, conditional on $r' = x$, the probability of $r \leq r'$ is $P_q(x) = (1 - e^{-qx})/(1 - e^{-q})$. Therefore

$$\begin{aligned} \langle a \rangle &= 1 - z + \frac{R-1}{2} + \frac{2-R}{2} z^2 - RI(q), \\ \langle B \rangle &= \frac{1}{2} - \frac{qe^{-q}}{2R} z(1-z). \end{aligned}$$

Adding the averaged quadrature, shifted-rank term $-(q/R)\langle a \rangle$, and $\langle B \rangle$ gives (69), using $2-R = e^{-q}$. It follows that

$$\langle V/b \rangle = \frac{q}{4}(1+z) + \frac{1}{2} - \frac{1}{q} + O(qe^{-q}).$$

The bundled formal-kernel audit gives the full continuum calculation and an independent numerical check. The limiting calculation does not bound its finite-state remainder uniformly in joint q, D regimes.

18.2 Formal transport and the scalar exponent

Away from the wrap and type-interface layers, the large- q bulk residual suggested by the discrete expansion is

$$V/b \approx qf(r, z), \quad f(r, z) = \begin{cases} \frac{1}{4} + \frac{r}{2}, & r < z, \\ \frac{r}{2}, & r > z. \end{cases}$$

Its rank average is $(1+z)/4$. At fixed q , write the profile-transformed jump generator as

$$(\mathcal{L}^\psi f)(x) = \sum_i \frac{\psi(q, x_i)}{\psi(q, x)} \{f(x_i) - f(x)\}.$$

Under the same formal large-state dynamics, the composition drift on the q scale is pointwise

$$\mathcal{L}^\psi z/b \approx \begin{cases} -(1+z)/2, & r < z, \\ (1-z)/2, & r > z, \end{cases}$$

whose rank average is $(1-3z)/2$. The factor $1/b$ converts the jump rates to the q time scale. Only the *averaged* composition drift equals $(1-3z)/2$.

Multiplying the profile by $\exp(qz/6)$ would formally combine these averaged contributions as

$$q \frac{1+z}{4} + q \frac{1-3z}{12} = \frac{q}{3}.$$

Integrating this scalar forcing suggests a factor $\exp(q^2/6)$. More explicitly, after the composition correction the leading mean-zero rank forcing is

$$g(r, z) = \frac{r}{2} - \frac{1}{4} + \frac{\mathbf{1}_{r < z} - z}{12}.$$

The frozen inverse (68) supplies a candidate rank correction of size $O(q^2/D)$. Its sign depends on the convention for conjugating the generator; its existence in the frozen model does not justify exponentiating it across all finite-state transitions.

If a sharp singular estimate $\log \mathcal{F}(t(q), x_0) \sim q^2/6$ and suitable coefficient transfer could both be proved, the scale $q \sim 2 \log n$ would suggest $h_n \sim (2/3)(\log n)^2$. This explains why the numerical candidate has structural motivation. Neither of those two required theorems has been established here.

[write] 2 October 2026. Part III proves the first of the two required theorems: $\log A'(t(q)) = q^2/6 + o(q^2)$, equation (72). Since $\mathcal{F}(t, x_0) = A'(t)$, this is the sharp singular estimate $\log \mathcal{F}(t(q), x_0) \sim q^2/6$. Its quantitative form has remainder $O(q^{3/2}\sqrt{\log q})$ (Theorem 28.1). Part III transfers it to the coefficients only in limsup and cumulative form; individual-coefficient transfer is still missing. Part III uses an explicit circle-kernel Poisson corrector, a global slow-profile patch and a first-exit decomposition, whose beta-integral factor cancels the large-state seed cost. That decomposition is how it overcomes the trapping at the root described in Section 18.3.

18.3 The nearly trapped root and rare growth paths

A naive rapid-mixing argument at the original root is false. For $x_0 = (1, 1, 1)$, the original ψ profile has rank tending to $1/3$. Its new-ascent child is $(2, 1, 2)$, with rank tending to $1/2$. Thus

$$\frac{\psi(q, (2, 1, 2))}{\psi(q, x_0)} = \exp(q/3 + O(1)), \quad \frac{1}{b(q)} \frac{\psi(q, (2, 1, 2))}{\psi(q, x_0)} = O(qe^{-q/6}).$$

The duplicate-descent child’s rate on this q scale is even smaller, $O(qe^{-2q/3})$. At a large frozen parameter, the transformed chain is therefore nearly trapped at its fixed root. It cannot be treated as if it were already in the large-state stationary rank regime, nor can unweighted typical trajectories be assumed to reach $D \gg q^2$ quickly.

A Feynman–Kac expectation could nevertheless be dominated by rare early growth paths rewarded by later potential. A useful proof would have to control their weighted contribution, possibly by a different conjugation or a quantitative seeding construction. It would also have to control returns to small states. The existing quadratic barriers are too coarse to show that this boundary contribution is $o(q^2)$.

The missing steps are therefore concrete: a uniform evolving-state residual through rank and type boundaries; control of finite differences of the exponentiated corrector; a weighted root-to-bulk estimate with subquadratic logarithmic cost; control of repeated small-state returns; and an individual-coefficient transfer theorem. A real-axis singular estimate by itself is not that transfer theorem.

19 Literature scope and further questions

The literature checks bundled with the research were made on 2 October 2026. They are scoped negative evidence, not an exhaustive novelty certificate.

Conway–Conway–Elvey Price–Guttmann [1] provide the recurrence, compaction, and numerical conjecture for the factorial constant. Their detailed Section 5 retains unresolved stretched-exponential and power factors, including numerical estimates $0.06 \leq \sigma \leq 0.1$ within the models investigated there. Those estimates are not rigorous bounds. The shorter expression in the introduction must not be promoted to an established equivalent. The upper theorem here rules out a positive fixed-power stretched-exponential factor, while leaving the limitations already stated.

Later results on additional forbidden patterns [4], selected length-four avoidance classes [5], and Fishburn matrix asymptotics [6] do not directly settle this single-000 restriction. In particular, no applicable multiplicity-preserving correspondence or suitable product generating function was identified in the checked sources.

The targeted regularity search likewise found no retrieved theorem proving log-concavity of $a_n/n!$. Real-rooted s -Eulerian polynomials [7] concern inversion sequences with prescribed coordinate bounds; the adaptive ascent bound and global letter cap here require an additional argument. Distributive-lattice methods [8] do not apply immediately to the non-meet-closed raw-word class. Generating series tracking total repetitions [9] or consecutive run lengths [10] do not record the complete multiplicity of every letter: for example, 0001 and 0011 have the same total repetitions, and a run-length cap would allow 01010 although one value appears three times.

An authorized indexed search of the default branch of the ProveIt repository [11] found no direct overlap under the sequence identifiers and the checked ascent-sequence, bounded-multiplicity, Fishburn, and constant keywords. The detailed queries and indexed commit are recorded in the bundled literature note. This is not a semantic audit of the entire repository or proof that the

indexed commit was the live tip. No publication or external repository change is part of this addendum.

The most focused next mathematical questions are:

1. Can one prove normalized log-concavity, or a weaker local coefficient regularity sufficient to reduce the lower extraction loss?
2. Can the circle-kernel Poisson inverse yield a uniform discrete corrector for rank and composition, with controlled boundary layers?
3. Can a weighted rare-seed estimate overcome the fixed-root trapping and control repeated excursions back to small states?
4. Can one establish a local limit or complex saddle estimate strong enough to transfer a sharp singular law to every coefficient?
5. After those steps, what are the next logarithmic, constant, and inverse corrections? A nonzero limiting amplitude and an all-orders expansion require additional estimates beyond a leading logarithmic-square law.

[write] **2 October 2026.** Status of these questions after the later Parts. Question 1 remains open; Part IV’s ratio theorem and its scans do not prove normalized log-concavity or the one-sided bound (204). Questions 2 and 3 are answered in Part III to the extent needed for the real-axis theorem. Its corrector is uniform at both interfaces and its nonlinear residual is controlled for $D \geq Kq$ (Sections 25 and 26). Its first-exit transfer (Section 27) bounds the boundary contribution, counting all returns below the boundary. Question 4 remains open: Part III transfers the singular law only to the cumulative sums and the limsup. Question 5 remains open.

20 Bundled sources and reproduction

This addendum is accompanied by its editable TeX and a two-pass `build.sh`. The build preserves the frozen report’s TeX-environment fallback and fixes the PDF metadata date for reproducibility. It requires pdfLaTeX with the packages used by the source. The illustration is already included as a PDF; the counting programs need not be rerun to build this document.

The directory `dependencies/frozen-foundation/` contains unchanged copies of the 1 October PDF, TeX, the radius and root-limit proof notes, verification scripts and recorded outputs, and the frozen audit notes. The source report directory and its archive were not rewritten.

The directory `support/fine-research/` preserves the research directory structure. Its principal items are:

- The signed and padded barrier proof, improved coarse lower proof, coefficient-obstruction note, and explicitly formal logarithmic-square route
- The kernel verification scripts and outputs, independent analytic audit, and formal-kernel audit
- Scoped literature notes and exact small-polynomial obstruction checks
- Exact integer coefficients through 400, floating normalized values through 1000, the published b-file through 176, C++ sources, analysis and fit scripts, diagnostics, and numerical reproduction instructions

Prebuilt counting binaries are not needed; compile the supplied source locally. The numerical README gives the C++17/GMP and Python dependencies and commands. The exact and floating

runs took approximately 62 and 278 seconds, respectively, in the recorded environment; these are observations, not performance guarantees.

The residual checker reports 86814 finite state–parameter cases and 616005 padding-transition mappings, all passing. Exact enumeration and all-state suffix checks corroborate the operator and the finite regularity observations. These computations are diagnostics: the infinite-state and all-parameter assertions are justified by the proofs in Sections 11–14, not by a grid. Recorded source hashes and the original provenance note support checking that copied foundation files are unchanged.

[write] **2 October 2026.** In this repository the delivery directories named above are flattened. The files of `support/fine-research/` and of the delivery root are shipped under the prefix `36-fine-` in `code/`, `data/` and the report root, and the figure is `figures/36-fine-correction-diagnostics.pdf`. The PDFs are not shipped, and neither are `dependencies/frozen-foundation.zip` and the nine files of `dependencies/frozen-foundation/` that are byte-identical to Part I’s delivery. The nine other files of that directory are shipped as `37-foundation-*`. The eighteen page renders `qa/page-NN.png` are not shipped; the report README explains how to regenerate them byte for byte. The README also maps every delivered name to its shipped name.

Part III

Logarithmic square growth for ascent sequences avoiding 000

Batch-77 manuscript 07, archive `a202058-lognormal-report.zip` (delivered as `a202058-lognormal.tex`, 23 pages), dated 2 October 2026. Delivered author line: “A self contained mathematical and reproducibility report”. Delivered Section k is Section $k + 20$ here.

Abstract of this Part (as delivered). Let a_n count ascent sequences in which every value occurs at most twice, let $A(t) = \sum_{n \geq 0} a_n t^n / n!$, and put $T = 3\pi^2/8$. We prove

$$\log A(t) \sim \frac{2}{3} \log^2 \frac{1}{T-t} \quad (t \uparrow T).$$

For $c_n = a_n T^n / n!$, positivity then gives

$$\log \sum_{j \leq n} c_j \sim \frac{2}{3} (\log n)^2, \quad \limsup_{n \rightarrow \infty} \frac{\log c_n}{(\log n)^2} = \frac{2}{3}.$$

The proof uses the exact compacted enumeration recurrence, an explicit circle-kernel Poisson correction uniform at both interfaces, a global slow-profile patch, and a first-exit integral whose beta factor cancels the large-state seed cost. The nonlinear residual is controlled for weighted state size $D \geq Kq$, even when the corrector itself is not small. All necessary characteristic and infinite-state comparison foundations are included. A parameter-uniform refinement gives the root remainder $O(q^{3/2} \sqrt{\log q})$. The full individual limit, a coefficient equivalent, a ratio limit, and an amplitude remain unproved. The article therefore resolves the leading real-axis and cumulative logarithmic-square scales, while leaving a specific coefficient-regularity problem open.

21 The result and its precise scope

An ascent sequence is a word $w_1 \cdots w_n$ of nonnegative integers with $w_1 = 0$ and

$$0 \leq w_i \leq 1 + \text{asc}(w_1 \cdots w_{i-1}) \quad (i \geq 2), \quad \text{asc}(w) = \#\{i : w_i < w_{i+1}\}.$$

Classical pattern avoidance of 000 means that no value occurs three times, whether or not those occurrences are consecutive. Let a_n count these words and set $a_0 = 1$. This is OEIS A202058, the cap-two column of A294220 [2, 3]. Its first terms are

$$1, 1, 2, 4, 10, 27, 83, 277, 1015, 4007, 17047.$$

Throughout, logarithms are natural. Define

$$T = \frac{3\pi^2}{8}, \quad \mu = T^{-1} = \frac{8}{3\pi^2}, \quad A(t) = \sum_{n \geq 0} a_n \frac{t^n}{n!}, \quad c_n = \frac{a_n T^n}{n!}, \quad S_N = \sum_{n=0}^N c_n. \quad (70)$$

The characteristic clock is

$$t(q) = \int_0^q \frac{v e^{-v/2}}{\sqrt{2 - e^{-v}}(1 - e^{-v})} dv, \quad q \geq 0, \quad (71)$$

where the integrand has value 1 at zero.

Theorem 21.1. *The function $t(q)$ increases from 0 to T , and*

$$\log A'(t(q)) = \frac{q^2}{6} + o(q^2), \quad (72)$$

$$\log A(t) \sim \frac{2}{3} \left(\log \frac{1}{T-t} \right)^2 \quad (t \uparrow T), \quad (73)$$

$$\log S_N \sim \frac{2}{3} (\log N)^2, \quad (74)$$

$$\limsup_{n \rightarrow \infty} \frac{\log c_n}{(\log n)^2} = \frac{2}{3}. \quad (75)$$

In particular A has radius T . No coefficient monotonicity or log-concavity hypothesis is used in these conclusions.

The recurrence and its compaction are due to Conway, Conway, Elvey Price and Guttmann [1, Sections 2.2 and 2.6]. Their Section 5 conjectured the factorial exponential constant $8/(3\pi^2)$ from numerical series analysis, while leaving finer factors unresolved. The present advance is the logarithmic-square theorem. The principal new work is a uniform corrected residual, its globalization to the *full* continuation function, and the boundary-to-root cancellation. The characteristic barriers are foundations from the preceding investigation, reproduced here so that the proof has no dependency on inaccessible companion notes.

[write] 2 October 2026. The “preceding investigation” is Parts I and II, which this manuscript does not cite. The characteristic barriers and the root limit are those of the 1 October report (Part I, Theorem 1.1). The padded comparison (96), the all-index enclosure (146) and its inverse (150) are Part II’s (43)–(50), Theorem 9.1 and Corollary 9.2. Part II’s Conjecture 16.1 is proved here in limsup and cumulative form (Theorem 21.1). The proof-input fingerprints of this manuscript (`data/07-lognormal-source-audit-proof-of-input-fingerprints.json`) include Part II’s padded-barrier proof note.

It is important to distinguish (75) from

$$\frac{\log c_n}{(\log n)^2} \longrightarrow \frac{2}{3}. \quad (76)$$

Equation (76) remains open. The theorem gives the upper envelope and a cumulative law, not a matching lower bound at every index. In particular it does not establish a coefficient transseries, a ratio limit, a limiting amplitude, or a full equivalent. Section 28 separately proves a quantitative $O(q^{3/2}\sqrt{\log q})$ remainder in (72); no optimal remainder or effective numerical constants are asserted. Section 30 records only inversions justified by proved statements. Section 31 describes the source and audit scope.

[write] 2 October 2026. Superseded: the ratio limit, listed here as not established, is proved in Part IV, Theorem 33.1, with rate $O(n^{-1/4}(\log n)^{3/4})$. The pointwise law (76), a coefficient transseries, a limiting amplitude and a full equivalent remain open.

22 The exact positive transfer model

Twice-used labels can be deleted from the active alphabet. The remaining once-used labels can be compacted to an initial segment without changing suffix counts. In the variables of [1], with

$G_0(a, \ell, s) = 1$, the compacted recurrence is

$$G_j(a, \ell, s) = \sum_{i=0}^{s-1} G_{j-1}(a + \mathbf{1}_{i>\ell} - 1, i - 1, s - 1) + \sum_{i=s}^{a+1} G_{j-1}(a + \mathbf{1}_{i>\ell}, i, s + 1), \quad a_n = G_{n-1}(0, 0, 1) \quad (n \geq 1). \quad (77)$$

The adjusted values a, ℓ may equal -1 . To recall why compaction is legitimate, take adjacent labels $i, i+1$ below or at the last value, with $i < \ell$, and interchange their remaining capacities. Reverse each maximal suffix block consisting of these two labels and interchange the labels inside it. This involution interchanges their multiplicities and preserves ascent counts within each block and at its outside comparisons. Both labels were already permitted, since $i+1 \leq \ell$; subsequent bounds see unchanged ascent totals. Thus a once-used and an unused label below the last value can be exchanged. After a new label is appended, it can be moved through unused labels until the once-used set is initial; after a repetition, the exhausted label is deleted. This is the binary-capacity instance of the compaction in [1], and gives (77). Canonical states preserve continuation counts; they need not encode every literal property of the original prefix.

Set $m = a + 2$, $k = \ell + 1$, $u = m - s$. We use the invariant state space

$$\mathcal{S} = \{x = (s, u, k) : s \geq 0, u \geq 1, 0 \leq k < s + u\}.$$

For every integer $0 \leq i < m = s + u$, the child is

$$x_i = \begin{cases} (s - 1, u, i), & i < s, i < k, \\ (s - 1, u + 1, i), & i < s, i \geq k, \\ (s + 1, u, i + 1), & i \geq s, i \geq k, \\ (s + 1, u - 1, i + 1), & i \geq s, i < k. \end{cases} \quad (78)$$

The first two cases are duplicate (D) steps, and the last two are new (N) steps. An ascent corresponds to $i \geq k$. An N descent from $u = 1$ would require $s \leq i < k \leq s$, so it cannot occur. There are no D steps when $s = 0$. Thus all children are valid states.

Define

$$(\mathcal{T}f)(x) = \sum_{i=0}^{m-1} f(x_i), \quad x_0 = (1, 1, 1), \quad F(t, x) = \sum_{j \geq 0} (\mathcal{T}^j \mathbf{1})(x) \frac{t^j}{j!}. \quad (79)$$

Transition multiplicities are retained. The series defines the minimal nonnegative continuation function, initially allowing $+\infty$. The recurrence gives $\mathcal{T}^j \mathbf{1}(x_0) = a_{j+1}$, hence

$$F(t, x_0) = A'(t). \quad (80)$$

The integer resource

$$H(x) = s + 2u \quad (81)$$

changes by $+1$ on an ascent and -1 on a descent. This exact property will be essential: using a clock-dependent resource at the first-exit step would lose the cancellation.

23 Characteristic foundations and global coarse bounds

23.1 The characteristic profile and endpoint

For $q \geq 0$ put

$$E = e^{-q}, \quad R = 2 - E, \quad p = \frac{q + \log R}{2}, \quad b = \frac{e^p(1 - E)}{q}, \quad (82)$$

with $b(0) = 1$. Then $t'(q) = 1/b(q)$ and $p'(q) = 1/R(q)$. For a state $x = (s, u, k)$ define

$$D = s + Ru, \quad h_0(y) = \min(y, s) + R \max(y - s, 0), \quad r(y) = h_0(y)/D, \quad r = r(k), \quad z = s/D, \quad (83)$$

and

$$\psi(q, x) = \exp\{ps + qu - qr\}. \quad (84)$$

Here $H/2 \leq D \leq H$, $0 \leq r < 1$, $0 \leq z < 1$, and $\psi(0, x) = 1$. We use one-sided limits when formulas extend to $z = 1$.

The substitution $y = \sqrt{2e^q - 1}$ gives

$$t(\infty) = \int_1^\infty \frac{2 \log((y^2 + 1)/2)}{y^2 - 1} dy.$$

Integration by parts, followed by $v = (y - 1)/(y + 1)$, yields

$$t(\infty) = -2 \int_0^1 \log v \frac{1 + v}{(1 - v)(1 + v^2)} dv = -2 \int_0^1 \log v \left(\frac{1}{1 - v} + \frac{v}{1 + v^2} \right) dv.$$

The boundary terms vanish. The first integral is $\pi^2/3$, and the second is $\pi^2/24$, by integration of the geometric series (or its Abel-regularized alternating version). Therefore $t(\infty) = 3\pi^2/8 = T$.

The explicit derivative in (71) also proves the two-sided tail estimate

$$c(q + 1)e^{-q/2} \leq \delta(q) := T - t(q) \leq C(q + 1)e^{-q/2} \quad (q \geq 0), \quad (85)$$

with fixed positive constants. Indeed, for $q \geq 1$ the derivative is between constant multiples of $qe^{-q/2}$; integrate to infinity and adjust on a compact interval. In particular

$$\log \frac{1}{\delta(q)} = \frac{q}{2} - \log(q + 1) + O(1). \quad (86)$$

Both directions in (85) are needed later.

23.2 Frozen weights and exact identities

For real $y \in [0, m)$ set

$$w(y) = \begin{cases} e^{-p+q\mathbf{1}_{y \geq k}}, & y < s, \\ e^{p-q\mathbf{1}_{y < k}}, & y \geq s, \end{cases} \quad g(y) = w(y)e^{-q(r(y)-r)}.$$

Direct integration on the intervals separated by s, k gives

$$\int_0^m g(y) dy = \lambda := Db/R. \quad (87)$$

One can also verify this identity by setting $f(y) = e^{-qh_0(y)/D}$: on the two type intervals $\lambda f' = -e^{-p}(e^q - 1)f$ and $\lambda f' = -e^p(1 - e^{-q})f$, respectively, and $f(m) = e^{-q}f(0)$. The resulting integral eigen-equation at $k = 0$ is

$$\frac{D}{q} \left[e^{q-p}(1 - e^{-qs/D}) + \frac{e^p}{R}(e^{-qs/D} - e^{-q}) \right] = \lambda,$$

since $e^{q-p} = e^p/R$; differentiation then verifies every k .

Each piece of g decreases, the breakpoints are integers, and $g \leq \sqrt{2}e^{q/2}$. Thus, with $S = \sum_i g(i)$,

$$0 \leq S - \lambda \leq 3\sqrt{2}e^{q/2}. \quad (88)$$

In the order of (78), the exact child ranks are

$$r_i = \frac{i}{D-1}, \quad \frac{i}{D+R-1}, \quad \frac{h_0(i)+1}{D+1}, \quad \frac{h_0(i)+1}{D+1-R}. \quad (89)$$

All relevant denominators are positive. Subtraction shows $|r_i - r(i)| \leq 4/D$; except for a duplicate ascent, $r_i \geq r(i)$. For that exceptional case,

$$0 \leq r(i) - r_i = \frac{i(R-1)}{D(D+R-1)} \leq \frac{1}{D}. \quad (90)$$

The exact transition ratio is $g(i)e^{-q(r_i - r(i))}$.

For a fixed integer state,

$$\partial_q \log \psi = D/R - B, \quad B = r + qE\partial_R r, \quad |B| \leq 1, \quad (91)$$

where

$$\partial_R r = \begin{cases} -r(1-z)/R, & r \leq z, \\ -z(1-r)/R, & r \geq z. \end{cases} \quad (92)$$

The formulas agree at the type interface, and $-1/(4R) \leq \partial_R r \leq 0$.

23.3 Padding and infinite-state comparison

Equations (88)–(91) imply

$$\frac{\mathcal{T}\psi}{b\psi} - \partial_q \log \psi \leq qe^{q/D} + 3\sqrt{2}(q+1)e^{q/D} + 1, \quad (93)$$

$$\partial_q \log \psi - \frac{\mathcal{T}\psi}{b\psi} \leq 4q + 1. \quad (94)$$

For the first inequality use $\mathcal{T}\psi/\psi \leq e^{q/D}S$, $\lambda/b = D/R$, $e^x - 1 \leq xe^x$, and $e^{q/2}/b \leq q+1$. For the second use $\mathcal{T}\psi/\psi \geq e^{-4q/D}\lambda$ and $1 - e^{-x} \leq x$.

For an integer $L \geq 0$, define $I_L(s, u, k) = (s+L, u, k+L)$. Each original transition i maps exactly to the padded transition $i+L$, and the remaining L transitions are nonnegative. Consequently

$$\mathcal{T}(f \circ I_L) \leq (\mathcal{T}f) \circ I_L \quad (f \geq 0). \quad (95)$$

Fix Q , take $L = \lceil Q \rceil$, and put

$$V(q) = \frac{e(1+3\sqrt{2})}{2}q^2 + (3\sqrt{2}e+1)q.$$

On $0 \leq q \leq Q$, padded states have $D \geq L+1$, so $e^{V(q)}\psi(q, I_L x)$ is a supersolution. Also $e^{-2q^2-q}\psi(q, x)$ is a subsolution. We claim

$$e^{-2q^2-q}\psi(q, x) \leq F(t(q), x) \leq e^{V(q)}\psi(q, I_L x), \quad 0 \leq q \leq Q. \quad (96)$$

Here and below an infinite positive operator needs an actual comparison argument. Give each of the m transitions rate 1. The associated continuous-time chain has generator $\mathcal{T} - mI$. Since each jump increases m by at most one, a Yule-process domination shows nonexplosion. Cancellation of the holding-time factors against the potential gives

$$F(t, x) = \mathbb{E}_x \exp \left(\int_0^t m(X_v) dv \right), \quad (97)$$

with infinity allowed: summing a fixed j -jump path contributes exactly $t^j/j!$. Stop on exiting $m \leq N$. Finite-state Dynkin comparison for a nonnegative supersolution, followed by discarding the nonnegative exit term and then letting $N \rightarrow \infty$, proves the upper inequality in (96) and finiteness for $t < T$.

For a subsolution at a fixed time $t < T$, choose $\varepsilon > 0$ with $t + \varepsilon < T$ and a single padded supersolution valid through that later time. Uniformly for $0 \leq v \leq t$,

$$\frac{e^{-2q(v)^2-q(v)}\psi(q(v), y)}{e^{V(q(v+\varepsilon))}\psi(q(v+\varepsilon), I_L y)} \leq C_{t,\varepsilon} e^{-\eta_{t,\varepsilon} m(y)}.$$

Indeed both $p(q(v+\varepsilon)) - p(q(v))$ and $q(v+\varepsilon) - q(v)$ have positive compact minima; rank terms and fixed padding contribute bounded factors. The stopped exit contribution is at most $Ce^{-\eta^N}$ times the later supersolution at the starting state, and vanishes. This proves the lower inequality.

Since $\log \psi(q, I_L x) - \log \psi(q, x) \leq Lp(q) + q$, we obtain the useful state-uniform comparison

$$-2q^2 - q \leq \log F(t(q), x) - \log \psi(q, x) \leq V(q) + \lceil q \rceil p(q) + q = O((q+1)^2). \quad (98)$$

The same argument permits any later subsolution whose growth, on each compact q interval, is bounded by a finite multiple of ψ . For comparison initialized at a positive time, apply the stopped formula with terminal data F at that time; the positive semigroup identity follows coefficientwise from (79). No uniqueness theorem for an unbounded system is being assumed.

24 The complete continuum forcing

The frozen continuum kernel becomes especially simple in weighted rank. Define

$$\kappa_q(d) = \frac{qe^{-qd}}{1-E} \quad (0 \leq d < 1), \quad (K_q f)(r) = \int_0^1 \kappa_q((\xi - r) \bmod 1) f(\xi) d\xi.$$

For a frozen target rank $\xi = r(y)$ put

$$\nu = \mathbf{1}_{\xi \geq z}, \quad a_* = \mathbf{1}_{\xi \geq r}, \quad c_* = 1 + (R-1)\nu, \quad \sigma = 2\nu - 1, \quad \Delta = -1 + (2-R)\nu + Ra_*. \quad (99)$$

Also write $a = \nu - \xi\Delta$, $c = \sigma - z\Delta$. At an actual lattice target $y = i$, direct substitution into (78) gives the exact identities

$$D_i = D + \Delta, \quad z_i - z = \frac{c}{D + \Delta}, \quad r_i - \xi = \frac{a}{D + \Delta}, \quad |\Delta| \leq 1. \quad (100)$$

The weak inequalities in (99) agree with the transition at $i = s$ and $i = k$, including coincidences. The last active label has rank strictly less than 1; the right endpoint is excluded.

The exact change of variables is

$$\frac{g(y)}{b} = \frac{c_*}{R} \kappa_q((\xi - r) \bmod 1), \quad dy = \frac{D}{c_*} d\xi. \quad (101)$$

Thus $g(y) dy / (Db/R) = \kappa_q d\xi$, proving directly that Lebesgue measure is stationary for K_q . In particular no exponential factor in q is lost when moving between label and rank coordinates. The rank mesh, including its closing gap, has spacing in $[1/D, 2/D]$.

The composition correction is $\exp(qz/6)$. At a fixed integer state,

$$z_q = -Ez(1 - z)/R.$$

Define the full first-order forcing by

$$h(q, r, z) = \frac{q}{R} \int_0^1 \kappa_q((\xi - r) \bmod 1) \left(\frac{c_*}{2} - a + \frac{c}{6} \right) d\xi + B - \frac{z}{6} - \frac{qz_q}{6}. \quad (102)$$

The first term includes the entire lattice Euler contribution, the exact rank shift, and the composition shift; the final terms subtract the fixed-state derivative. We next compute its mean without discarding rank-wrap or type-interface layers.

Lemma 24.1. *Let*

$$P_q(\xi) = \frac{1 - e^{-q\xi}}{1 - E}, \quad I(q) = \int_0^1 \xi P_q(\xi) d\xi = \frac{1/2 - [1 - (1 + q)E]/q^2}{1 - E}, \quad m(q) = qI(q) - \frac{q}{3R} + \frac{1}{2}. \quad (103)$$

Then

$$\bar{h}(q, z) := \int_0^1 h(q, r, z) dr = m(q) - \frac{qEz}{6R(1 - E)} = \frac{q}{3} + \frac{1}{2} - \frac{1}{q} + O(qe^{-q}), \quad (104)$$

uniformly for $q \geq 1$ and $0 \leq z \leq 1$.

Proof. Under the joint measure $dr \kappa_q((\xi - r) \bmod 1) d\xi$, the target ξ is uniform and $\mathbb{P}(a_* = 1 \mid \xi) = P_q(\xi)$. Integrating the two indicator types gives

$$\begin{aligned} \langle a \rangle &= 1 - z + \frac{R - 1}{2} + \frac{2 - R}{2} z^2 - RI(q), \\ \langle B \rangle &= \frac{1}{2} - \frac{qE}{2R} z(1 - z), \\ \langle \Delta \rangle &= 1 - Ez + \frac{RE}{1 - E} - \frac{R}{q}, \\ \langle c \rangle &= 1 - 3z + Ez^2 - \frac{RzE}{1 - E} + \frac{Rz}{q}. \end{aligned}$$

Since $\langle c_* \rangle = R - (R - 1)z$, the uncorrected part $q\langle c_*/2 - a \rangle / R + \langle B \rangle$ simplifies to $qI(q) + q(z - 1)/(2R) + 1/2$. Adding $q\langle c \rangle / (6R) - z/6 - qz_q/6$ proves the exact formula. Expanding only exponentially small rational factors proves the final uniform estimate. This calculation uses the frozen continuum measure exactly, not a stationarity assumption for the discrete state process. $\square$

25 A uniform rank corrector

Put

$$v = h - \bar{h}, \quad Jv(r) = \int_0^r v(u) du, \quad \eta(q, D, r, z) = -\frac{R}{D} \{qJv(r) - v(r)\}, \quad \phi = \psi \exp(qz/6 + \eta). \quad (105)$$

The suppressed parameters of Jv are q, z . Since v has mean zero, Jv is continuous and periodic, with $Jv(0) = Jv(1) = 0$. Its own mean need not vanish. Use the right-hand value at the periodic seam $r = 0$.

Theorem 25.1 (Uniform residual). *There are absolute constants K_0, C such that, for $q \geq 1$ and $D \geq K_0 q$,*

$$\left| \frac{\mathcal{T}\phi}{b\phi} - \partial_q \log \phi - m(q) \right| \leq C \left(\frac{q^2}{D} + qe^{-q} \right). \quad (106)$$

In particular

$$\left| \frac{\mathcal{T}\phi}{b\phi} - \partial_q \log \phi - \frac{q}{3} \right| \leq C \left(1 + \frac{q^2}{D} \right). \quad (107)$$

The derivative fixes the integer state. The bounds hold at rank zero, at the type interface, through wrap, and as composition approaches either endpoint. They do not require $\eta = o(1)$ or $D \gg q^2$.

25.1 Bounded variation and the moving interface

We give the regularity details because an unquantified smooth bulk expansion would not imply Theorem 25.1. Write

$$w_z(\xi) = \frac{1}{3} + \frac{3R-7}{6} \mathbf{1}_{\xi \geq z} + [-1 + (2-R)\mathbf{1}_{\xi \geq z}](\xi - z/6),$$

and

$$A_0(q, r) = \frac{1 - e^{-q(1-r)}}{1 - E}, \quad A_1(q, r) = \frac{r + 1/q - (1 + 1/q)e^{-q(1-r)}}{1 - E}.$$

Then (102) becomes exactly

$$h = \frac{q}{R} K_q w_z + q(A_1 - zA_0/6) + B - z/6 + \frac{qEz(1-z)}{6R}. \quad (108)$$

The supremum and periodic total variation of w_z are uniformly bounded for $R \in [1, 2]$, $z \in [0, 1]$. Its distributional z derivative is a bounded density plus one atom of bounded mass at z . The kernel obeys

$$\|\kappa_q\|_\infty \leq Cq, \quad \|\kappa_q\|_1 = 1, \quad \|\partial_q \kappa_q\|_1 \leq C/q, \quad \text{TV}(\kappa_q) \leq Cq.$$

Convolving the finite measure Dw_z gives both a supremum $O(q)$ and an L^1 norm $O(1)$ for $\partial_r K_q w_z$. The same two bounds hold for $\partial_z K_q w_z$, including the moving atom. Translation continuity of the L^1 kernel proves continuity across $r = z$.

The explicit derivatives

$$\partial_r A_0 = -\frac{qe^{-q(1-r)}}{1 - E}, \quad \partial_r A_1 = \frac{1 - (q+1)e^{-q(1-r)}}{1 - E}$$

have supremum $O(q)$ and L^1 norm $O(1)$; also $|\partial_q A_j| \leq C/q$. The last bound follows by integrating $\partial_q \kappa_q$ against 1 or ξ over $[r, 1]$. The piecewise bilinear factor in B has bounded one-sided derivatives and agrees at $r = z$; its multiplier qE/R and its q derivative are bounded. Thus

$$|h| \leq Cq, \quad |h_q| \leq C, \quad |h_r| + |h_z| \leq Cq^2, \quad \int_0^1 (|h_r| + |h_z|) dr \leq Cq. \quad (109)$$

The integral bound is essential for controlling $\partial_z Jv$.

At the periodic seam, direct evaluation of (108) gives

$$h(q, 1-, z) - h(q, 0, z) = qz/6 + \frac{qE}{1-E}. \quad (110)$$

Indeed the convolution has no jump, the qA terms have jump $-1 + qE/(1-E) + qz/6$, and B has jump 1. The interface $r = z$ is continuous, including coincident endpoints by one-sided limits.

It follows from (105)–(110) that

$$\begin{aligned} |\eta| &\leq Cq^2/D, & |\eta_r| + |\eta_z| &\leq Cq^2/D, \\ |\eta(1-) - \eta(0)| &\leq Cq/D, & \left| \frac{d\eta}{dq} \right|_{\text{fixed state}} &\leq Cq/D. \end{aligned} \quad (111)$$

For the derivative, first hold (D, r, z) fixed. Equation (109) gives $\eta_q = O(q/D)$. Along a fixed integer state, $D_q/D, r_q, z_q = O(E)$, and their extra contributions are $O(q^2 E/D) \leq Cq/D$. The sign of $r - z$ is the sign of $k - s$ and never changes as q varies. If $k = s$, the path stays on the continuous common interface and the tangential derivatives agree.

If $d = (\xi - r) \bmod 1$, integration along the clockwise arc, adding at most one seam jump, gives

$$|\eta(q, D, \xi, z) - \eta(q, D, r, z)| \leq C \left(\frac{q^2 d}{D} + \frac{q}{D} \right). \quad (112)$$

The q/D term is needed only across the seam, but retaining it everywhere is harmless and convenient.

25.2 The pointwise Poisson identity

Integration by parts in the forward kernel, using the continuity and periodicity of Jv , gives

$$K_q v = q(K_q - I)Jv.$$

Consequently

$$(K_q - I)(qJv - v) = v, \quad \frac{D}{R}(K_q - I)\eta = -v. \quad (113)$$

This is the concrete form of $(K_q - I)^{-1} = q\partial_r^{-1} - I$ on mean-zero functions. It is valid pointwise with the selected right-hand representative at zero: convolution ignores individual point values, and the same value of v is used in both identity terms. Thus no almost-everywhere assertion is substituted at an actual lattice state.

25.3 Euler correction and linear discrete errors

On an original-label interval of integer length, all type and ascent indicators are fixed. The frozen weight is geometric with exponent $\alpha = qc_*/D$. Its left sum divided by its integral is exactly $\alpha/(1 - e^{-\alpha})$. Uniformly for $D \geq K_0 q$,

$$\frac{\alpha}{1 - e^{-\alpha}} = 1 + \frac{\alpha}{2} + O(\alpha^2).$$

After summing intervals and using (101),

$$\frac{S}{b} = \frac{D}{R} + \frac{q}{2R} K_q[c_*](r) + O(q^2/D). \quad (114)$$

The remainder is at most $C(q^2/D^2)(D/R)$. Empty intervals or coincident integer breakpoints contribute no extra term. In particular all wrap and interface Euler terms are retained in the convolution in (114).

Let $\tau = -a + c/6$. The logarithmic change of $\psi e^{qz/6}$ relative to the frozen ratio is exactly

$$\ell_i = \frac{q\tau_i}{D + \Delta_i} = \frac{q}{D}\tau_i + O(q/D^2).$$

The function τ is bounded and piecewise affine. On the original label intervals the variation of $(g/b)\tau$ is $O(q)$, including the finite jumps. The left Riemann error is bounded by that variation. Since the total frozen weight is $O(D)$, it follows that

$$\sum_i \frac{g_i}{b} \ell_i = \frac{q}{R} \int_0^1 \kappa_q((\xi - r) \bmod 1) \tau(\xi) d\xi + O(q^2/D). \quad (115)$$

Replacing $D + \Delta$ by D costs only $O(q/D)$ in the summed expression, which is absorbed for $q \geq 1$.

For the frozen rank increment $\Delta_r \eta = \eta(q, D, \xi, z) - \eta(q, D, r, z)$, the smooth variation of $(g/b)\Delta_r \eta$ is bounded by

$$C \int_0^1 (q\kappa_q(d)|\Delta_r \eta| + \kappa_q(d)|\eta_r(\xi)|) d\xi \leq Cq^2/D.$$

Here (112) and the exponential moments of κ_q were used. A type-weight jump obeys the same bound, since

$$\sup_{0 \leq d < 1} \kappa_q(d)(q^2 d/D + q/D) \leq Cq^2/D.$$

At the interior kernel jump $\xi = r$, the increment itself is zero. At the seam, the corrector jump is $O(q/D)$ and the kernel endpoint is $O(q)$, again giving $O(q^2/D)$. Thus the genuine lattice estimate is

$$\sum_i \frac{g_i}{b} [\eta(q, D, \xi_i, z) - \eta(q, D, r, z)] = \frac{D}{R} (K_q - I)\eta + O(q^2/D), \quad (116)$$

where $\xi_i = r(i)$ denotes the frozen target rank and ℓ_i the logarithmic increment.

Changing (D, ξ_i, z) to the actual child's coordinates in (100) changes η by $O(q^2/D^2)$: $\eta_D = -\eta/D$, and the rank and composition changes are $O(1/D)$. The straight segment between the two ranks stays in $[0, 1)$, so it does not cross the seam. The type interface is continuous with bounded one-sided derivatives. Summing against $O(D)$ frozen weight gives another $O(q^2/D)$ error.

For clarity, the Riemann principle just used is elementary. For each unit interval with its right-continuous endpoint convention,

$$\left| f(i) - \int_i^{i+1} f(y) dy \right| \leq \text{Var}(f; [i, i+1)).$$

All discontinuities here are at integer breakpoints. Summing smooth variations and one-sided jumps proves the bounds even when type and rank breakpoints coincide.

25.4 The nonlinear exponential error

Let

$$\chi_i = \ell_i + \eta(q, x_i) - \eta(q, x).$$

The preceding estimates imply, for $d_i = (r(i) - r) \bmod 1$,

$$|\chi_i| \leq Cq/D + Cq^2 d_i/D. \quad (117)$$

Choose K_0 large enough that $Cq^2/D \leq q/2$ when $D \geq K_0 q$. For $j = 0, 1, 2$ the weighted rank mesh gives

$$\sum_i \frac{g_i}{b} d_i^j e^{qd_i/2} \leq CDq^{-j}. \quad (118)$$

To prove this, each bin $[\ell/q, (\ell+1)/q)$ contains at most $C(D/q+1)$ points; $g_i/b \leq Cqe^{-qd_i}$ by (101). Sum $Cq(D/q+1)q^{-j}(\ell+1)^j e^{-\ell/2}$ over $\ell \geq 0$ and use $D \geq q$. This includes the bin containing $d_i = 0$.

Now $|e^x - 1 - x| \leq x^2 e^{|x|}/2$ and (117)–(118) yield

$$\sum_i \frac{g_i}{b} |e^{\chi_i} - 1 - \chi_i| \leq C \left(\frac{q^2}{D^2} D + \frac{q^4}{D^2} \frac{D}{q^2} \right) \leq Cq^2/D. \quad (119)$$

This is the decisive uniform estimate. A supremum bound $e^{O(q^2/D)}$ would not suffice in the range needed for root transfer. The exponentially decaying clockwise kernel instead controls rare large rank increments.

Combining (114), (115), (116), and (119), and subtracting the fixed-state derivative of $\log \phi$, gives

$$\frac{\mathcal{T}\phi}{b\phi} - \partial_q \log \phi = h + \frac{D}{R}(K_q - I)\eta - \frac{d\eta}{dq} + O(q^2/D) = \bar{h} - \frac{d\eta}{dq} + O(q^2/D).$$

Equations (104) and (111) prove Theorem 25.1.

26 Global barriers for the full continuation function

A local sharp estimate does not bound the full function if paths can return to small states. We now construct global barriers, including those excursions, rather than compare only with a killed process.

Lemma 26.1 (Slow profile). *There is an absolute q_* such that for every $A \geq 1$,*

$$S_A(q, x) = e^{Aq^2} \psi(q/2, x), \quad (\partial_q - \mathcal{T}/b(q))S_A \geq (q + H/8)S_A \quad (q \geq q_*).$$

Proof. At a parameter v , every frozen ratio is at most $\sqrt{2}e^{v/2}$. Only a duplicate ascent lowers the true rank below the frozen rank, and (90) sharpens to

$$\frac{i(R-1)}{D(D+R-1)} \leq \frac{s(R-1)}{(s+R)^2} \leq \frac{R-1}{4R} \leq \frac{1}{8}.$$

Thus each exact child ratio at v is at most $\sqrt{2}e^{5v/8}$. There are $m \leq H$ children, and $b(q) \asymp e^{q/2}/q$, so

$$\frac{\mathcal{T}\psi(q/2, x)}{b(q)\psi(q/2, x)} \leq CHqe^{-3q/16} \leq H/8$$

for large absolute q . Meanwhile (91) gives $\partial_q \log \psi(q/2, x) \geq H/4 - 1/2$. Hence the residual of S_A is at least $2Aq + H/8 - 1/2 \geq q + H/8$ after enlarging q_* . $\square$

The slow profile alone does not dominate F initially over arbitrarily large states. It is used only as a patch combined with a full- q term.

Fix $0 < \varepsilon < 1/6$ and set

$$\kappa_{\pm} = 1/6 \pm \varepsilon, \quad P_{\pm} = e^{\kappa_{\pm} q^2} \phi.$$

By (107), choose a fixed K sufficiently large and then $q_0 \geq q_*$ sufficiently large, depending only on ε , so

$$(\partial_q - \mathcal{T}/b)P_+ \geq \varepsilon q P_+, \quad (\partial_q - \mathcal{T}/b)P_- \leq -\varepsilon q P_- \quad \text{when } H \geq Kq. \quad (120)$$

Here $D \geq H/2$ has absorbed a fixed factor into K .

26.1 The upper cutoff

Choose a smooth function $0 \leq \theta \leq 1$, zero on $[0, 1]$ and one on $[2, \infty)$, with bounded derivative. Write $\theta_q(x) = \theta(H/(Kq))$, and try

$$U(q, x) = C_+ \{ \theta_q(x) P_+(q, x) + S_A(q, x) \}.$$

Outside the enlarged strip $Kq - 1 \leq H \leq 2Kq + 1$, every child cutoff lies on the same plateau as its parent. Inside the strip, the exact defect is

$$\begin{aligned} (\partial_q - \mathcal{T}/b)(\theta P_+) &= \theta(\partial_q - \mathcal{T}/b)P_+ + (\partial_q \theta)P_+ \\ &\quad - \sum_i \frac{P_+(x_i)}{b} [\theta(x_i) - \theta(x)]. \end{aligned}$$

The first term is nonnegative whenever $\theta > 0$, by (120); it vanishes when $\theta = 0$.

Here is a uniform bound for the remaining terms that does not assume any mixing. On the strip, $H_i \leq 2Kq + 2$, $D_i \geq H_i/2$, and (111) is global. Thus

$$\begin{aligned} \log P_+(q, x_i) &\leq \kappa_+ q^2 + H_i q/2 + C(H_i + q + q^2/D_i), \\ \log S_A(q, x) &\geq Aq^2 + Hq/4 - q/2. \end{aligned}$$

The ratios of $P_+(x)$ and $P_+(x_i)$ to $S_A(x)$ are at most

$$\exp\{(\kappa_+ - A + K/2)q^2 + O_K(q)\}.$$

There are at most $2Kq + 1$ children, $|\partial_q \theta| = O(1/q)$, and $1/b \leq e^{O(q)}$. Choose $A > \kappa_+ + K/2 + 2$ and then enlarge q_0 . The entire negative cutoff defect is at most $qS_A/2$, which the strict slow margin absorbs. Hence U is a global supersolution.

26.2 The positive-part lower barrier

Increase A if necessary. For $H \leq Kq$, using $D \geq H/2 \geq 1$ and (111),

$$\log(P_-/S_A) \leq (\kappa_- - A + K/4 + C/2)q^2 + O_K(q).$$

Thus $P_- \leq S_A$ on the small-state region for $q \geq q_0$. Set

$$L(q, x) = c_- \{ P_-(q, x) - S_A(q, x) \}_+.$$

Where $L > 0$, the sharp inequality applies and

$$\partial_q(P_- - S_A) \leq \mathcal{T}(P_- - S_A)/b \leq \mathcal{T}(P_- - S_A)_+/b.$$

The positive part is locally absolutely continuous. Almost everywhere on its zero set its derivative is zero, while its transfer is nonnegative. Hence L is a global subsolution in the absolutely continuous comparison sense, including zero crossings.

26.3 Initial data and comparison at infinity

At fixed q_0 , (98) shows that F/ψ is bounded above and below uniformly in all states. The same is true of ϕ/ψ by (111). Choose $c_- > 0$ small enough that $L(q_0) \leq F$. For $H \geq 2Kq_0$ the sharp term lets a fixed C_+ dominate F ; there are only finitely many states with $H < 2Kq_0$, and increasing C_+ covers them by the positive slow term. Thus $U(q_0) \geq F$ globally.

On every compact q interval, $|\eta| \leq Cq^2$, $e^{qz/6} \leq e^{q/6}$, and $\psi(q/2, x)/\psi(q, x) \leq e^q$. Every trial function is consequently bounded by a finite scalar multiple of $\psi(q, x)$. A padded supersolution at a strictly later original time dominates these trial functions by an additional e^{cH} factor. The stopped comparison and vanishing-exit argument of Section 23.3 applies verbatim, including the positive part and cutoff. We have proved

$$c_-(P_- - S_A)_+ \leq F(t(q), x) \leq C_+(\theta_q P_+ + S_A) \quad (q \geq q_0). \quad (121)$$

These are bounds for full F and already include returns to small states.

Corollary 26.2 (Large-state theorem). *Uniformly as $q \rightarrow \infty$ and $H(x)/q \rightarrow \infty$,*

$$\log F(t(q), x) - \log \psi(q, x) = q^2/6 + o(q^2). \quad (122)$$

Proof. The identity $p(q) = q/2 + \log R(q)/2$ and monotonicity of R give

$$\log \psi(q, x) - \log \psi(q/2, x) \geq qH/4 - q. \quad (123)$$

Thus

$$\log(S_A/P_\pm) \leq (A - \kappa_\pm)q^2 - qH/4 + q + Cq^2/D,$$

which tends to $-\infty$ uniformly in the specified joint limit. Moreover $qz/6 + \eta = O(q) + O(q^2/D) = o(q^2)$, and the cutoff eventually equals one. Squeeze in (121) and then let $\varepsilon \downarrow 0$. $\square$

27 First-exit transfer and the fixed root

A large-state bound alone does not prove a fixed-root result. In particular ordinary typical growth of the root-conjugated process is not required here. Instead we use the exact resource H and a positive first-exit decomposition.

Proposition 27.1 (Finite boundary transfer). *Let $B \geq 5$ be an integer, $Q \geq 2$, $K \geq 0$, $E_0 \geq 0$. If for every state $H(x) = B$ and all $0 \leq v \leq Q$,*

$$F(t(v), x) \leq e^{Kv^2 + E_0} \psi(v, x), \quad (124)$$

then

$$\log F(t(Q), x_0) \leq KQ^2 + E_0 + CB \log(Q + 2) + CQ + C. \quad (125)$$

If $x_B = (B - 2, 1, B - 2)$ satisfies

$$F(t(Q - 1), x_B) \geq e^{K(Q-1)^2 - E_0} \psi(Q - 1, x_B), \quad (126)$$

then

$$\log F(t(Q), x_0) \geq KQ^2 - E_0 - B \log \frac{B}{Q + 1} - C_K(B + Q + 1). \quad (127)$$

The upper constant is absolute; the lower one may depend on fixed K .

Proof. Let M be the transfer matrix restricted to the finite set $H < B$, and C_B its exit matrix to $H = B$. Splitting paths at their first exit gives, coefficientwise and hence for all $t < T$,

$$F_{\text{in}}(t) = e^{tM} \mathbf{1} + \int_0^t e^{sM} C_B F_{\text{boundary}}(t-s) ds. \quad (128)$$

The boundary function is full F , counting arbitrary subsequent paths, including returns below B . All outgoing row sums inside are at most B . Since $H(x_0) = 3$, a first exit requires at least $h = B - 3$ edges. Thus $(M^j C_B \mathbf{1})(x_0) = 0$ for $j < h - 1$, and

$$(e^{sM} C_B \mathbf{1})(x_0) \leq B \sum_{j \geq h-1} \frac{(Bs)^j}{j!} \leq \frac{B^h s^{h-1} e^{Bs}}{(h-1)!}. \quad (129)$$

No norm depending on the growing matrix dimension enters this estimate. The unexited term is at most e^{BT} .

On $H = B$,

$$\psi(v, x) \leq e^{CB} e^{Bv/2}, \quad e^{v/2} \leq \frac{C(v+1)}{\delta(v)},$$

by (85). At $t = t(Q)$, let $t - s = t(v)$; then $0 \leq v \leq Q$ and $\delta(v) = \delta(Q) + s$ exactly. Keeping s as the integration variable entails no time-change Jacobian. Hypothesis (124) gives

$$F_{\text{boundary}}(t-s) \leq e^{KQ^2 + E_0 + CB} \left[\frac{C(Q+1)}{\delta(Q) + s} \right]^B.$$

Insert this and (129) into (128), bound $e^{Bs} \leq e^{BT}$, and extend the positive integral to infinity. Its value is exact:

$$\int_0^\infty \frac{s^{h-1}}{(\delta+s)^B} ds = \delta^{-3} \frac{\Gamma(h)\Gamma(3)}{\Gamma(B)}. \quad (130)$$

The $\Gamma(h) = (h-1)!$ factor cancels the path denominator. The exit contribution is at most

$$e^{KQ^2 + E_0 + CB} [C(Q+1)]^B \frac{2B^{B-3}}{\Gamma(B)} \delta(Q)^{-3}. \quad (131)$$

Stirling gives $\log(B^{B-3}/\Gamma(B)) = B - \frac{5}{2} \log B + O(1) = O(B)$. The *lower* bound in (85) ensures $\log(1/\delta(Q)) = O(Q)$. Add the unexited term to prove (125). This beta cancellation removes the otherwise prohibitive $BQ/2$ cost.

For the lower bound, repeatedly choose the largest new label. The root reaches x_B by $h = B - 3$ unit-weight edges, so $\partial_t^h F(t, x_0) \geq F(t, x_B)$. Taylor positivity between $t(Q-1)$ and $t(Q)$ gives

$$F(t(Q), x_0) \geq \frac{\Delta_t^h}{h!} F(t(Q-1), x_B), \quad \Delta_t = t(Q) - t(Q-1).$$

The explicit derivative gives $\log \Delta_t \geq -Q/2 + \log(Q+1) - C$, and $\log \psi(Q-1, x_B) \geq (B/2-1)(Q-1)$. With $\log h! \leq h \log h$, the logarithm of this lower bound is at least

$$KQ^2 - E_0 + h \log \frac{Q+1}{h} - Ch - B/2 + 1 + (1/2 - 2K)Q + K.$$

Finally

$$h \log \frac{Q+1}{h} - B \log \frac{Q+1}{B} = B \log B - (B-3) \log(B-3) - 3 \log(Q+1) \geq -3 \log(Q+1),$$

which proves (127) with all constants uniform in B, Q . $\square$

27.1 Boundary uniformity on the entire time interval

Choose $B = B(Q) = \lceil Q^{3/2} \rceil$ and fix ε and all the constants in (121). Every boundary state has $D \geq B/2$. For $0 \leq v \leq q_0$, use the fixed padding $L = \lceil q_0 \rceil$ in (96). It gives

$$\log F(t(v), x) - \log \psi(v, x) \leq C_\varepsilon$$

uniformly in all states, with no factor depending on B .

For $q_0 \leq v \leq Q$, the sharp cutoff is one for sufficiently large Q . Equation (123) gives, uniformly on $H = B$,

$$\log(S_A/P_\pm) \leq (A - \kappa_\pm)v^2 - Bv/4 + v + Cv^2/D \leq -Bv/8$$

for all large Q , because $B/Q \rightarrow \infty$. Also $|vz/6 + \eta| \leq Q/6 + 2CQ^2/B$. Consequently (124) holds throughout $0 \leq v \leq Q$ with $K = \kappa_+$ and $E_\varepsilon(Q) = O_\varepsilon(Q) = o(Q^2)$. At $v = Q - 1$, $x = x_B$, the slow term is exponentially negligible in the lower barrier, so (126) holds with $K = \kappa_-$ and the same harmless error size.

Both losses $B \log(Q + 2)$ and $B |\log(B/(Q + 1))|$ are $o(Q^2)$. Proposition 27.1 therefore gives

$$\frac{1}{6} - \varepsilon \leq \liminf \frac{\log F(t(Q), x_0)}{Q^2} \leq \limsup \frac{\log F(t(Q), x_0)}{Q^2} \leq \frac{1}{6} + \varepsilon.$$

First let $Q \rightarrow \infty$, then $\varepsilon \downarrow 0$. This proves (72). Constants in this construction can deteriorate with ε ; the following separate argument supplies the uniform parameter control needed for a quantitative remainder.

28 A parameter uniform quantitative refinement

The fixed- ε construction proves the leading constant but does not itself provide a rate. A separate uncut patch keeps the dependence on the accuracy parameter uniform and yields a quantitative bound.

Theorem 28.1. *There are absolute finite constants C, K_* such that whenever $K_* \leq K \leq Q - 1$,*

$$\left| \log F(t(Q), x_0) - \frac{Q^2}{6} \right| \leq C \left(\frac{Q^2}{K} + KQ \log(Q + 2) + K^2 \right). \quad (132)$$

In particular

$$\log A'(t(Q)) = \frac{Q^2}{6} + O(Q^{3/2} \sqrt{\log Q}). \quad (133)$$

The constants are independent of K, Q ; no numerical effective value or optimality is asserted.

28.1 Global bounds for the corrector and the small state defect

Let C_r, C_η, K_0 be absolute constants such that (107) holds with C_r and

$$|\eta| \leq C_\eta q^2/D, \quad |d\eta/dq|_{\text{fixed state}} \leq C_\eta q/D.$$

The latter estimates hold for every state and $q \geq 1$, irrespective of $D \geq K_0 q$: they follow from the explicit forcing, the L^1 derivative bounds and the chain rule in Section 25. Only the sharp residual requires the large-size hypothesis.

Fix an absolute $a \geq \max(1, 4C_r)$, take $K_* \geq \max(1, q_*, 2K_0, 12a)$, and for $K \geq K_*$ define

$$\kappa_{\pm} = 1/6 \pm a/K, \quad P_{\pm} = e^{\kappa_{\pm} q^2} \phi, \quad q_0 = K.$$

Then $\kappa_{\pm} \in [1/12, 1/4]$. When $q \geq K$, $H \geq Kq$, $D \geq Kq/2$ and $C_r(1 + q^2/D) \leq 3C_r q/K$, so with $\mathcal{L} = \partial_q - \mathcal{T}/b$,

$$\mathcal{L}P_+ \geq (aq/K)P_+, \quad \mathcal{L}P_- \leq -(aq/K)P_-. \quad (134)$$

Additional absolute lower bounds on K_* can be imposed below.

In the complementary region $H \leq Kq$ we need only a coarse defect bound, with an absolute constant independent of K :

$$\mathcal{L}P_+ \geq -e^{C_d q^2} P_+ \quad (q \geq K \geq K_*). \quad (135)$$

Indeed the individual child ratio of ψ is at most $\sqrt{2}e^{5q/8}$, $b^{-1} \leq Cqe^{-q/2}$, and the composition and corrector changes give

$$\frac{\mathcal{T}P_+}{bP_+} = \frac{\mathcal{T}\phi}{b\phi} \leq CHq \exp(7q/24 + 2C_{\eta}q^2).$$

The exponent $7q/24$ is $5q/8 - q/2 + q/6$. Moreover $|\partial_q \log P_+| \leq C(H + q)$ by the global derivative bound, (91), and $z_q = -Ez(1 - z)/R$. Since $H \leq Kq \leq q^2$, all polynomial factors can be absorbed in $e^{C_d q^2}$ with absolute C_d , proving (135).

28.2 The uncut patch and its initial normalization

The exact characteristic identity gives both inequalities

$$Hq/4 - q \leq \log \frac{\psi(q, x)}{\psi(q/2, x)} \leq Hq/4 + (H/2) \log 2 + q/2. \quad (136)$$

Choose absolute $C_b = C_{\eta} + 2$ and $c_A \geq C_d + C_b + 2$, and set $A = c_A K$, $S = e^{Aq^2} \psi(q/2, x)$. On $H \leq Kq$, $q \geq K$,

$$\begin{aligned} \log(S/P_+) &\geq (A - \kappa_+ - K/4 - C_{\eta})q^2 - (\log 2)Kq/2 - 2q/3 \\ &\geq [(c_A - 1/4)K - C_b]q^2 \geq (C_d + 1)q^2. \end{aligned}$$

Together with the slow margin and (135), this proves $\mathcal{L}(P_+ + S) \geq 0$ in the small-state region. In the large-state region both summands are supersolutions by (134). Thus $P_+ + S$ is a global supersolution *without a cutoff*. Also $P_- \leq P_+ \leq S$ in the small-state region. The same positive-part argument as before makes $(P_- - S)_+$ a global subsolution.

At $q = K$, the global corrector bound implies

$$e^{-C_p K^2} \leq P_{\pm}(K, x)/\psi(K, x) \leq e^{C_p K^2}$$

uniformly over every state. The coarse comparison (98) therefore yields an absolute C_i such that

$$e^{-C_i K^2} P_-(K, x) \leq F(t(K), x) \leq e^{C_i K^2} P_+(K, x).$$

Initial comparison uses the sharp components alone. Although S itself contains $e^{c_A K^3}$ at $q = K$, its large positive size never forces a multiplicative normalization $e^{O(K^3)}$.

Full-state comparison now gives

$$e^{-C_i K^2} (P_- - S)_+ \leq F(t(q), x) \leq e^{C_i K^2} (P_+ + S), \quad q \geq K. \quad (137)$$

For each fixed K and compact clock interval these functions are bounded by a finite multiple of ψ , so the later padded supersolution eliminates stopped exit contributions exactly as before. Its auxiliary finite constants vanish with the exit limit; they do not alter C_i . The positive semigroup identity justifies starting at the positive time $t(K)$.

28.3 Moving boundary estimates with uniform constants

Choose once and for all $\beta \geq 8(c_A + C_\eta + 4)$ and $B = \lceil \beta K Q \rceil$. For $H = B$ we have $D \geq B/2$. For $K \leq v \leq Q$,

$$\begin{aligned} \log(S/P_\pm) &\leq (c_A K - \kappa_\pm)v^2 - Bv/4 + v + 2C_\eta v^2/B \\ &\leq -(\beta/4 - c_A)KQv + (1 + 2C_\eta/(\beta K))v \leq -KQv. \end{aligned}$$

Hence $S/P_\pm < 1/2$ on this whole interval. The sharp-profile logarithm satisfies $|vz/6 + \eta| \leq Q/6 + 2C_\eta Q^2/B \leq C(Q + Q/K)$. It follows from (137) that with $E_0 = C_e(K^2 + Q)$, for an absolute C_e ,

$$F(t(v), x) \leq e^{\kappa_+ v^2 + E_0} \psi(v, x) \quad (H = B, K \leq v \leq Q).$$

For $0 \leq v \leq K$ the same assertion follows from the uniform coarse bound with error $O(K^2)$, since $\kappa_+ \geq 0$. At $v = Q - 1 \geq K$, the lower half of (137) also gives

$$F(t(Q - 1), x_B) \geq e^{\kappa_- (Q-1)^2 - E_0} \psi(Q - 1, x_B).$$

Thus both hypotheses of Proposition 27.1 hold with constants genuinely independent of growing K .

Its lower-transfer constant is uniform for $\kappa_- \in [1/12, 1/4]$, as the explicit proof only adds a term linear in that parameter. Since $B \leq CKQ$ and $K \leq Q - 1$,

$$B \log \frac{B}{Q + 1} \leq CKQ \log(K + 2) \leq CKQ \log(Q + 2).$$

The upper and lower transfer bounds prove (132). Finally $K = \sqrt{Q/\log(Q + 2)}$ is admissible for large Q and balances Q^2/K with $KQ \log(Q + 2)$; its K^2 term is smaller. This proves (133).

The refinement quantifies the root real-axis theorem. It does not give an all-index lower bound at the logarithmic-square scale or imply any coefficientwise regularity.

29 What positivity transfers to coefficients

Because $F(t, x_0) = A'(t)$ is nondecreasing,

$$A(t(Q)) \leq 1 + t(Q)F(t(Q), x_0), \quad A(t(Q)) \geq [t(Q) - t(Q - 1)]F(t(Q - 1), x_0).$$

The interval length has logarithm $-Q/2 + O(\log Q)$, so $\log A(t(Q)) \sim Q^2/6$. Equation (86) proves (73). It also proves divergence at T . Finiteness below T and positivity then establish that the radius is exactly T .

Set $C(z) = A(Tz) = \sum c_n z^n$ and $\alpha = 2/3$. For $s = -\log(t(Q)/T)$, we have $s \sim \delta(Q)/T$ and $\log(1/s) = Q/2 - \log Q + O(1)$. The increasing characteristic covers all sufficiently small positive s , not merely a sequence. Thus

$$\log C(e^{-s}) \sim \alpha(\log(1/s))^2 \quad (s \downarrow 0). \tag{138}$$

Proposition 29.1 (Positive coefficient transfer). *For any nonnegative sequence satisfying (138),*

$$\log S_N \sim \alpha(\log N)^2, \quad \limsup \frac{\log c_n}{(\log n)^2} = \alpha.$$

If in addition c_n is eventually nondecreasing, the limsup is a full limit.

Proof. Positivity gives $S_N \leq eC(e^{-1/N})$, proving the upper cumulative bound. For the lower bound take $s = (\log N)^3/N$. The weighted tail is at most

$$\sum_{k>N} c_k e^{-sk} \leq e^{-sN/2} C(e^{-s/2}) = o(1),$$

since its logarithm is bounded above by $-\frac{1}{2}(\log N)^3 + O((\log N)^2)$. The weighted prefix is at most S_N , so $S_N \geq C(e^{-s}) - o(1)$. Since $\log(1/s) = \log N - 3\log \log N$ and $C(e^{-s}) \rightarrow \infty$, this proves the matching lower cumulative asymptotic.

Also $c_n \leq eC(e^{-1/n})$, giving the limsup upper bound. If the limsup were less than α , choose $\max(0, \limsup) < \beta < \alpha$. For all large n , $c_n \leq e^{\beta(\log n)^2}$, whence $S_N \leq O(1) + (N+1)e^{\beta(\log N)^2}$, a contradiction. All the coefficients here are positive; the general proof may set $\log 0 = -\infty$. Finally eventual nondecreasingness gives $S_N \leq O(1) + (N+1)c_N$, and $\log(N+1) = o((\log N)^2)$ supplies the individual lower bound. $\square$

The quantitative refinement propagates to the same positive-coefficient statements. If $L = \log(1/s)$, then (86) gives $Q = 2L + O(\log L)$, so the integration bounds and Theorem 28.1 give

$$\log C(e^{-s}) = \frac{2}{3}L^2 + O(L^{3/2}\sqrt{\log L}). \quad (139)$$

In the preceding proof, the lower choice $s = (\log N)^3/N$ changes L from $\log N$ by $O(\log \log N)$. The induced $O(\log N \log \log N)$ error is smaller than the remainder in (139). Hence, as $N \rightarrow \infty$,

$$\log S_N = \frac{2}{3}(\log N)^2 + O((\log N)^{3/2}\sqrt{\log \log N}). \quad (140)$$

This is still a cumulative statement; it does not provide a lower bound for an arbitrary individual c_n .

This completes Theorem 21.1. The sufficient extra regularity condition has a concrete form:

$$c_{n+1} \geq c_n \iff \frac{a_{n+1}}{a_n} \geq (n+1)\mu. \quad (141)$$

Neither this eventual inequality nor eventual log-concavity of c_n has been proved. The latter would be sufficient, once the normalized root limit is known: decreasing eventual ratios cannot ever fall below one, since then all later ratios are bounded above by a fixed number below one, contradicting $c_n^{1/n} \rightarrow 1$. This implication is not a proof of its premise.

29.1 A weaker one sided regularity criterion

Eventual monotonicity is stronger than necessary. The following abstract criterion identifies a more permissive sufficient target.

Proposition 29.2. *Let $c_n > 0$, $S_n = \sum_{j \leq n} c_j$, and $\log S_n \sim \alpha(\log n)^2$ with $\alpha > 0$. If for some fixed $C \geq 0$ and all sufficiently large n ,*

$$\log(c_{n+1}/c_n) \geq -C \log n/n, \quad (142)$$

then $\log c_n \sim \alpha(\log n)^2$.

Proof. The upper bound follows from $c_n \leq S_n$. If the lower conclusion fails, there are $\delta > 0$ and infinitely many n with $\log c_n \leq (\alpha - \delta)(\log n)^2$. Choose fixed $0 < \beta < 1$ so close to one that $\frac{C}{2}(1 - \beta^2) < \delta/2$. Summing (142) backwards, uniformly for $n^\beta \leq k \leq n$,

$$\begin{aligned} \log c_k &\leq \log c_n + C \sum_{j=k}^{n-1} \frac{\log j}{j} \\ &\leq [\alpha - \delta + \frac{C}{2}(1 - \beta^2) + o(1)](\log n)^2 \leq [\alpha - \delta/2 + o(1)](\log n)^2. \end{aligned}$$

Integral comparison for $(\log x)/x$ makes the error uniform on this whole interval. Therefore

$$S_n \leq S_{\lfloor n^\beta \rfloor} + (n+1) \exp\{[\alpha - \delta/2 + o(1)](\log n)^2\}.$$

The first summand has logarithmic coefficient $\alpha\beta^2 < \alpha$; the second has coefficient at most $\alpha - \delta/2 < \alpha$. Taking the logarithm of their sum adds only $O(1)$, and its factor $n+1$ adds only $O(\log n)$, contradicting the cumulative asymptotic. $\square$

For A202058 a sufficient model-specific bound is consequently

$$\frac{a_{n+1}}{a_n} \geq \mu(n+1) - L \log n \quad \text{eventually, for some fixed } L \geq 0. \quad (143)$$

Divide by $\mu(n+1)$ and use $\log(1-x) \geq -2x$ for sufficiently small $x \geq 0$ to obtain (142). This bound is unproved for the model. A ratio limit alone, or a finite numerical fit, would not supply its quantitative one-sided premise.

[write] 2 October 2026. Still unproved after Part IV. Its raw-ratio error $O(n^{3/4}(\log n)^{3/4})$ is far from the $O(\log n)$ that (143) requires; see Part IV, Section 40.

30 All index coarse control and safe inversions

The new cumulative theorem cannot simply be inverted as though it were an individual coefficient equivalent. We include the older all-index control, with its proof, to make the distinction explicit.

30.1 An all index lower enclosure

[write] 2 October 2026. Second route. The enclosure (146) below is Theorem 9.1 of Part II, and the inverse (150) is its Corollary 9.2. The root limit derived at the end of this subsection is Theorem 1.1 of Part I. The argument is the growing-seed Chernoff extraction of Part II, Section 14, built on this Part's own comparison (98). It is printed in full because Part IV cites (144), (145) and (146). Part IV improves the lower side of the enclosure to $-O(\sqrt{n}(\log n)^{3/2})$.

For $x_N = (N, 1, N)$, (98) gives

$$\log F(t(q), x_N) = Np(q) + O((q+1)^2), \quad (144)$$

uniformly in $N \geq 1$. Put $d(q) = (\log t(q))'$ and $M(q) = p'(q)/d(q) = t(q)p_t(t(q))$. Explicit differentiation gives

$$M(q) \sim \frac{Te^{q/2}}{\sqrt{2q}}, \quad d(q) = \frac{qe^{-q/2}}{\sqrt{2T}}(1 + o(1)),$$

and M is eventually strictly increasing. For $b_j = (\mathcal{T}^j \mathbf{1})(x_N)$, use the tilted distribution

$$\mathbb{P}(J = j) = \frac{b_j t(q)^j}{j! F(t(q), x_N)}.$$

Then for all large q , $0 < \epsilon < 1/2$,

$$\mathbb{P}(|J - NM(q)| > \epsilon NM(q)) \leq 2 \exp\{-cN\epsilon^2 + C(q+1)^2\}. \quad (145)$$

For completeness, with $H_q(v) = p(q+v) - p(q) - M(q)[\log t(q+v) - \log t(q)]$, we have $H_q(0) = H'_q(0) = 0$ and, uniformly for $|v| \leq 1$,

$$H''_q(v) \rightarrow \frac{1}{4}e^{-v/2}, \quad M(q)d(q+v) \rightarrow \frac{1}{2}e^{-v/2}.$$

Thus $|H_q(v)| \leq v^2/2$ and $M(q)|\log t(q \pm \eta) - \log t(q)| \geq \eta/4$ for large q and $0 \leq \eta \leq 1$. Comparing the tilted series at $q \pm \eta$ using (144), with $\eta = \epsilon/8$, bounds each tail exponent by $-3N\epsilon^2/128 + C(q+1)^2$, proving (145).

Take

$$N = \lceil Kn^{2/3}(\log n)^{2/3} \rceil, \quad \epsilon = 2N/n, \quad NM(q) = (1 - 4\epsilon)n,$$

with fixed sufficiently large K . Then $q = \frac{2}{3} \log n + \frac{2}{3} \log \log n + O(1)$ and $N\epsilon^2 = 4N^3/n^2 \geq 4K^3(\log n)^2$. The central window in (145) has mass at least $1/2$. It contains an index j with

$$b_j \frac{t(q)^j}{j!} \geq \frac{F(t(q), x_N)}{4\epsilon NM(q) + 6}, \quad j \geq n - 6\epsilon n, \quad N + j \leq n.$$

The root reaches x_N by $N - 1$ new-maximal steps, hence $a_{N+j} \geq b_j$. Any ascent sequence extends injectively by appending one plus its current number of ascents. This value is new: inductively the maximum previously used value is at most the current ascent count. Thus $a_n \geq a_{N+j}$, and Stirling or the factorial product bound gives

$$-C_1 n^{2/3}(\log n)^{5/3} \leq \log c_n \leq C_2(\log(n+2))^2. \quad (146)$$

For the lower bound the factorial-extension loss is $O(N \log n)$, while the quadratic error and the pigeonhole loss are smaller; eventually $1 < t(q) < T$, so $-j \log t(q) \geq -n \log T$. The upper bound follows already from the main theorem and coefficient positivity (or from (98)). In particular $c_n^{1/n} \rightarrow 1$, proving the full factorial-normalized root limit. The lower exponent in (146) reflects the broad-window method and is not a proposed true correction scale.

30.2 Three justified inverse statements

For the cumulative sequence define $N_S(y) = \min\{N : S_N \geq e^y\}$. Since S_N is increasing and diverges, (140) and the adjacent-integer squeeze give

$$\log N_S(y) = \sqrt{3y/2} + O(y^{1/4}\sqrt{\log y}). \quad (147)$$

For the strictly increasing real-axis function, let t_y solve $A(t_y) = e^y$. Then (73) gives

$$\log \frac{1}{T - t_y} \sim \sqrt{3y/2}, \quad T - t_y = \exp\{-\sqrt{3y/2}(1 + o(1))\}. \quad (148)$$

The same quantitative argument gives $\log(1/(T - t_y)) = \sqrt{3y/2} + O(y^{1/4}\sqrt{\log y})$. To see either inverse error, the leading scale first implies its logarithmic variable is of order $\sqrt{y}$; divide the error $O(y^{3/4}\sqrt{\log y})$ by the leading quadratic derivative of order $\sqrt{y}$. Adjacent integer values in the cumulative inverse have a vanishing logarithmic gap and obey the same bound. These are bounds on logarithmic variables, not equivalents after exponentiation or exact integer rounding formulas.

For the original integer counts put $N_a(y) = \min\{n \geq 1 : \log a_n \geq y\}$ and

$$x = \frac{y}{W(y/(eT))}, \quad x \log \frac{x}{eT} = y, \quad (149)$$

where W is the positive real Lambert function. Stirling and (146) give

$$x - O(\log x) \leq N_a(y) \leq x + O(x^{2/3}(\log x)^{2/3}). \quad (150)$$

Indeed the main smooth term is $f(x) = x \log(x/(eT))$ with derivative $f'(x) = \log(x/T)$. The upper coefficient error $O(\log^2 x)$ divided by $f'(x)$ gives the left displacement $O(\log x)$; the lower error gives the right displacement in (150). Taylor's theorem on these $o(x)$ windows and integer rounding complete the monotone inverse comparison. There is no justified logarithmic-square correction to N_a from the limsup theorem alone.

[write] **2 October 2026.** Part IV, (155), improves the right displacement in (150) to $O(\sqrt{x \log x})$. The left displacement $O(\log x)$ is unchanged.

31 Source scope and mathematical verification

The enumeration recurrence is attributed to [1]. Its Section 5 uses coefficients through roughly order 395 and discusses a numerical ansatz with factorial, exponential, possible stretched-exponential and power factors. That discussion does not prove a full equivalent; the shorter introductory expression must not be read as such a theorem. The rigorous elementary lower bound there is $a_{2n} \geq n!$, which by itself does not determine the exponential-generating-function radius.

A scoped primary-source check on 2 October 2026 covered A202058, A294220, the single-000 restriction, bounded multiplicity, and the factorial growth constant. No retrieved external source established the logarithmic-square theorem proved here. Adjacent work on simultaneous avoidance of several length-three patterns [4], length-four patterns [5], or general Fishburn matrices [6] does not supply the missing result for this exact class. In particular a theorem about inversion sequences or modified ascent sequences cannot be transferred without a multiplicity-preserving correspondence.

A read-only indexed keyword search of the default branch of the ProveIt repository [11] found no A202058 result. The latest returned indexed links used the commit

4b874cea0012c51a6841ad9c58f6e20fa57c71da

The related search “logarithmic ascent” returned unrelated material. This records the index observed, not a certified live branch tip. These are scoped negative findings, not an exhaustive semantic repository audit or a worldwide-priority certificate. No external repository or publication was modified.

The mathematical verification had distinct stages. The original positive-operator and padded-comparison foundations were checked separately. The new uniform rank residual, interface and seam estimates, nonlinear moment bound, and global patch received an independent analytical audit. A separate analytical audit checked the first-exit transfer and positive-coefficient reductions. A separate independent analytical audit checked the parameter-uniform quantitative refinement and the weaker one-sided ratio criterion. The exact versions of the proof inputs were verified against SHA-256 records before transcription. The audit corrected a two-sided-tail exposition point and restricted an optional return-process argument to its actual stopped domain; the optional argument is not used in this article. The delivered reproducibility archive records source fingerprints and the integrated article review status without redistributing third-party papers or working correspondence.

These are mathematical review and reproducibility checks, not a claim of formal proof-assistant verification. The finite tests described next corroborate exact identities and implementations; they do not establish uniform estimates or asymptotic limits.

32 Reproduction and further questions

The accompanying archive includes this editable TeX source, the compiled PDF, portable shell entry points, Python verification scripts, expected results, and checksums. The core rational test uses only the Python standard library. It checks 73,984 exact rational state transitions for $m \leq 16$ and $R \in \{1, 3/2, 7/4, 2\}$, including the resource, child coordinates and one-sided rank shift. A direct word enumerator compares the compacted recurrence with literal ascent words at small lengths. A numerical residual test uses NumPy and 64-point Gaussian quadrature with finite differences at forty states spanning seam, type interface, and extreme compositions. Its tolerance is a finite sanity check, not a certified value of the constant in Theorem 25.1.

After extracting with Python's ZIP module, run the scripts explicitly with `bash`; executable permission preservation is unnecessary. The standard-library checks run without scientific Python. The full numerical test requires NumPy, and PDF rebuilding requires a standard pdfLaTeX installation with the packages named in the source. No network access, credentials, compiler extension, or third-party paper is needed. The replay procedure checks archive member hashes before running tests and writes regenerated outputs separately from the frozen results.

[write] 2 October 2026. In this repository the scripts are shipped as `code/07-lognormal-*` and the frozen results as `data/07-lognormal-results-*`. The source-audit records are shipped as `data/07-lognormal-source-audit-*` and `07-lognormal-source-audit-verification-summary.md`. The compiled PDF and the SHA256SUMS manifest are not shipped. The shipped replay scripts still use the delivery names; the report README gives working commands.

The principal next mathematical question is the weaker one-sided ratio bound (143), eventual normalized monotonicity (141), or another theorem strong enough to transfer cumulative growth to every coefficient. A suitable local coefficient estimate could serve instead. Beyond that lie sharper quantitative residual propagation, a controlled next-order real-axis expansion, complex-variable transfer, ratios, and amplitudes. The exact scalar mean $m(q) = q/3 + 1/2 - 1/q + O(qe^{-q})$ is suggestive, but integrating its smaller terms and declaring them to be coefficients of a transseries would ignore the global and root-transfer errors. None of those stronger claims follows from the present argument.

The completed part is substantial and sharply delimited: the real-axis singularity has logarithmic-square constant $2/3$, the cumulative factorial-normalized coefficients have the same constant, and the individual upper envelope has exactly that constant. The remaining individual lower-bound problem is a separate, explicit open step.

Part IV

Ratio asymptotics for ascent sequences avoiding 000

Batch-77 manuscript 08, archive `a202058-ratio-report.zip` (delivered as `a202058-ratio.tex`, 14 pages), dated 2 October 2026. Delivered author line: “A sequel with complete proofs and reproducibility materials”. The delivered “predecessor” is Part III. Delivered Section k is Section $k + 32$ here.

Abstract of this Part (as delivered). Let a_n count ascent sequences avoiding 000, let $T = 3\pi^2/8$, and put $c_n = a_n T^n / n!$. We prove

$$\frac{a_{n+1}}{(n+1)a_n} = \frac{8}{3\pi^2} + O\left(n^{-1/4}(\log n)^{3/4}\right).$$

The underlying local fact is exact: the raw ratios $R_n = a_{n+1}/a_n$ satisfy $R_{n+1} \geq R_n - 1$. An elementary excursion argument already turns the earlier factorial-normalized root limit into the ratio limit. A second exact identity for exponentially tilted transfer paths controls the difference between path length and terminal degree. Keeping all continuations from sufficiently large terminal degrees improves the all-index lower bound to

$$-C\sqrt{n}(\log n)^{3/2} \leq \log c_n \leq C'(\log n)^2.$$

This supplies the stated ratio rate and an upper inverse error $O(\sqrt{x \log x})$ around the Lambert W approximation. The analytic inputs inherited from the earlier report are explicitly identified and the coarse characteristic and concentration estimates are derived here. The full pointwise law $\log c_n \sim \frac{2}{3}(\log n)^2$ remains open. An exact reachable-state counterexample rules out one proposed childwise invariant; extensive finite log-concavity checks are recorded only as evidence, never as a proof of an asymptotic statement.

33 Results and the dependency boundary

An ascent sequence is a word $w_1 \cdots w_n$ of nonnegative integers with $w_1 = 0$ and

$$0 \leq w_i \leq 1 + \text{asc}(w_1 \cdots w_{i-1}), \quad \text{asc}(w) = \#\{i : w_i < w_{i+1}\}.$$

Avoiding 000 means that each value occurs at most twice, with no requirement that equal occurrences be consecutive. Let $a_0 = 1$ and let a_n count these words. The sequence is OEIS A202058, also the cap-two column of A294220 [2, 3]. Its initial values are

$$1, 1, 2, 4, 10, 27, 83, 277, 1015, 4007, 17047.$$

Throughout, logarithms are natural. The normalization is

$$T = \frac{3\pi^2}{8}, \quad \mu = \frac{1}{T} = \frac{8}{3\pi^2}, \quad c_n = \frac{a_n T^n}{n!}, \quad R_n = \frac{a_{n+1}}{a_n}. \quad (151)$$

In particular the factor 3 in T is essential.

Theorem 33.1. *There are constants $C_1, C_2 > 0$ such that, for all sufficiently large n ,*

$$-C_1\sqrt{n}(\log n)^{3/2} \leq \log c_n \leq C_2(\log n)^2, \quad (152)$$

$$\frac{a_{n+1}}{(n+1)a_n} = \mu + O\left(n^{-1/4}(\log n)^{3/4}\right). \quad (153)$$

Define $N_a(y) = \min\{n \geq 1 : \log a_n \geq y\}$, and let

$$x = \frac{y}{W(y/(eT))}, \quad x \log \frac{x}{eT} = y, \quad (154)$$

where W denotes the positive real Lambert function. As $y \rightarrow \infty$,

$$x - O(\log x) \leq N_a(y) \leq x + O(\sqrt{x \log x}). \quad (155)$$

No effective numerical constants or exact integer rounding formula are asserted.

[write] 2 October 2026. This theorem improves Part II's Theorem 9.1 and Corollary 9.2 (and their second proof in Part III). It also supersedes the statements that the ratio limit is unresolved or unproved: Part I, abstract and Section 8; Part II, abstract; Part III, abstract and the scope paragraph after Theorem 21.1.

The predecessor (Part III of this report) is the research report *Logarithmic square growth for ascent sequences avoiding 000*, dated 2 October 2026. It is included unchanged, as PDF and editable TeX, in the accompanying archive. It is a preceding investigation, not an external published theorem. It established, with $A(t) = \sum a_n t^n / n!$ and $S_n = \sum_{j \leq n} c_j$,

$$\log A(t) \sim \frac{2}{3} \log^2 \frac{1}{T-t}, \quad t \uparrow T, \quad (156)$$

$$\log S_n = \frac{2}{3} (\log n)^2 + O((\log n)^{3/2} \sqrt{\log \log n}), \quad (157)$$

$$\limsup_{n \rightarrow \infty} \frac{\log c_n}{(\log n)^2} = \frac{2}{3}, \quad (158)$$

$$-C n^{2/3} (\log n)^{5/3} \leq \log c_n \leq C' (\log n)^2. \quad (159)$$

For these results, see Theorem 21.1 of the predecessor, its cumulative bound in Section 29, equation (140), and its all-index enclosure in Section 30.1, equation (146). The underlying state-uniform comparison is in Section 23.3, equation (98); its large-seed specialization and Chernoff estimate are in Section 30.1, equations (144) and (145).

[write] 2 October 2026. The predecessor is printed as Part III. Its unchanged PDF and TeX, which this manuscript carried under `prior/`, are byte-identical to manuscript 07 and are not shipped. The delivered bibliography entry for the predecessor adds: “The earlier statement that a ratio limit was unproved is superseded by Theorem 33.1 of this sequel.” The enclosure (159) was first proved in Part II, Theorem 9.1; Part III re-proves it.

This sequel proves the ratio limit left open there, and improves the lower enclosure and its inverse consequence. Section 36 first derives the ratio limit from the old enclosure and an exact one-step inequality. Sections 37 and 38 then give the improved enclosure and rate. To make the dependency transparent, Section 37 reproduces the characteristic estimates needed for that proof, including the infinite-state comparison and concentration argument. The more delicate earlier real-axis and cumulative theorems are imported only for context and for the discussion of the remaining pointwise problem.

The present conclusion is strictly weaker than

$$\log c_n \sim \frac{2}{3} (\log n)^2. \quad (160)$$

Neither (160), eventual normalized monotonicity or log-concavity, a full coefficient equivalent, nor a limiting amplitude is proved here. The recurrence is due to Conway, Conway, Elvey Price and Guttmann [1, Sections 2.2 and 2.6]. Their Section 5 gives numerical evidence for the constant $8/(3\pi^2)$; it is not a proof of the ratio theorem.

34 The exact transfer model

Twice-used labels are removed from the active alphabet, and the once-used labels are compacted to an initial segment. In the published recurrence, with $G_0(a, \ell, s) = 1$,

$$G_j(a, \ell, s) = \sum_{i=0}^{s-1} G_{j-1}(a + \mathbf{1}_{i>\ell} - 1, i - 1, s - 1) + \sum_{i=s}^{a+1} G_{j-1}(a + \mathbf{1}_{i>\ell}, i, s + 1), \quad a_n = G_{n-1}(0, 0, 1) \quad (n \geq 1).$$

The adjusted a and ℓ may equal -1 . Compaction preserves suffix counts and transition multiplicities. Its elementary bijection exchanges the remaining capacities of adjacent labels $i, i + 1$ below or at the last value ℓ , with $i < \ell$: reverse each maximal suffix block in the two labels and interchange the labels within it. Multiplicities are exchanged while ascent counts and the outside comparisons are preserved. Both labels are already permitted, so all subsequent ascent-sequence bounds remain valid. Repeated exchanges place once-used labels first, and exhausted labels can be deleted. See [1] for the original compacted recurrence.

Set $m = a + 2$, $k = \ell + 1$, and $u = m - s$. The invariant state space is

$$\mathcal{S} = \{(s, u, k) : s \geq 0, u \geq 1, 0 \leq k < s + u\}.$$

Write $m(x) = s + u$. For each integer $0 \leq i < m$, the child is

$$x_i = \begin{cases} (s - 1, u, i), & i < s, i < k, \\ (s - 1, u + 1, i), & i < s, i \geq k, \\ (s + 1, u, i + 1), & i \geq s, i \geq k, \\ (s + 1, u - 1, i + 1), & i \geq s, i < k. \end{cases} \quad (161)$$

The first two cases repeat a label; the last two use a new label. An ascent corresponds to $i \geq k$. The potentially invalid last case with $u = 1$ would require $s \leq i < k \leq s$, and therefore never occurs. There are no repeat transitions when $s = 0$.

Define

$$(\mathcal{T}f)(x) = \sum_{i=0}^{m(x)-1} f(x_i), \quad F(t, x) = \sum_{j \geq 0} (\mathcal{T}^j \mathbf{1})(x) \frac{t^j}{j!}. \quad (162)$$

The formal pre-empty state $x_* = (0, 1, 0)$ has the unique child $(1, 1, 1)$. Consequently

$$a_n = (\mathcal{T}^n \mathbf{1})(x_*), \quad n \geq 0. \quad (163)$$

Each transition changes m by at most one, and changes the integer resource $H = s + 2u$ by exactly 1 or -1 . The state

$$x_N = (N, 1, N)$$

is reached from x_* by N successive largest-new-label steps. We shall use the continuation series

$$G_N(t) = F(t, x_N) = \sum_{j \geq 0} b_j \frac{t^j}{j!}, \quad b_j = (\mathcal{T}^j \mathbf{1})(x_N). \quad (164)$$

Paths, rather than distinct terminal states, are always counted. A merger of states therefore does not identify different prefixes.

35 Two exact moment identities

Put $d(x) = u - k$. The number of children with degree $m + 1$ minus the number with degree $m - 1$ is

$$[m - \max(s, k)] - \min(s, k) = m - s - k = u - k.$$

There are m children in total. Hence

$$\mathcal{T}m = m^2 + d, \quad d \geq 2 - m, \quad |d| \leq m. \quad (165)$$

35.1 Fixed length paths and the slow decrease of ratios

Choose one of the a_n transfer paths of length n from x_* uniformly, with multiplicity. Write $\mathbb{E}_n$ and Var_n for the terminal-state expectation and variance. Since every terminal state has m children,

$$R_n = \mathbb{E}_n m, \quad R_{n+1} = \frac{\mathbb{E}_n(\mathcal{T}m)}{\mathbb{E}_n m}.$$

Substitution of (165) gives the exact identity

$$R_{n+1} = R_n + \frac{\text{Var}_n(m) + \mathbb{E}_n d}{R_n}. \quad (166)$$

In particular, for every $n \geq 0$,

$$1 \leq R_n \leq n + 1, \quad R_{n+1} \geq R_n - 1 + \frac{2}{R_n} \geq R_n - 1. \quad (167)$$

The degree starts at 1 and can increase by at most one per step, which proves the upper bound; every state has at least one child, which proves the lower bound. This exact local regularity property is sufficient to turn a sublinear bound on $\log c_n$ into a ratio limit.

[write] **2 October 2026.** The identity (166) first appears as (66) in Part II, which writes M_n for R_n . The variance form (205) of Section 40 is Part II's (67). New here are the consequence $R_{n+1} \geq R_n - 1$ in (167) and its use in Lemma 36.1.

35.2 Tilted paths and terminal degree

Fix any starting state x and $0 < t < T$. Assign each individual length- j path probability $t^j/(j!F(t, x))$. Let J be its length and let m, d refer to its terminal state. Finiteness of F below T is proved in Section 37. All differentiations below are justified by positivity and finiteness at any slightly later time $t' < T$: $m \leq m(x) + J$, $|d| \leq m$, and all moments of J are finite in the interior of the convergence disk.

The transfer identities and termwise differentiation give

$$\begin{aligned} \mathbb{E}m &= F'/F, & \mathbb{E}J &= tF'/F, \\ \mathbb{E}(m^2 + d) &= F''/F, & \mathbb{E}(Jm) &= tF''/F, \\ & & \mathbb{E}J^2 &= t^2F''/F + tF'/F. \end{aligned} \quad (168)$$

Expanding the square cancels the second derivatives exactly:

$$\mathbb{E}(J - tm) = 0, \quad \mathbb{E}[(J - tm)^2] = \mathbb{E}J - t^2\mathbb{E}d \leq (1 + t)\mathbb{E}J. \quad (169)$$

The last inequality uses $d \geq -m$ and $t\mathbb{E}m = \mathbb{E}J$. No independence, log-concavity assumption, Gaussian approximation or local limit theorem is involved.

36 An elementary slow decrease lemma

Lemma 36.1. *Let $0 < \mu < 1$, $a_n > 0$, $c_n = a_n \mu^{-n}/n!$, and $R_n = a_{n+1}/a_n$. Suppose $R_n \leq n+1$ and $R_{n+1} \geq R_n - 1$. If $|\log c_n| \leq E(n)$ eventually, where $E(n) \geq 1$, $E(n) = o(n)$, and $E(k) = O(E(n))$ uniformly for $n/2 \leq k \leq 2n$, then*

$$R_n = \mu(n+1) + O\left(\sqrt{nE(n)} + 1\right). \quad (170)$$

For the qualitative conclusion $R_n/[\mu(n+1)] \rightarrow 1$, it is enough that $\log c_n = o(n)$.

Proof. Put $D_n = R_n - \mu(n+1)$ and $L = 1 + \mu$. Then

$$D_{n+1} \geq D_n - L, \quad \log c_{k+1} - \log c_k = \log \left(1 + \frac{D_k}{\mu(k+1)}\right). \quad (171)$$

First suppose $D_n = D > 0$, and set $h = \lfloor D/(2L) \rfloor$. If D is bounded there is nothing to prove. Otherwise $h \geq cD$, and, for $0 \leq j < h$,

$$D_{n+j} \geq D - Lj \geq D/2.$$

Because $D \leq (1 - \mu)(n+1)$, all these indices lie in $[n/2, 2n]$ for large n . Also $D_{n+j}/[\mu(n+j+1)]$ belongs to a fixed bounded nonnegative interval, since $R_{n+j} \leq n+j+1$. On that interval $\log(1+v) \geq c_\mu v$. Therefore

$$\log c_{n+h} - \log c_n \geq c \frac{hD}{n} \geq c' \frac{D^2}{n}. \quad (172)$$

The left side is $O(E(n))$, giving the required bound.

If $D_n = -D < 0$, set the same $h = \lfloor D/(2L) \rfloor$. Telescoping the one-sided increment bound backwards gives

$$D_{n-j} \leq -D + Lj \leq -D/2, \quad 1 \leq j \leq h.$$

Positivity of R_n implies $D < \mu(n+1)$, so $n-h \geq n/2$ for large n . For these indices the argument of the logarithm in (171) is positive. Using $\log(1-v) \leq -v$ for $0 \leq v < 1$,

$$\log c_{n-h} - \log c_n = - \sum_{k=n-h}^{n-1} \log \left(1 + \frac{D_k}{\mu(k+1)}\right) \geq c' \frac{D^2}{n}. \quad (173)$$

Again the left side is $O(E(n))$. Bounded D and the floor cases contribute only the additional $O(1)$ in (170).

For the qualitative assertion, any subsequence with $|D_n| \geq \eta n$ for fixed $\eta > 0$ would, by the same forward or backward excursion, force a difference of order n between logarithms at comparable indices. This contradicts $\log c_k = o(k)$. $\square$

Applying the lemma to (159) and (167) already proves

$$\frac{a_{n+1}}{(n+1)a_n} = \mu + O\left(n^{-1/6}(\log n)^{5/6}\right). \quad (174)$$

Thus the ratio limit itself does not depend on the new lower enclosure. The remainder of the proof improves its quantitative rate.

37 Characteristic estimates and concentration

This section derives the analytic input used by the new extension argument. These estimates reproduce the coarse foundations and the large-seed specialization in Sections 23.3 and 30.1 of Part III; they are not claimed as new in this sequel.

37.1 The clock and its exact endpoint

For $q \geq 0$, set

$$E = e^{-q}, \quad \mathcal{R} = 2 - E, \quad p = \frac{q + \log \mathcal{R}}{2}, \quad b = \frac{e^p(1 - E)}{q}, \quad (175)$$

with $b(0) = 1$, and define

$$t(q) = \int_0^q \frac{ve^{-v/2}}{\sqrt{2 - e^{-v}}(1 - e^{-v})} dv. \quad (176)$$

The integrand at zero is its continuous value 1. Direct differentiation gives $t' = 1/b$ and $p' = 1/\mathcal{R}$. The substitution $y = \sqrt{2e^q - 1}$ yields

$$t(\infty) = \int_1^\infty \frac{2 \log((y^2 + 1)/2)}{y^2 - 1} dy.$$

Integration by parts followed by $v = (y - 1)/(y + 1)$ gives

$$\begin{aligned} t(\infty) &= -2 \int_0^1 \log v \frac{1 + v}{(1 - v)(1 + v^2)} dv \\ &= -2 \int_0^1 \log v \left(\frac{1}{1 - v} + \frac{v}{1 + v^2} \right) dv = \frac{\pi^2}{3} + \frac{\pi^2}{24} = \frac{3\pi^2}{8} = T. \end{aligned}$$

Boundary terms vanish; integrating the geometric and alternating geometric series gives the two displayed values. Moreover

$$T - t(q) \asymp (q + 1)e^{-q/2}, \quad \log \frac{1}{T - t(q)} = \frac{q}{2} - \log(q + 1) + O(1), \quad (177)$$

since for $q \geq 1$ the derivative in (176) is between fixed positive multiples of $qe^{-q/2}$. Compact intervals are absorbed into the constants.

37.2 A state uniform comparison

For $x = (s, u, k)$, introduce

$$D = s + \mathcal{R}u, \quad h_0(y) = \min(y, s) + \mathcal{R} \max(y - s, 0), \quad r(y) = \frac{h_0(y)}{D},$$

and write $r = r(k)$, $z = s/D$. Define the characteristic profile

$$\psi(q, x) = \exp\{ps + qu - qr\}. \quad (178)$$

Here $H/2 \leq D \leq H$, $0 \leq r, z < 1$, and $\psi(0, x) = 1$. For real $0 \leq y < m$ set

$$w(y) = \begin{cases} e^{-p+q\mathbf{1}_{y \geq k}}, & y < s, \\ e^{p-q\mathbf{1}_{y < k}}, & y \geq s, \end{cases} \quad g(y) = w(y)e^{-q(r(y)-r)}.$$

Direct integration on the intervals separated by s, k gives

$$\int_0^m g(y) dy = \lambda := Db/\mathcal{R}. \quad (179)$$

For example, at $k = 0$ the integral is

$$\frac{D}{q} \left[e^{q-p}(1 - e^{-qs/D}) + \frac{e^p}{\mathcal{R}}(e^{-qs/D} - e^{-q}) \right] = \lambda,$$

using $e^{q-p} = e^p/\mathcal{R}$; differentiating in the cut point k verifies the same identity for every k . Each smooth piece of g is decreasing, breakpoints are integers, and $g \leq \sqrt{2}e^{q/2}$. Thus, with $S = \sum_i g(i)$,

$$0 \leq S - \lambda \leq 3\sqrt{2}e^{q/2}. \quad (180)$$

In the order of the four transitions (161), the child ranks computed at the same q are

$$r_i = \frac{i}{D-1}, \quad \frac{i}{D+\mathcal{R}-1}, \quad \frac{h_0(i)+1}{D+1}, \quad \frac{h_0(i)+1}{D+1-\mathcal{R}}.$$

All denominators in transitions that actually occur are positive. Subtraction gives $|r_i - r(i)| \leq 4/D$. Except for a repeat ascent, $r_i \geq r(i)$; in that exceptional case,

$$0 \leq r(i) - r_i = \frac{i(\mathcal{R}-1)}{D(D+\mathcal{R}-1)} \leq \frac{1}{D}.$$

The exact transition ratio is $\psi(q, x_i)/\psi(q, x) = g(i)e^{-q(r_i - r(i))}$. For the derivative at a fixed integer state,

$$\begin{aligned} \partial_q \log \psi &= D/\mathcal{R} - B, & B &= r + qE\partial_{\mathcal{R}}r, & |B| &\leq 1, \\ \partial_{\mathcal{R}}r &= \begin{cases} -r(1-z)/\mathcal{R}, & r \leq z, \\ -z(1-r)/\mathcal{R}, & r \geq z. \end{cases} \end{aligned}$$

The formulas agree at the interface, and $-1/(4\mathcal{R}) \leq \partial_{\mathcal{R}}r \leq 0$. Combining these exact expressions with (180) yields

$$\frac{\mathcal{T}\psi}{b\psi} - \partial_q \log \psi \leq qe^{q/D} + 3\sqrt{2}(q+1)e^{q/D} + 1, \quad (181)$$

$$\partial_q \log \psi - \frac{\mathcal{T}\psi}{b\psi} \leq 4q + 1. \quad (182)$$

Indeed $\mathcal{T}\psi/\psi \leq e^{q/D}S$ and $\mathcal{T}\psi/\psi \geq e^{-4q/D}\lambda$; use $e^x - 1 \leq xe^x$, $1 - e^{-x} \leq x$ and $e^{q/2}/b \leq q+1$.

For $L \geq 0$ an integer, put $I_L(s, u, k) = (s+L, u, k+L)$. Each original transition i is exactly the padded transition $i+L$, with L additional nonnegative transitions. Hence

$$\mathcal{T}(f \circ I_L) \leq (\mathcal{T}f) \circ I_L \quad (f \geq 0). \quad (183)$$

Fix Q , choose $L = \lceil Q \rceil$, and set

$$V(q) = \frac{e(1+3\sqrt{2})}{2}q^2 + (3\sqrt{2}e+1)q.$$

For $0 \leq q \leq Q$, padded states have $D \geq L + 1$, so $e^{V(q)}\psi(q, I_L x)$ is a supersolution of $\partial_q F = \mathcal{T}F/b$. Equation (182) makes $e^{-2q^2-q}\psi(q, x)$ a subsolution. Both equal 1 at $q = 0$.

For completeness, comparison on the infinite state space requires more than a finite matrix inequality. Give each of the m transitions rate 1. The resulting chain has generator $\mathcal{T} - mI$, and is nonexplosive by domination of its degree by a Yule process. Expanding jump paths and cancelling holding-time factors gives the positive representation

$$F(t, x) = \mathbb{E}_x \exp \left(\int_0^t m(X_v) dv \right), \quad (184)$$

with infinity initially allowed. Stop at the exit from $m \leq M$. Finite-state comparison, discarding a nonnegative supersolution exit term and then letting $M \rightarrow \infty$, proves the upper bound below and finiteness for $t < T$.

For the lower comparison at fixed $t < T$, choose $\eta > 0$ with $t + \eta < T$ and a padded supersolution valid through that later time. Uniformly for $0 \leq v \leq t$, the ratio of the subsolution at v to this supersolution at $v + \eta$ is at most $Ce^{-cm(x)}$. To see this, both $p(q(v + \eta)) - p(q(v))$ and $q(v + \eta) - q(v)$ have strictly positive compact minima. Rank and fixed padding terms contribute bounded factors, while the s, u terms give the exponential decay. In the stopped representation the lower exit term is therefore at most Ce^{-cM} times the later supersolution at the initial state. It vanishes as $M \rightarrow \infty$. Consequently

$$e^{-2q^2-q}\psi(q, x) \leq F(t(q), x) \leq e^{V(q)}\psi(q, I_{\lceil Q \rceil} x), \quad 0 \leq q \leq Q. \quad (185)$$

There is no appeal to uniqueness for an unbounded differential system. Since $\log \psi(q, I_L x) - \log \psi(q, x) \leq Lp(q) + q$, taking $Q = q$ gives

$$\log F(t(q), x) = \log \psi(q, x) + O((q + 1)^2), \quad (186)$$

with constants independent of the state x .

For $x_N = (N, 1, N)$, the profile logarithm is $Np + q\mathcal{R}/(N + \mathcal{R})$, so the uniform estimate becomes

$$\log G_N(t(q)) = Np(q) + O((q + 1)^2), \quad N \geq 1. \quad (187)$$

The same comparison at x_* , where $F(t, x_*) = A(t)$ and $\log \psi(q, x_*) = q$, also gives the upper coefficient bound directly. Choose $q = 2 \log n + 2 \log \log n + O(1)$, so that $T - t(q) = O(1/n)$ by (177). Positivity implies

$$c_n \leq A(t(q))(T/t(q))^n,$$

whose logarithm is $O(q^2) + O(1) = O((\log n)^2)$. Thus the upper half of (152) uses only the coarse comparison, not the earlier sharp cumulative theorem.

37.3 Concentration for the large seed

Let $\ell(q) = (\log t(q))'$ and

$$M(q) = \frac{p'(q)}{\ell(q)} = t(q)p_t(t(q)) = \frac{t(q)e^{p(q)}(1 - e^{-q})}{q\mathcal{R}(q)}. \quad (188)$$

Explicit differentiation and (177) give

$$M(q) \sim \frac{Te^{q/2}}{\sqrt{2q}}, \quad \ell(q) \sim \frac{qe^{-q/2}}{\sqrt{2T}}, \quad (189)$$

and M is eventually strictly increasing. Under the tilted law

$$\mathbb{P}(J = j) = \frac{b_j t(q)^j}{j! G_N(t(q))},$$

we claim that, for all sufficiently large q and $0 < \epsilon < 1/2$,

$$\mathbb{P}(|J - NM(q)| > \epsilon NM(q)) \leq 2 \exp\{-cN\epsilon^2 + C(q+1)^2\}. \quad (190)$$

Here and below constants are independent of N .

To prove this, define

$$H_q(v) = p(q+v) - p(q) - M(q)[\log t(q+v) - \log t(q)].$$

Then $H_q(0) = H'_q(0) = 0$ and, uniformly for $|v| \leq 1$,

$$H''_q(v) \longrightarrow \frac{1}{4}e^{-v/2}, \quad M(q)\ell(q+v) \longrightarrow \frac{1}{2}e^{-v/2}.$$

For large q , these imply $|H_q(v)| \leq v^2/2$ and

$$M(q)|\log t(q \pm \eta) - \log t(q)| \geq \eta/4, \quad 0 \leq \eta \leq 1.$$

Apply the exponential Markov inequality by comparing the positive series at q and $q \pm \eta$. Formula (187) bounds either tail exponent by

$$N\eta^2/2 - N\epsilon\eta/4 + C(q+1)^2.$$

With $\eta = \epsilon/8$ this is $-3N\epsilon^2/128 + C(q+1)^2$, proving (190).

38 A quadratic cost extension bound

A path ending at degree m has at least

$$(m-h)^h \quad (191)$$

continuations of length h whenever h is a nonnegative integer and $m > h$. Indeed the degree before each successive extension is at least $m-h$, because a transition decreases it by at most one. The stronger falling product $m(m-1) \cdots (m-h+1)$ is available but unnecessary. The useful point is to retain all these continuations, rather than one prescribed continuation at each step.

38.1 Choosing the seed and controlling the mean length

For large n , choose

$$N = \left\lceil \sqrt{n \log n} \right\rceil, \quad \epsilon = \frac{D_0 \log n}{\sqrt{N}}, \quad NM(q) = (1-4\epsilon)n, \quad (192)$$

where D_0 is a fixed sufficiently large constant. Eventual monotonicity of M gives a solution q on its increasing branch. Equation (189) implies

$$q = \log n + \log \log n + O(1), \quad \epsilon \rightarrow 0, \quad N = o(\epsilon n), \quad (q+1)^2 = o(N). \quad (193)$$

The Chernoff interval $(1 \pm \epsilon)(1 - 4\epsilon)n$ lies inside $[n - 5\epsilon n, n - 3\epsilon n]$. Since $N\epsilon^2 = D_0^2(\log n)^2$, choosing D_0 sufficiently large in (190) ensures that

$$\mathcal{C} = \{n - 5\epsilon n \leq J \leq n - 3\epsilon n\} \quad \text{has probability at least } 7/8. \quad (194)$$

A high-probability central event by itself does not bound the mean. For that additional step, $v \mapsto \log G_N(e^v)$ is convex, with derivative $\mathbb{E}J$ at $v = \log t(q)$. The forward secant and (187) give

$$\begin{aligned} \mathbb{E}J &\leq \frac{N[p(q+1) - p(q)] + C(q+2)^2}{\log t(q+1) - \log t(q)} \\ &\leq C[N + (q+1)^2]M(q) = O(n). \end{aligned} \quad (195)$$

Here $p(q+1) - p(q) = O(1)$ and the denominator is comparable to $1/M(q)$, as follows directly by integrating (189) over a unit interval. The last equality uses (192) and $q^2 = o(N)$.

Now (169), (195) and Markov's inequality for the square imply, for a fixed sufficiently large A_0 ,

$$\mathcal{D} = \{|J - t(q)m| \leq A_0\sqrt{n}\} \quad \text{has probability at least } 7/8. \quad (196)$$

Therefore $\mathbb{P}(\mathcal{C} \cap \mathcal{D}) \geq 3/4$.

38.2 Extracting a useful coefficient and retaining its degree

Write $t = t(q)$. For each integer j in the central interval, let K_j count length- j paths from x_N ending at degree $m \geq j/t - A_0\sqrt{n}/t$. As t is eventually bounded away from zero, a fixed B_0 can replace A_0/t . There are at most $n+1$ possible central integers. Hence some j satisfies

$$\frac{K_j t^j}{j!} \geq \frac{G_N(t)}{2(n+1)}, \quad m \geq j/t - B_0\sqrt{n} \quad \text{for every counted path.} \quad (197)$$

This follows by summing the tilted mass over $\mathcal{C} \cap \mathcal{D}$ and then applying the pigeonhole principle.

Set $h = n - N - j$. Equations (193) and (194) show $h \geq 0$ and $h \leq 5\epsilon n = o(n)$ for all sufficiently large n . Also $j \sim n$ and $t \rightarrow T$, so the base below is positive. Prefix each path by the fixed length- N increasing path and then use (191). This gives the exact counting inequality

$$a_n \geq K_j (j/t - B_0\sqrt{n} - h)^h. \quad (198)$$

Different transfer paths remain distinct after prefixing and extension, even when their terminal states agree. Thus (198) does not assume individual coefficient regularity or extract a local coefficient asymptotic from the cumulative theorem.

38.3 Cancellation of factorial costs

Since $j = n - N - h$, write

$$j/t - B_0\sqrt{n} - h = \frac{n}{t}(1 - z), \quad z = \frac{N + (1+t)h + B_0t\sqrt{n}}{n} = o(1). \quad (199)$$

Use $\log(1 - z) \geq -2z$ eventually and $\log(n!/j!) \leq (N + h)\log n$. Combining (187), (197) and (198) gives

$$\begin{aligned} \log c_n &\geq Np(q) - N\log n + n\log(T/t) + N\log t \\ &\quad - C \left((q+1)^2 + \log(n+1) + \frac{hN}{n} + \frac{h^2}{n} + \frac{h}{\sqrt{n}} \right). \end{aligned} \quad (200)$$

For clarity, before simplification the lower bound is

$$\log G_N(t) - \log(2(n+1)) + \log(j!/n!) + n \log T - j \log t + h \log(j/t - B_0\sqrt{n} - h).$$

The $h \log n$ term from the many extensions cancels that portion of the factorial loss, leaving only $-N \log n$ in (200). The last positive terms $n \log(T/t)$ and $N \log t$ may be discarded, since eventually $1 < t < T$.

The remaining seed cost has a useful exact cancellation as well. Equations (188) and (192) give

$$\begin{aligned} p(q) - \log n &= \log(q/N) + \log \frac{(1-4\epsilon)\mathcal{R}(q)}{t(q)(1-e^{-q})} \\ &= \log(q/N) + O(1). \end{aligned} \tag{201}$$

Thus

$$\log c_n \geq -C \left(N \log(N/q) + N + q^2 + \log n + \frac{hN}{n} + \frac{h^2}{n} + \frac{h}{\sqrt{n}} \right). \tag{202}$$

Every loss is at most a constant multiple of $\sqrt{n}(\log n)^{3/2}$. More explicitly,

$$\begin{aligned} N \log(N/q) &= O\left(\sqrt{n}(\log n)^{3/2}\right), \\ h^2/n &= O(\epsilon^2 n) = O\left(n(\log n)^2/N\right) = O\left(\sqrt{n}(\log n)^{3/2}\right), \\ hN/n &= O(\sqrt{N} \log n) = O\left(n^{1/4}(\log n)^{5/4}\right), \\ h/\sqrt{n} &= O\left(\sqrt{n/N} \log n\right) = O\left(n^{1/4}(\log n)^{3/4}\right), \\ q^2 + \log n &= O((\log n)^2). \end{aligned}$$

Together with the upper bound from Section 37, this proves (152). Lemma 36.1 with $E(n) = C\sqrt{n}(\log n)^{3/2}$ now gives

$$R_n = \mu(n+1) + O\left(n^{3/4}(\log n)^{3/4}\right).$$

Dividing by $n+1$ proves (153).

39 The improved inverse enclosure

The counts a_n are nondecreasing because each transfer state has at least one child. They tend to infinity by the lower enclosure and Stirling's formula. Write

$$f(v) = v \log \frac{v}{eT}, \quad \log a_n = f(n) + e_n.$$

Stirling's formula and (152) imply

$$-C\sqrt{n}(\log n)^{3/2} \leq e_n \leq C'(\log n)^2 \tag{203}$$

for sufficiently large n ; the Stirling correction $O(\log n)$ is absorbed. Equation (154) gives $f(x) = y$, and $f'(v) = \log(v/T)$.

For fixed sufficiently large K_1 , put $n_- = \lfloor x - K_1 \log x \rfloor$. On this $o(x)$ window, $f'(v) \sim \log x$, and therefore

$$f(n_-) \leq y - (K_1 + o(1)) \log^2 x.$$

The upper bound in (203) makes $\log a_{n_-} < y$ when K_1 exceeds the corresponding error constant. Monotonicity gives $N_a(y) > n_-$.

Similarly, for a fixed sufficiently large K_2 , put $n_+ = \lceil x + K_2\sqrt{x \log x} \rceil$. On this $o(x)$ window,

$$f(n_+) \geq y + (K_2 + o(1))\sqrt{x}(\log x)^{3/2}.$$

The lower error bound in (203) then gives $\log a_{n_+} \geq y$, and $N_a(y) \leq n_+$. This proves (155), including integer rounding. It improves the previous right displacement $O(x^{2/3}(\log x)^{2/3})$; the left displacement remains $O(\log x)$. It does not justify a logarithmic-square correction to the integer inverse, because the full individual law (160) is still open.

40 The remaining pointwise problem

One sufficient route from the earlier cumulative law (157) to (160) would be the sharper one-sided estimate

$$R_n \geq \mu(n+1) - O(\log n). \quad (204)$$

It would imply $\log(c_{n+1}/c_n) \geq -C \log n/n$ eventually. For completeness, this weak regularity condition forces the individual logarithmic-square law for any positive sequence with $\log S_n \sim \alpha(\log n)^2$, $\alpha > 0$. If along a subsequence $\log c_n \leq (\alpha - \delta)(\log n)^2$, choose fixed $\beta < 1$ sufficiently close to 1 that $C(1 - \beta^2)/2 < \delta/2$. Summing the one-sided increment bound backwards gives, uniformly for $n^\beta \leq k \leq n$,

$$\log c_k \leq [\alpha - \delta/2 + o(1)](\log n)^2.$$

Consequently

$$S_n \leq S_{\lfloor n^\beta \rfloor} + (n+1) \exp\{[\alpha - \delta/2 + o(1)](\log n)^2\},$$

whose logarithmic-square coefficient is smaller than α , a contradiction. The present raw-ratio error $O(n^{3/4}(\log n)^{3/4})$ is much larger than $O(\log n)$, so it supplies no such implication.

There is also an exact variance formulation. Factorial-normalized log-concavity at adjacent indices is equivalent to

$$\text{Var}_n(m) + \mathbb{E}_n(u - k) \leq \frac{R_n^2}{n+1}. \quad (205)$$

Indeed $a_n/n!$ is log-concave at the center $n+1$ exactly when $R_{n+1}/(n+2) \leq R_n/(n+1)$; substitute (166). If (205) held eventually, $R_n/(n+1)$ would be nonincreasing and, by the proved ratio limit, never below μ thereafter. Thus c_n would be eventually nondecreasing. Then $S_n \leq O(1) + (n+1)c_n$, and (157) would prove (160). The premise (205) remains unproved.

[write] 2 October 2026. This is Part II's target (67), rewritten with $R_n = M_n$. The conditional chain from normalized log-concavity to $\log c_n \geq 0$ is also in Part II, Section 17. The same chain from eventual monotonicity of c_n to the pointwise law is in Part III, (141).

40.1 An exact counterexample to a childwise strengthening

Let $f_r(x) = (\mathcal{T}^r \mathbf{1})(x)$. One tempting invariant would require, for every transition $x \rightarrow y$,

$$r f_r(y) f_{r-1}(x) \leq (r+1) f_{r-1}(y) f_r(x). \quad (206)$$

It is false. Take $r = 19$, $x = (5, 1, 4)$ and $y = (4, 2, 4)$, where the transition selects the repeat label $i = 4$. Exact recurrence evaluation gives

$$\begin{aligned} f_{18}(x) &= 328610526895548, & f_{19}(x) &= 2835761210180706, \\ f_{18}(y) &= 1125633749884382, & f_{19}(y) &= 10225957430377930. \end{aligned}$$

The two sides of (206) are respectively

$$\begin{aligned} L &= 63846787924950777938074214657160, \\ R &= 63840570495847624656576542673840, \end{aligned}$$

and

$$L - R = 6217429103153281497671983320 > 0.$$

This is a reachable-state obstruction: six largest-new-label steps reach $(6, 1, 6)$; selecting the repeat label $i = 4$ reaches x , and selecting it again reaches y . The archive’s independent memoized recurrence reproduces all four counts and the exact integer difference. The counterexample rules out this childwise assertion. It does not refute the statewise log-concavity conjectures below.

[write] 2 October 2026. Scope of the counterexample. The childwise inequality (206) is not stated in Parts I–III, nor in any other report of the collection; this manuscript introduces it as “one tempting invariant”. The counterexample therefore refutes no delivered and no repository claim. The statewise property that Part II, Section 17, tests on 204 states with $j \leq 15$ is (208) below, now scanned much further.

40.2 Finite evidence and its limits

Two exact integer scans found no violation, each in 48,510,450 inequalities, over all valid states and center indices satisfying

$$1 \leq r \leq 179, \quad H(x) + r + 1 \leq 200, \quad H = s + 2u. \quad (207)$$

The first tested factorial-normalized log-concavity,

$$(r + 1)f_r(x)^2 \geq rf_{r-1}(x)f_{r+1}(x). \quad (208)$$

The second tested the stronger rising-factorial normalization,

$$(r + m)f_r(x)^2 \geq (r + m - 1)f_{r-1}(x)f_{r+1}(x), \quad m = s + u. \quad (209)$$

Here $(m)_r = m(m + 1) \cdots (m + r - 1)$, so (209) is log-concavity of $f_r(x)/(m)_r$. For $m \geq 1$ it is at least as strong as (208); the two agree when $m = 1$, in particular at x_* . At $r = 1$, the slack in (209) is exactly $m(s + k)$, with equality when $s = k = 0$.

The resource restriction is a safe finite domain, not an uncontrolled truncation. Every step changes H by ± 1 . To compute all $f_j(x)$ with $H(x) \leq 200 - j$, the preceding layer through $H \leq 201 - j$ therefore contains every child needed. At layer $j = r + 1$, the three values in a test have been computed exactly. All arithmetic in these two scans is Python integer arithmetic. Their root arrays through a_{180} agree. These finite results suggest useful targets, but they prove neither eventual log-concavity nor any asymptotic estimate.

41 Reproduction and source scope

The archive contains this PDF and editable TeX source, the unchanged predecessor PDF and TeX, portable build and replay entry points, verification scripts, frozen exact outputs and SHA-256 checksums. It contains no third-party paper or private working correspondence. Its README gives the required software and exact commands.

Run `bash replay.sh core` after extraction. The core replay checks the manifest, exact transfer identities on a finite state set, forward moments and root counts, the childwise counterexample,

and a coefficientwise version of the tilted square identity. The separate full replay, `bash replay.sh full`, regenerates both scans in (207) and compares the outputs with the frozen files. Replay outputs go to a separate build directory; they do not overwrite frozen results. All mathematical verification scripts use only the Python standard library. Rebuilding the PDF uses standard pdfLaTeX packages listed in the source and requires no network access.

The new moment identities, the excursion proof, and the improved extension argument have received independent analytical review. This is not a formal proof-assistant certificate. Computational checks corroborate the exact recurrence implementations and finite examples; the uniform and asymptotic estimates are proved by the arguments in the text.

[write] **2 October 2026.** In this repository the scripts are shipped as `code/08-ratio-*`, the frozen outputs as `data/08-ratio-results-*`, and the provenance note as `08-ratio-PROVENANCE.md`. The PDF, the SHA256SUMS manifest and the `prior/` copy of Part III are not shipped. The shipped `replay.sh` still uses the delivery layout and `python3`; the report README gives working commands. The two full scans were not rerun at intake.

The primary external attribution is [1] for the recurrence and its compaction. Sequence identifiers come from [2, 3]. The earlier analytic work is separately attributed to the predecessor, Part III of this report. No worldwide novelty or exhaustive literature-search claim is made, and no external repository, publication or sequence entry has been modified. The open problem is precisely the all-index lower bound at the logarithmic-square scale; the proved ratio limit does not close it.

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